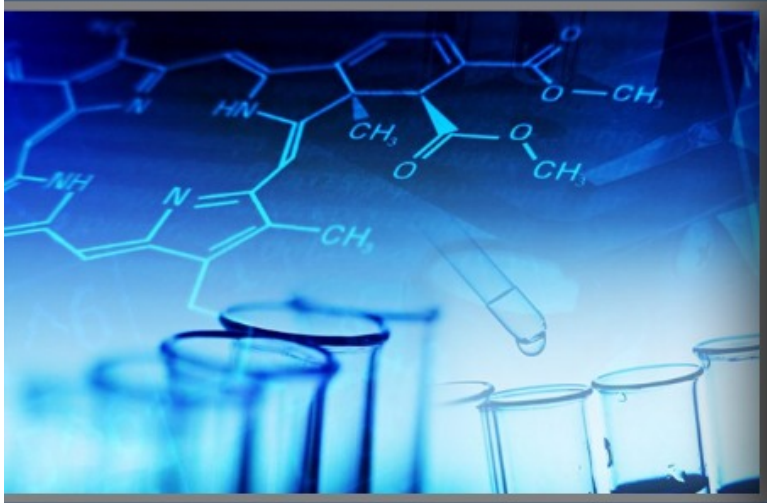
A Study Material for Applied Chemistry





Dr. R. S. S. Srikanth Vemuri
Dr. Madhavi Konni

A Study Material for Applied Chemistry

First Edition

Authors

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Preface

"A Study Material For Applied Chemistry" book is exclusively dedicated to First Year B. Tech Graduate students (CSE, ECE & IT branches) under JNTU Kakinada affiliated colleges R20 regulation. This book prepared as per the syllabus prescribed by JNTU Kakinada. This material is well written and easy to understand the concepts. This book is essential for Semester exam preparation. All topics covered and suggested important questions which are coming in semester end exam of Applied Chemistry Paper.

This book contains 5 chapters (Units) as per the prescribed syllabus by JNTU Kakinada. First chapter (Unit) is Polymer technology, the outcome of this chapter is at the end of this chapter student can be able to understand the importance of plastics and knows the fabrication process of plastics. In this chapter, authors explained bio medical, bio – degradable and bio polymers separately.

Second Chapter (unit) is Electrochemical Cells & Corrosin, the outcome of this chapter is at the end of this chapter student can be able to identify different types of batteries and apply the corrosion control methods in real life. This Chapter devides into two parts. Part A deals about electrochemical cells and different types of batteries and Part B deals about corrosion factors and types of corrosion and its control methods.

Third Chapter (Unit) is Material Chemistry, This unit devides as part A & B. Because **Mid I exam portion is upto part A of this chapter**. Part A consists of Semiconductors preparations and applications, Magnetic materials and Hall Effect, and Electrical insulators and their applications. **Mid II exam portion starts from part B**. Part B contains Nano materials, Carbon nanotubes, Fullerenes, Graphene, Liquid Crystals and Super conductors. At the end of this chapter student can be able to learn about various materials which are used in different fields like electrical and medical and chemical industries.

Fourth Chapter (Unit) is Spectroscopic techniques and Non conventional energy sources. At the end of this chapter student can be able to learn advanced analytical techniques like UV-Vis, FT- IR, NMR spectroscopic techniques and understand the alternate energy sources for non – conventional energy sources. Solar cells and Ocean Thermal Energy Conversation are promise techniques in future energy.

Fifth Chapter (Unit) is Advanced Topics in Chemistry deals Introduction Computational Chemistry and Molecular machines. These new topics are introduced in this syllabus from last year. Especially introduction to computational chemistry is very new and to these level students, so far there is no sufficient information available for this level student hence this book is very helpful. At the end of this chapter students will learn the basic information about computational chemistry and molecular mechanics.

Acknowledgement

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Unit – 1

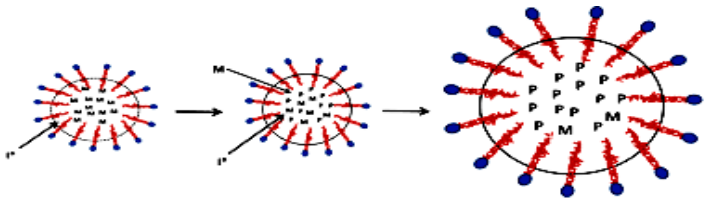
Polymer Technology

1.1 Emulsion Polymerization method

In this method the monomers & water soluble initiators forms the emulsion with water. The surfactants such as soap, detergents, etc are add which hold the monomers & initiators (H_2O_2 or persulphate) in the form of a micelles. Polymerization takes place inside the micelle on exposing to heat or radiation. De-emulsifying agents (3% solution of $Al_2(SO_4)_3$)& Filtration can isolate the obtained polymer.

Monomer + Initiator + surfactant ⟶ polymer

(Dissolved in inert solvent) (water soluble) (Emulsion in water)



Advantages

- High molecular weight polymers can be obtained
- Polymer process is fast
- Final product can directly use for fabrication
- High purity product is obtained.
- This method is used to prepare synthetic rubbers like BuNa-S, acryl rubber, Teflon, paints or adhesives etc.,

1.2 Suspension Polymerization Method (Pearl Polymerization method)

In this method water insoluble monomers are suspended in water as tiny droplets by continuously agitating (shaking). Initiators are soluble in monomers. Monomers and initiator forms heterogeneous mixture with the water. Polymerization is initiator by heating or exposing to radiation.

As the polymerization takes place, the polymer is obtained as spherical pearls or beads; hence it is also called as pearl polymerization methods.

Since the product is insoluble in water. It can be easily separated just by filtration & the product needs no further purification.

Monomer + Initiator + suspending agent ⟶ polymer

(Suspension in water) (dissolved in monomer) (suspended in water as beads)

Advantages

- Viscosity build up is negligible
- Better heat transfer can be achieved.
- High purity product is obtained.
- Isolation of the product is easy and it does not require any purification.

Disadvantages

- Few monomers are water soluble.

Uses

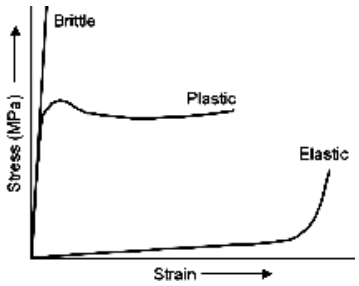
- This method is used for preparing PVC, PVA (Polyvinyl acetate), styrene, divinyl benzene (ion exchange)

1.3 Mechanical Properties of Polymers

Mechanical Properties

1. **Toughness:** Below the Tg temperature the polymer break with a brittle fracture and they become tougher. The size of the crystalline structure also determines the toughness. The smallest spherical crystalline masses increase toughness.

2. **Strength:** The strength of a polymer depends on length of polymer chain, branching or cross-linking which are controlled by elastomeric properties, crystallinity and branching of chain. The strength of a polymer increases with increase in molecular weight. The intermolecular forces of attraction, presence of polar groups and chain length increases the strength of the polymers. A polymer strength is determined by stress-strain test



3. **Elasticity:** Elasticity is the property of recovering original shape after the removal of deforming stress. Natural rubbers possess high elasticity due to coiled structure.

1.4 Compounding of Plastics

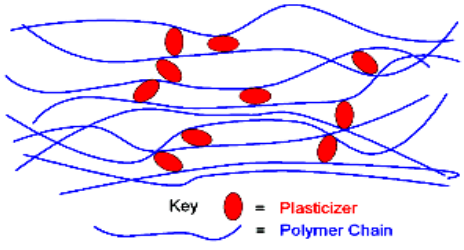
During moulding, different constituents are added to impart the required property. These constituents that are added to plastic during its moulding process are called moulding constituents.

Ingredients used in compounding of plastics

Binder (or) resin: Most important component for compounding. Binder holds different constituents together. Large molecular weight compounds are usually used as binders.
Eg: resin, glue, gum etc.

Fillers or Extenders: Fillers are generally added to thermosetting plastics to increase tensile strength, hardness and crack resistance. Fillers also reduces the cost of final of the polymers.
Eg: wood flour, Asbestos, Mica, Cotton, Carbon black, ZnO, PbO etc.

Plasticizers: Plasticizers are substances added to enhance the plasticity and flexibility of the material and to reduce the cracking on the surface. Plasticizers also increase the flow property of the plastics. Plasticizer neutralizes the intermolecular forces of attraction between polymer chains.



Eg: Dibutytyle oxalate, Castor oil and Tri cresylphosphate, vegetable oil, camphor etc.,

Lubricants: These are added to prevent the plastics from sticking to the moulds and they impart glassy finish to the final product
Eg: Oils, Waxes and Soaps etc.

Stabilizers: They are added during compounding to prevent various adverse effects during processing or during end use. For example PVC undergoes decomposition and de colorization at moulding temperature. So during their moulding, stabilizers are used.

a) **Antioxidants:** These are can be used to prevent oxidation eg: di-tertiary butyl - p- cresol.

b) **Heat stabilizers:** To prevent thermal degradation. eg: white lead, lead chromate, red lead etc.

c) **Radiation absorbers**: Most of the polymer degrades quickly when exposed to UV radiation. Eg: carbon balck, titanium oxide.

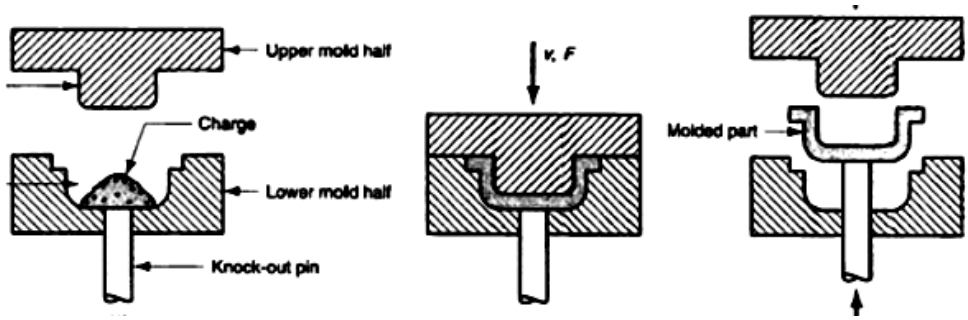
Colorants: These are added to impart the desired color to the plastics and give decorative effect. eg: organic dyes, Carbon black- black; Titanium oxide (white), CdS (yellow) and Pb_3O_4 (red)

Fabrication of plastics (or) Moulding of Plastics

Fabrication of plastics is the manufacturing process of plastic articles. The methods, usually depends upon the types of resins used i.e., whether thermosetting or thermoplastic. Different fabrication techniques are described below.

1.5 Compression Moulding

- This method is applied to **thermosetting resins.**
- The predetermined quantity of plastic ingredients in proper proportions is filled between the two half –pieces of mould which are capable of being moved relative to each other.
- The heat and pressure are than applied according to specifications.
- Two halves are closed very slowly.
- Under the influence of heat, the compacted material gets cured and hardened.
- After curing the moulded article is taken out by opening the mould parts.
- It is used for manufacturing rubber tyres, shoes, electric plug sockets, pot handles etc.
- Advantages: Low expensive and low maintenance
- Disadvantages: Longer cycle time.

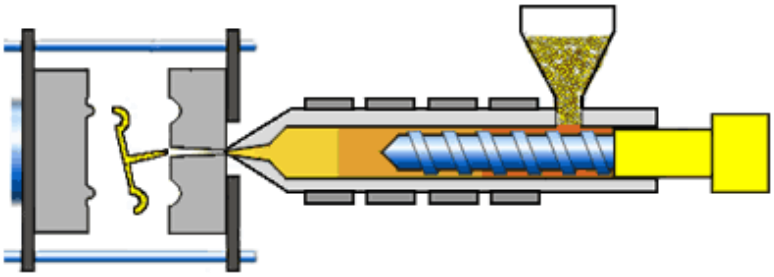


1.6 Injection Moulding

- It is carried out for producing articles made of ***thermoplastics*** *material*
- In this process, a correct amount of plastic material as granules, pellets or powder is added from the hopper into the cylinder.
- The plunger pushes the plastic material in to the heating chamber where the material is converted to viscous liquid.
- Again with the help of the plunger the molten plastic material is

injected through the nozzle into the mould cavity maintained at room temperature.

- When the materials have been cured sufficiently, half of the mould is opened to get the moulded article.



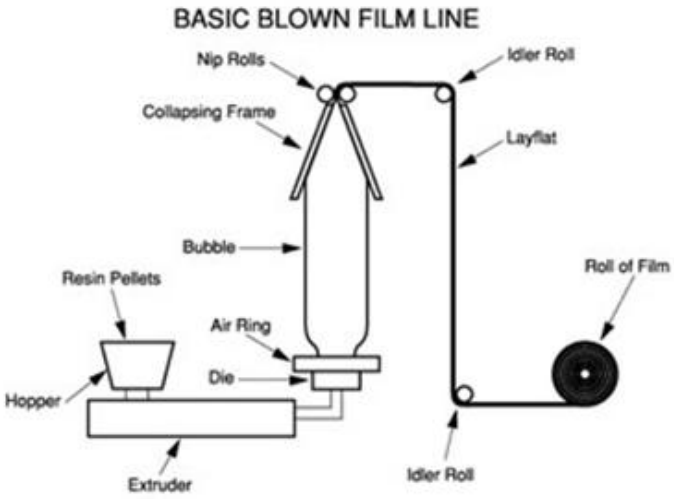
Advantages

- high speed production and low mould cost
- Loss of material will be less.

Disadvantage

- Large number of cavities cannot be filled, so there is a limitations to designed articles

1.7 Blown Film Method



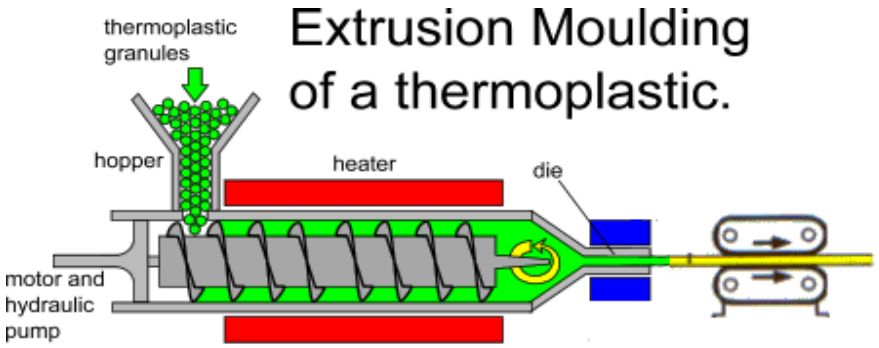
Blown Film is a manufacturing process that produces films of plastic sheeting and tubes. These sheets or tubes can either be fully cylindrical, or have other object inserted into them so they form useful objects like shopping bags, rubbish bags or cling wrap

The process of blown film is reasonably simple. First you must take plastic pellets and heat them to their melting point (like injection moulding). The plastic is then forced through a cylindrical die and forms a thin walled bubble. This bubble is sustained by air, and is fed up through a scaffold where the tube continues to rise. Over this length, the plastic begins to cool. Once it has cooled enough, it is collapsed down on itself, forming a flattened tube. This tube is passed through various rollers and spun onto a spool. The edges of the tube can be trimmed and it can also be punched, cut or printed on before going on the spool.

Blown film manufacturing is great for producing products that require uniform properties along the length and width of the product. It's also a great high volume production method; however it requires massive capital cost to purchase a blown film machine. Blown film can produce up to 130 metres of products in a single minute. The sheets and film they can produce can be as fine as 10-20 microns (0.01mm – 0.02mm) to 250 microns (0.25mm) thick. Most Blown film is either High or Low Density Polyethylene (HDPE or LDPE) but there are exceptions. The downside to blown film is that it can be only product sheets or tubes.

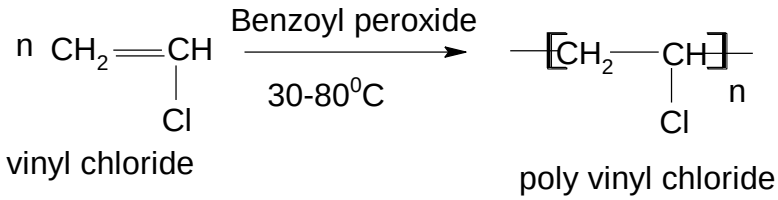
1.8 Extrusion Moulding

- Extrusion moulding is used mainly for continuous moulding of ***thermoplastic*** materials into articles of uniform cross section like tubes, rods, strips, insulated electric cables.
- In this process, a correct amount of plastic material as granules, pellets or powder is added from the hopper into the cylinder.
- The plunger pushes the plastic material in to the heating chamber where the material is converted to viscous liquid.
- The molten mass pushed by means of a rotating screw into a die, having the required outer shape of the article to be the manufactured.
- The material extruded out of the die is quite hot and carried over the conveyor belt.
- Here the plastic mass gets cooled, due to the atmospheric exposure (or artificially by airjets).
- Advantage: High production rate.



1.9 Poly Vinyl Chloride

Poly Vinyl Chloride is produced by free radical chain polymerization of vinyl chloride in presence of a small amount of benzoyl peroxide or hydrogen peroxide.



{Vinyl chloride, so needed is generally prepared by treating acetylene at 1 to 1.5 atmospheres with hydrogen chloride at 60°C to 80°C in the presence of metal chloride as catalyst.}

$$CH \equiv CH + HCl \longrightarrow CH_2 = CHCl$$

Acetylene — Vinylchloride

Properties

- It is a colorless, non-inflammable, chemically inert material.
- It is having low density and high softening point 148°C
- It is resistant to light, atmospheric oxygen, inorganic acids and alkalis.
- It is not stable to heat and UV radiation.
- Strong and brittle.

Uses

- It is mainly used as cable insulation, tank-lining, packing and toys.
- It is used for manufacturing of safety helmets, refrigerator components and floor covering.
- PVC pipes are used for carrying corrosive chemicals in petrochemical factories.

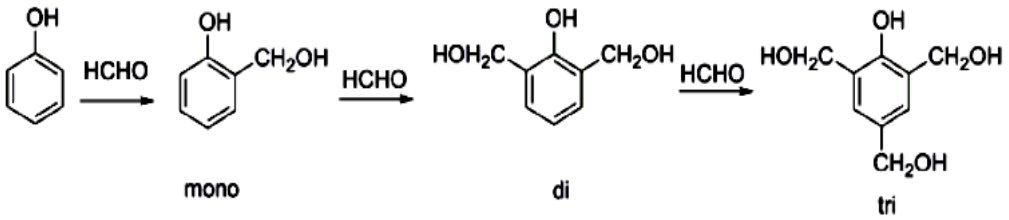
- Water carrying pipes, pipes for drainage and making frames for doors and windows.

1.10 Bakelite (Phenol Formaldehyde Resin)

It is prepared by condensing phenol with formaldehyde in presence of acidic/alkaline catalyst.

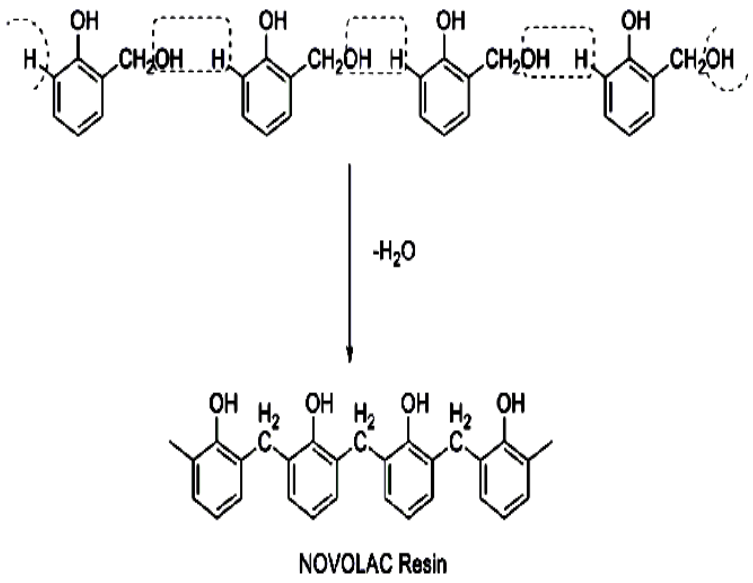
Preparation

Step1: Phenol reacts with formaldehyde in basic condition to form mono, di or tri methanolated Phenol (methylol phenol), depending on the formaldehyde ratio taken (1:1, 1:2 or 1:3)



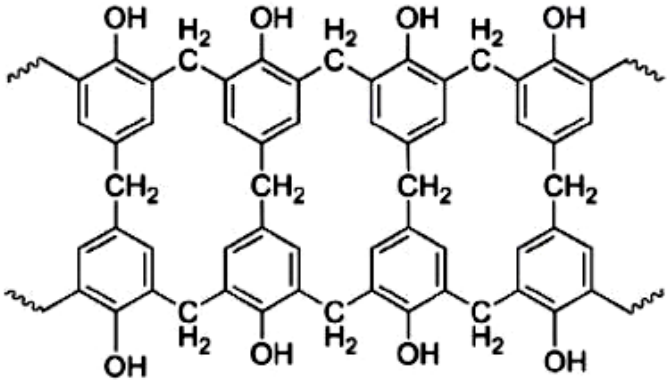
Step2: Preparation of PF resin

Mono-methanolated phenol derivative condense under acid catalysis to give a low molecular weight polymer called resin (NOVOLAC).



Step 3: Bakelite formation

This stage is done during moulding. NOVOLAC resin is mixed with additives and curing agent hexamethylenetetramine (HMTA) and heated. Methylene Bridge cross linking occurs between NOVOLAC resins. A hard insoluble material called Bakelite is formed.



Bakelite

Properties and Uses

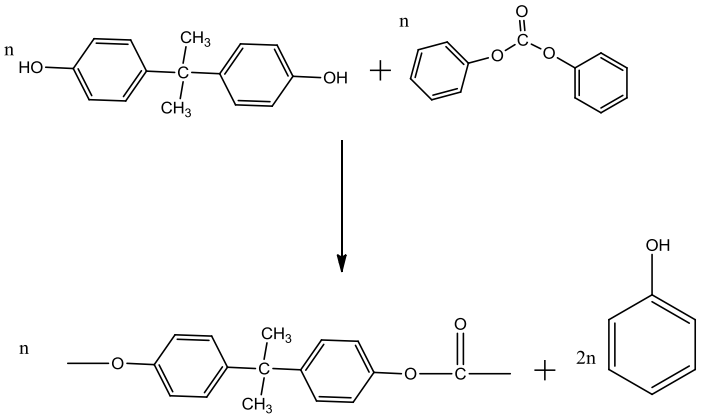
- It is electrically non-conducting, they are insulators. Used in the manufacture of circuit boards, sockets, plug points
- They are heat resistant. Used in the manufacture of handles for kitchen ware, heaters etc.
- They are resistant to oxidizing acids. Used for the manufacture of acid storage cans.
- It is a good adhesive. Used as adhesive for grinding wheels.
- It is a good ion exchanging resin so used in water purification by ion exchange method
- It has good chemical and corrosive resistance.

1.11 Polycarbonates

Polycarbonate (PC), known by the trade names lexan, makrolon, makroclear. These are a group of thermoplastic resins containing carbonate group in their structure and it is a epoxy resin. They contain [-O–CO–O-] linkage.

Synthesis

They are also prepared by condensation reaction of di phenyl carbonate and Bisphenol-A.



Properties

- Polycarbonate is a transparent and good electrical insulator.
- Polycarbonate has a high melting point, around 265°C.
- It has a high tensile strength and impact resistance.
- It is resistant to water and many organic compounds but alkalis slowly hydrolyze it.

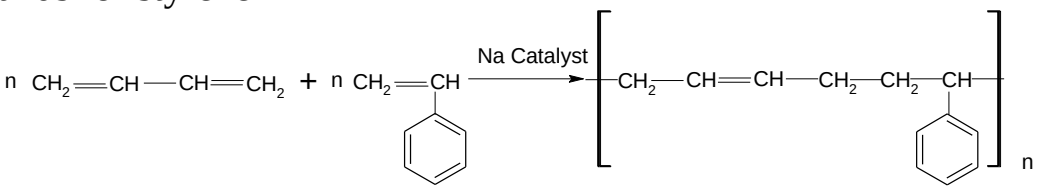
Applications

- It is used for making hair driers, electric razors
- It finds application in making many useful articles such as safety goggles, windows in automobiles, bullet proof glasses ,police safety shields, Telephone parts and automobile light lenses etc.,
- Polycarbonates are mainly used in CD-ROMs, DVD and baby bottles, syringes are well known examples

Elastomers

1.12 BuNa-S (styrene butadiene rubber) / SBR / GR-S

Buna – S prepared by the copolymerization of butadiene with styrene using sodium as a catalyst.
Bu stands for butadiene - monomer
Na stands for sodium –catalyst
S- Stands for styrene



Properties

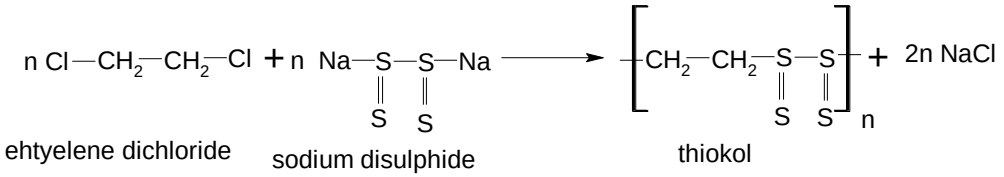
- It is a strong and tough polymer.
- It is vulcanized by sulphur monochloride or sulphur.
- It resembles natural rubber in processing characteristics and quality of finished products.
- It has excellent abrasion resistance and also good insulator.
- It is resistant to chemicals but swells in oils and attacked by even traces of ozone present in atmosphere.
- It has high load bearing capacity.

Applications

- Major applications of styrene rubber are in the manufacture of tyres.
- It is used in the foot wear industry for making shoe soles and foot wear components and
- It is also used for making wires and cables insulations.
- The other applications of styrene rubber are for the production of floor tiles, tank linings in chemical industries and as an adhesive.

1.13 Thiokol Rubber (or) GRP (or) Polysulphide Rubber

Thiokol rubber was prepared by co-poymerisation of sodiumpolysulphide (Na_2S_4) and ethylene dichloride produces Thiokol rubber.



Properties

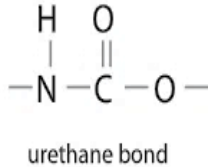
- It has strength and impermeability to gases.
- This rubber cannot be vulcanized and it cannot form hard rubber.
- It possesses extremely good resistance to mineral oils, fuels, oxygen, solvents, ozone and sunlight.
- Thiokol rubber possesses low abrasion resistance.

Applications

- Fabrics coated with thiokol are used for barrage balloons, life rafts and jackets which are inflated by CO2.
- Thiokol are used for lining hoses for conveying gasoline and oil.
- It is used for making gaskets and seals for printing rolls.

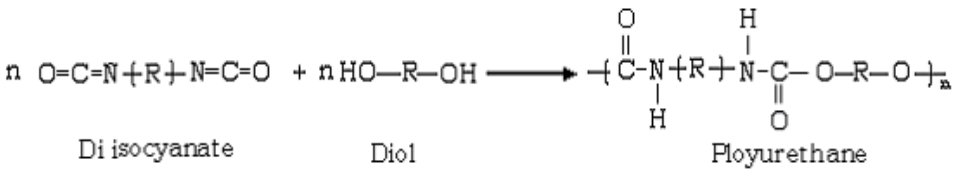
1.14 Polyurethane

Polyurethanes are produced by the polymerization of di-isocyanate and diol or triol (or the addition reaction between p-phenyl di-isocyanate with glycol).



urethane bond

Synthesis



Properties

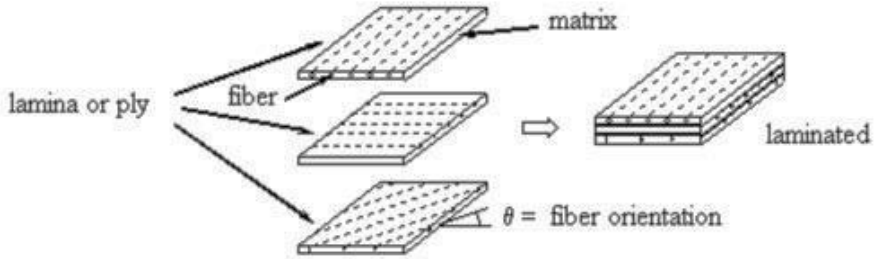
- These can be obtained in the form of foams, fibers, Elastomers, Coatings etc.
- They have high strength, hardness.
- Good resistance to gas, oil
- Excellent resistance to oxygen and ozone.

Uses

- Flexible foams are used for cushions in automobiles & furniture.
- Rigid foams are used to reinforce hallow structural units.
- Fibers of Polyurethanes are used in lightweight garments and swim suits because of their stretching property.
- These are used to coat gymnasium floor and dance floor.
- These are used in tyre treads and industrial wheels.

1.15 Composite Materials

Combination of two or more structurally different components combine to form a new class of material suitable for structural applications is referred as composite material. When one of the component is a polymer, the resulting composite is called a polymer composite.



Constituents of Composites

Two essential constituents of composites are

1. **Matrix Phase:** It is the continuous body constituents which encloses the composite and give its bulk form. Matrix phase may be metal, ceramics (or) polymers. Composites using these matrixes are known as metal matrix composites (MMC)**,** Ceramic Matrix Composites (CMC) and Polymer Matrix Composites (PMC) respectively.

Functions of Matrix Phase

- It binds the dispersed phase together.
- Acts as a medium to transmit and distribute an externally applied load to the dispersed phase.
- It protects the dispersed phase from chemical action and keeps in proper position and orientation during application of load.
- It prevents propagation of brittle cracks due to its plasticity and softness.

2. **Dispersed Phase:** It is the structural constituent, which determines the internal structure of composite. Important dispersed phases of composites are fibers, particles, whiskers, or flakes.

Some example of composite materials

1.16 Fiber Reinforced Plastics (FRP)

- FRP are produced by reinforcing a plastic matrix with a high strength fiber as dispersed phase.
- Natural fibers such as sisal, asbestos are used for reinforcement
- Depending on the desired properties of final composite the nature of fiber used is decided.
- FRP are produced by following methods.
  - Filament winding technique
  - Hand lay-up technique
  - Spray lay-up technique
  - Continuous lamination method
- Some important types of fiber reinforced composite are glass fiber reinforced composite, carbon fiber reinforced composite, aramid fiber reinforced composite etc.

1. Fiber glass (glass fiber reinforced composite)

- Glass fiber is most extensively used reinforced fiber because of durability, acid proof, water proof and fire proof nature of glass.
- Glass is drawn into threads or fibers into the form of filaments
- Then the filaments are woven in the form of mats
- The fiber material is suitably bonded with plastic material to be reinforced
- The common plastic resins used are polyester, epoxy, silicon, vinyl derivatives.

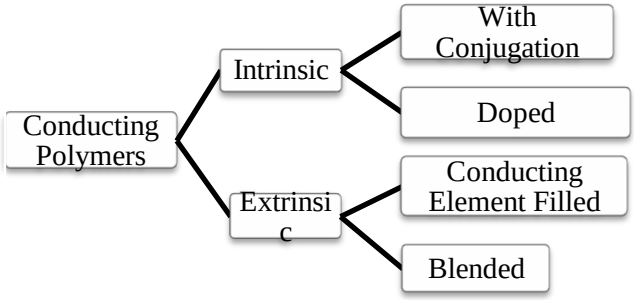
Applications composite materials

- Automobile parts and storage tanks are made from fiber glass
- It is used for making industrial floorings, plastic pipes
- It is used for making transport vehicles to reduce vehicle weight and boost fuel efficiency.

1.17 Conducting Polymers

Those polymers which conduct electricity are called conducting polymers. The conduction of polymers may be due to unsaturation or due to the presence of externally added ingredients in them.

The conducting polymers can be classified in the following ways.

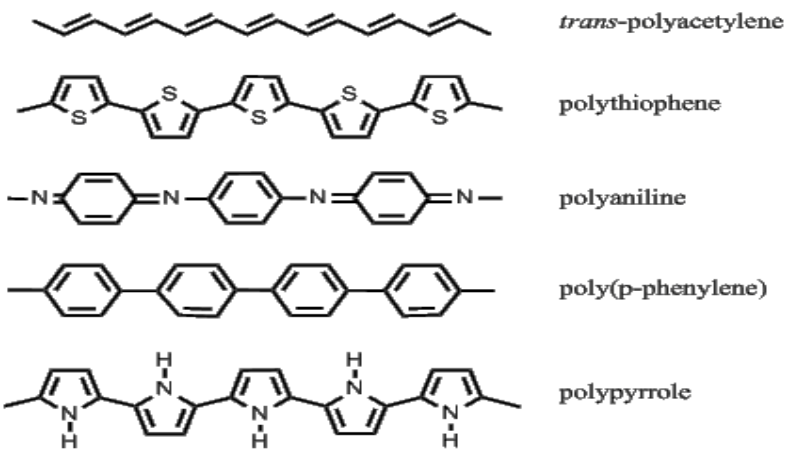


A. Intrinsic Conducting Polymers

Intrinsic conducting polymers have extensive conjugation in the backbone which is responsible for conduction. These are two types.

1. **Conducting polymers having conjugation:** These polymers having conjugated double bonds in their backbone. In such case, overlapping of orbitals having conjugated π electrons over the entire backbone forms the valence band as well as conduction band which were separated by a significant Fermi energy gap. The electrical conductivity takes place only after thermal or photolytic activation of the electrons, which give them sufficient energy to jump the electrons from valence band to conduction band.

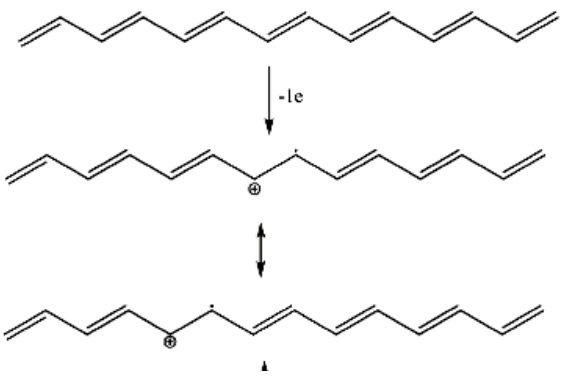
   Eg: polyacetylene, polyaniline, polypyrrole and polybutadienylene.



2. **Doped conducting polymers**: Intrinsic conducting polymers can be easily oxidized or reduced because of their low ionization energy and high electron affinities. The conductivity such polymers can be increased by creating either positive or negative charge on the polymer backbone by oxidation or reduction. This process is referred as doping, which is the following two types.

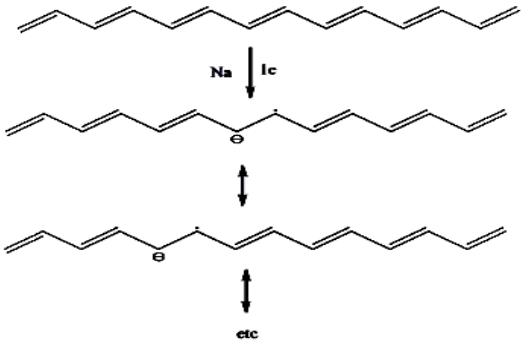
   a. **P-type doping:** Creating positive charge on polymer backbone is called p-doping. It is done by oxidation of a conducting polymer with a suitable oxidizing agent such as a lewis acid ($AlCl_3$, $FeCl_3$) or iodine vapor.

   $$\text{-(-}CH_2\text{= }CH_2\text{-)}_n\text{+ }FeCl_3 \longrightarrow (CH_2{=}CH_2)_n + FeCl_4^- + FeCl_2$$



   b. **n-type doping:** Creating negative charge on polymer backbone is called n-doping. It is done by reduction of a conducting polymer with a suitable reducing agent such as a lewis base (naphthylamine, sodium napthalide, etc.)

   $$\text{-(-}CH_2{=}CH_2\text{-)}_n\text{- + }Na^+(C_{10}H_8)^- \longrightarrow (\ CH_2{=}CH_2\ )^-_n\ Na^+ + C_{10}H_8$$



B. Extrinsic Conducting Polymers

The conductivity of these polymers is due to the presence of externally added ingredients in them. These are further two types.

a. **Conducting element filled polymers:** In such case, polymer acts as a binder to hold the conducting such as carbon black, metallic fibers and metallic oxides etc. The minimum concentration of conducting filler is added so that the polymer starts conducting. This minimum concentration of conductive filler is called percolation threshold. At this concentration of filler, a conducting path is formed in polymeric material.
Advantages: These polymers possess quite good bulk conductivity, light in weight, mechanically durable, can be easily molded and are of low cost.

b. **Blended conducting polymers:** These are obtained by blending a conventional polymer with a conducting polymer. Such polymers possess better physical, chemical, electrical and mechanical properties.

Factors affecting conductivity

1. **Band gap:** Band gap increases, conductivity decreases.
2. **Doping level:** Conductivity increases with increase in doping level
3. **Temperature:** The conductivity of a conducting polymer decreases with decrease in temperature**.**
4. **Conjugation length:** Conductivity increases with increase in conjugation (or) chain length.

Applications of conducting polymers

- Conducting polymers find their applications in rechargeable batteries.
- Conducting polymers also used for making analytical sensors for pH, O_2, NO_X, SO_2, NH_3 and glucose.
- Conducting polymers also used for making ion exchangers.
- Conducting polymers are used for making electronic displays and optical fibers.
- Conducting polymers are applicable in photovoltaic devices.

1.18 Biodegradable Polymers

Biodegradable plastics are plastics that are capable of being decomposed by bacteria or other microorganisms.
Eg: Aliphatic polyesters are biodegradable due to their enzymatically hydrolysable ester bond. These are classified as

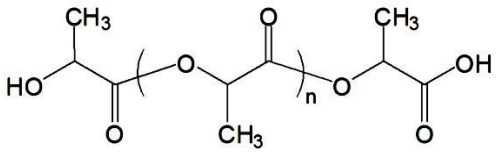
1. Natural

1. Polysaccherides-**Eg:** starch and cellulose.
2. Proteins – **Eg:** gelatin, casein, silk, wool
3. Polyesters- **Eg:** polyhydroxyalkanoates
4. Others- **Eg:** lignin, natural rubber.

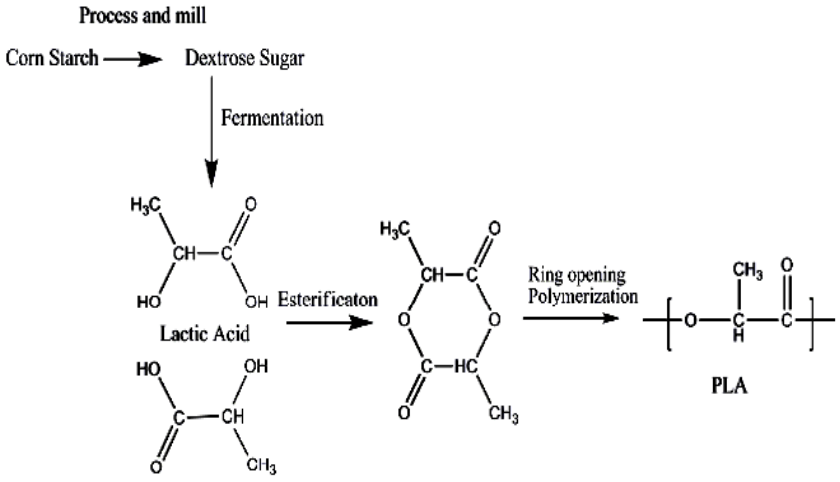
2. Synthetic
E.g: PVA, PLA,PHA, PHB etc.,

- **Polylactic acid/polylactide (PLA):**

- This is a biodegradable thermoplastic polyester belonging to the class of polyhydroxyalkanoates.

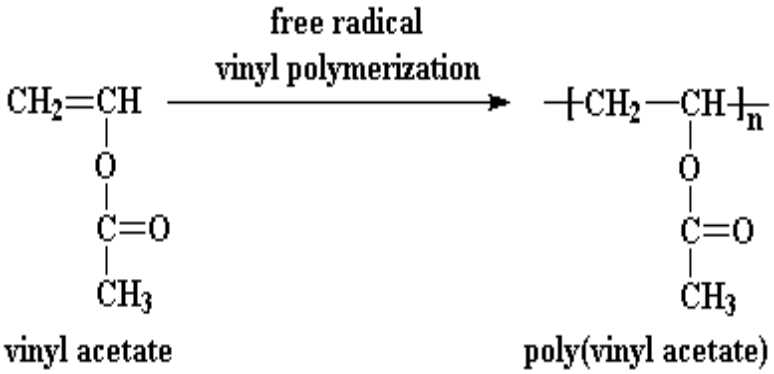


- It is derived from renewable resources (corn starch, tapioca roots, sugarcane etc.)



- **Polyvinyl acetate (PVA)**

- Polyvinyl acetate is a water soluble and biodegradable polymer, possessing excellent mechanical properties and compatibility with starch.



Properties of biodegradable polymers

- Moisture sensitive
- Sensitive towards Acids and bases
- Dissolves in chlorinated organic solvents
- Good oxygen permeability.
- PLA is biocompatible

Application of biodegradable polymers

- It is used in number of biomedical applications such as drug delivery, stents, dialysis media, medical implants like anchors, screws, pins, mesh, surgical threads etc.,

- Disposable table wares, packages for food and flushable sanitary products, diapers etc.
- Used in food industry as a packing materials, food storage and compostable bags, chewing gums.
- Adhesive for wood, paper & cloth for book bindings, envelops, wall paper, etc.

1.19 Biopolymers

- Biopolymers are natural polymers developed from the cells of living organisms. Biopolymers consist of large number of monomers which are covalently bonded.
- Examples of biopolymers are Proteins, Carbohydrates, DNA, RNA, Lipids, Nucleic acids, Peptides, Polysaccharides (glycogen, starch and cellulose), Natural rubber (polymer of isoprene), Suberin and Lignin (complex of polyphenolic polymers), Cutin and Cutan (complex polymers of long-chain fatty acids) and Melanin.
- Biopolymers are classified as three main classes according to the monomers involved in the structure of biopolymer formed.

1. **Polynucleotides** : Like RNA and DNA are long polymers composed of 13 or more nucleotide monomers.

2. **Polypeptides**: These are polymers of amino acids and some major examples include collagen, actin and fibrin.

3. **Polysaccharides**: These are linear or branched polymeric carbohydrates and examples include starch, cellulose and alginate.

Biopolymer Structure

Unlike synthetic polymers, Biopolymers have a well-marked structure. These polymers have a uniformly distributed set of molecular mass and appear as a long chain of worms or a curled up string ball under a microscope. This type of polymer is differentiated based on their chemical structure.

Biopolymer Uses

These polymers play an essential role in nature. They are extremely useful in performing functions like storage of energy, preservation and transmittance of genetic information and cellular construction.

- Sugar based polymers, such as Polyactides, naturally degenerate in the human body without producing any harmful side effects. This is the reason why they are used for medical purposes. Polyactides are commonly used as surgical implants.
- Starch based biopolymers can be used for creating conventional plastic by extruding and injection molding.
- Biopolymers based on synthetic are used to manufacture substrate mats.
- Cellulose based Biopolymers, such as cellophane, are used as a packaging material.
- These chemical compounds can be used to make thin wrapping films, food trays and pellets for sending fragile goods by shipping.

1.20 Biomedical Polymers

Biomedical polymers are similar to biopolymers which are macromolecular compounds obtained from natural origin. Examples are polysaccharides, protein and bacterial polyesters. Biomedical polymers can be classified as natural and synthetic polymers.

Properties

- These are light weight polymers and possess high flexible in nature.
- Show resistance towards biochemical attack and also shows biocompatibility.
- These substances and available in a wide variety of compositions with adequate physical and mechanical properties.
- These can be easily manufactured into products with the desired shape.
- Host response: The response of the host organism (local and systemic) to be implemented polymeric material or device.
- The ability of a material to perform (biocompatibility) with an appropriate host response in a specific application.
- Non – toxic materials.
- They show mechanical properties of biomedical polymers like tensile strength, yield strength and hardness etc.

Applications

- Cardiovascular applications: Heart valves and vascular prostheses, artificial heart and heart pump designs. Eg: silicone rubber, poly(tetrafluoroethylene)

- Bones, Joints and teeth: Repaired with the use of polyurethanes, epoxy resins and rapid curing vinyl resins. For elbow joints repair with vinyl polymers and nylon. For teeth poly methacrylate were used.
- Contact lenses and intraocular lenses
- Artificial kidney and Hemodilaysis materials
- Oxygen transport membrane
- Surgical sutures done by polyglycolic acid.
- Tissue ingrowth polymers
- Controlled release of drugs

Important Questions

1. Explain emulsion polymerization.
2. Write a note on suspension polymerization.
3. Write preparation, properties and applications of Bakelite.
4. Mention the synthesis, properties and applications of polycarbonate and polyurethane.
5. Explain compression moulding.
6. Discuss injection moulding.
7. Write a note on blown film method with the help of neat sketch.
8. Write the procedure of extrusion moulding technique for fabrication of plastics.
9. Define composite material. Write about fibre reinforced plastics.
10. What are conducting polymers? Explain.
11. List out applications of biomedical polymers.
12. Write a note on biodegradable polymers.
13. Discuss biopolymers.
14. Mention preparation and properties of Buna-S, Thiokol and PVC.

Beyond the syllabus (Not applicable for SEM & MID exams)

1.21 Physical Properties of Polymers

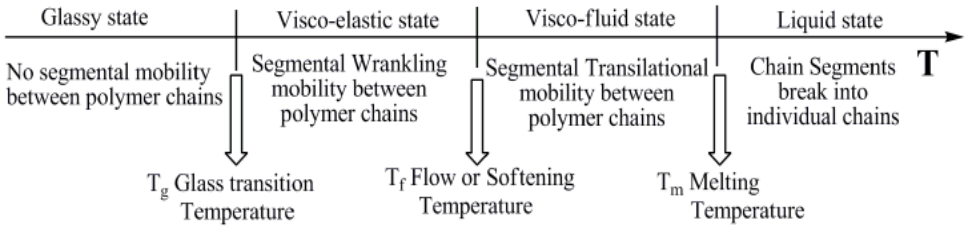
1. **Crystallinity:** Polymeric molecules does not have a definite structure like solids (cubic, ruby, face centered cube, body centered cube) structures. A polymer is a mixture of 60% crystalline and 40%amorphous. As the crystallinity increases, brittleness, strength and chemical resistance increases.

2. **Molecular weight:** The polymer molecule has different number of monomeric units hence different molecular weights. As a result the molecular weight of polymer should be expressed as average molecular weight ($\bar{M}$). There are two types of average molecular weights (M_n & M_w).

$$M_n = \frac{\sum M_i N_i}{\sum N_i}, \quad M_w = \frac{\sum M_i^2 N_i}{\sum M_i N_i},$$

3. **Solubility:** When a polymer is dissolved in a suitable solvent, the polymer absorbs water and swells in size. Slowly polymer goes into the solution a viscous polymer solution which is heterogeneous.

4. **Effect of heat on Polymer**



At room temperature all polymers are solids existing in rubbery or visco elastic state. As the temperature increases the polymer melts and this is called melting point of polymer (Tf). As the temperature further increases to boiling (Tb) the polymer decomposes to monomers which are in vapor phase. When the temperature is lowered below zero degree the polymer becomes hard, brittle and glassy and the polymer behaves like glass.

5. **Glass transition temperature:** The temperature at which the polymer transforms from rubbery state to glassy state is known as glass transition temperature. The rotation about the inter unit bonds in the polymer is effectively frozen at Tg and below Tg.

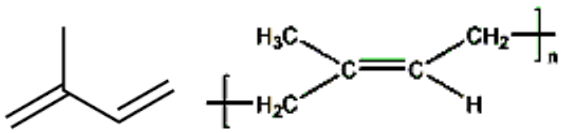
   Eg: Tg of **LDPE** = -110°C; Tg of **HDPE** = -90 ^{O}C

6. **Diffusion and permeability:** Crystalline polymers resist diffusion of molecules through voids or other gaps and tend to resist diffusion because of much greater degree of molecular packing. The diffusion rates are higher in rubber.

1.22 Natural Rubber

Rubber is a natural elastic polymer of isoprene. It is obtained from the milk of rubber called 'Latex' of Heveabresiliensis type of plants.

- Trans poly isoprene is called gutta-percha



Preparation of Natural Rubber from Latex

- The main composition of natural rubber is poly isoprene.
- Latex is rubber milk containing about 30 to 45% of rubber.
- The rubber milk is diluted with water and allowed to stand for some time.
- The clear liquid from the top is treated with acetic acid or formic acid to precipitate rubber.
- The precipitated rubber is collected and passed through rollers to get sheets of rubber. This is called 'Creep rubber'.
- In the second type of process, the rubber sheets are hung in smoked chambers for 3-4 days at 40-50 °C. This is called smoked rubber.

Defects of natural rubber

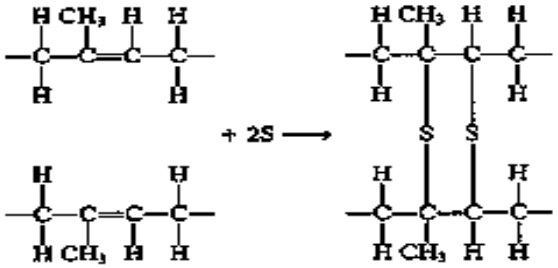
The natural rubber obtained from latex cannot be used in industries because it has some defects.

- It becomes soft and sticky during summer and it became hard and brittle during winter.

- It is attacked by atmospheric O_2,CO_2, moisture, light and gets less durability
- It swells up in oils.
- It flows plastically due to prolonged stress.
- Chemicals easily affect natural rubber.
- It possess poor tensile strength

1.23 Vulcanization of Rubber

- Vulcanization is compounding of rubber with sulphur.
- Vulcanization is done by heating rubber with sulphur at 140°C in CO_2 atmosphere
- Sulphur adds to the double bonds present in rubber to provide cross links between the polymer chains.
- 2 to 4%Sulphur addition gives soft elastic rubber. When sulphur content is more than30%,we get hard rubber called 'Ebonite'.



Properties of Vulcanized Rubber

- Vulcanized rubber has good tensile strength.
- It has good resistance to organic solvents.
- It acts as good electrical insulator.
- It has low water absorption tendency.
- It has good impact resistance.
- The elasticity of vulcanized rubber is very low.

Unit 2
Electrochemical Cells and Corrosion

2.1 Single Electrode Potential

Single electrode potential: The potential of half cell i.e., the potential difference between the metal and its salt solution in which it is dipped is called single electrode potential. It cannot be directly measured.

$Zn|ZnSO_4 \| CuSO_4|Cu$
←half cell → ←half cell→
← Single electrode → ←Single electrode→
(anode) (cathode)
$E_{cell} = E_{anode} + E_{cathode}$
$E_{cell} = E_{Ox}+E_{Red}$
$E_{cell} = E_R-E_L$

E_R of $Cu^{2+/}Cu$ is 0.337 V and E_L of Zn/Zn^{2+} is -0.763 V. Ecell = 0.337-(-0.763) = 1.1 V

Direction of spontaneous reaction: If Ecell > 0, the cell reaction has a tendency to go from left to right. If however the Ecell <0 the cell reaction will have a tendency to go from right to left. The reaction is spontaneous or feasible only when Ecell is positive for that case the standard free energy of the reaction (ΔG^0) would be negative ($\Delta G^0 = -nFE^0$).

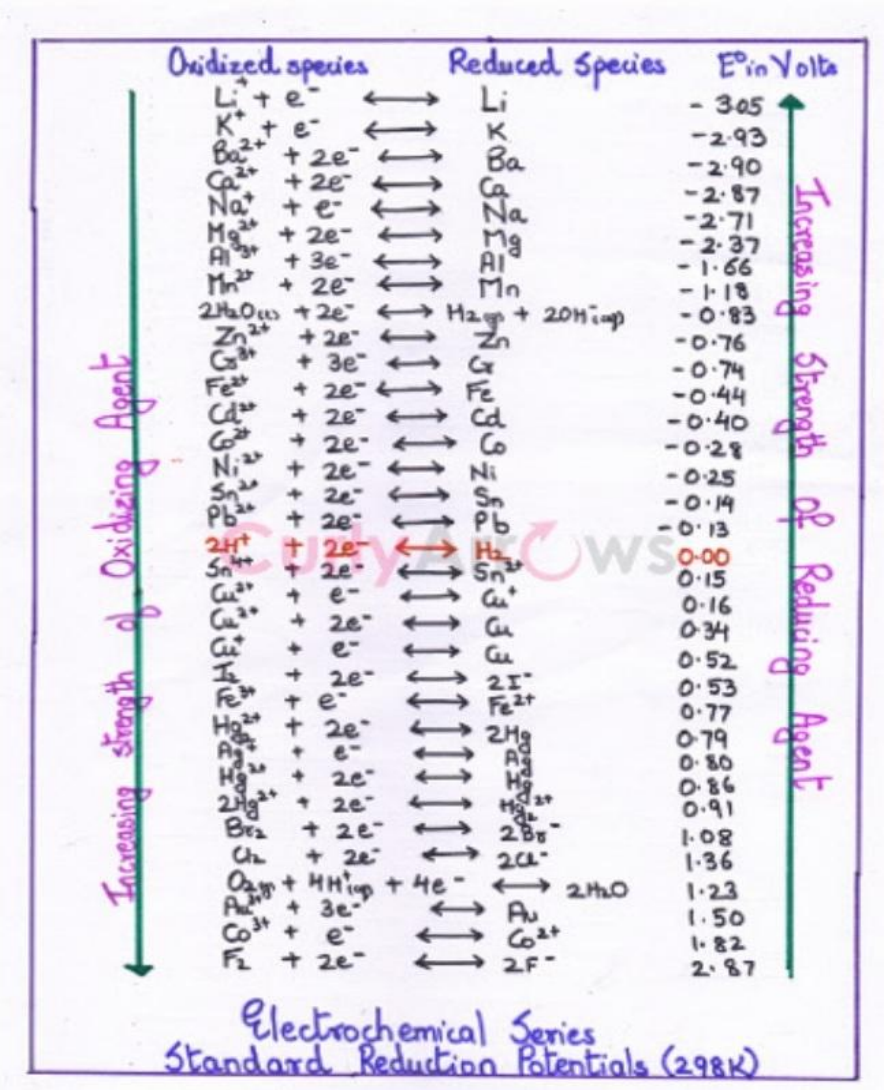
2.2 Electrochemical Series and uses of Series

A series of electrodes or half cells arranged in order of their increasing standard oxidation potentials or in the decreasing of their reduction potentials is called as electromotive force series (EMF series) or electrochemical series.

Characteristics electrical series

1. In this series, all reduction potentials are given on scale and hydrogen E^0 is taken as zero.
2. The standard reduction potential of an element is a measure of the tendency of that element to get reduced.

3. The element which has greater reduction potential gets reduced easily, while the elements with low reduction potential will get easily oxidized.
4. Elements that lose electron more easily have lower reduction potential and those which lose electrons with greater difficulty or instead of losing they accept electrons more easily have a higher reduction potential.
5. In EMF series elements having higher +ve reduction potential is placed at the bottom and while those having lower –ve reduction potential is placed at the top. SHE has the middle position in the electrochemical series.
6. The substances which are stronger oxidising agents than hydrogen are placed below the hydrogen in the series and have +ve standard reduction potential.



7. The substances which are weaker reducing agents than hydrogen are placed above the hydrogen in the series.

Applications/Uses

1. **For comparison of the relative oxidizing and reducing power of various metals:** A positive number indicate stronger reduction potential of the electrode, and it functions as a stronger oxidizing agent. A negative number means weaker reduction potential of the electrode, and hence it is a more powerful reducing agent.

   Eg: The F_2 electrode with the standard reduction potential of +2.87V is a strong oxidizing agent and Li^+ ion with the standard reduction potential value of -3.05 V is a strong reducing agent.

2. **For comparison of the relative activities of metals towards displacement reactions based on their placement in the activity series:** Any metal that is higher in the activity series can displace other metal that is placed lower in the activity series, from its compound.

   Eg: $Al + Fe_2O_3 \rightarrow Al_2O_3 + Fe$

   Al (-1.66 V) has higher oxidation potential than Fe (0.4 V) to form Al_2O_3

3. **To calculate the standard EMF of any electrochemical cell (E^0_{cell}):** The standard EMF of the cell is an addition product of standard electrode potential of two half-cells: Reduction half-cell and oxidation half-cell. $E^0_{cell} = E^0_{red} + E^0_{ox}$

   The standard oxidation potential (E^0_{ox}) = - standard reduction potential (E^0_{red})

   Therfore,

   E^0_{cell} = (standard reduction potential of the reduction half reaction)
   – (standard reduction potential of oxidation half reaction)

   $E^0_{cell} = E^0_{cathode} - E^0_{anode}$

   Eg: For a reaction $Zn (s) + Pb^{2+} (aq) \rightarrow Zn^{2+} (aq) + Pb (s)$

   The standard reduction potentials given are $Pb^{2+} (aq)|Pb(s) = -0.125$ V & $Zn^{2+}(aq)|Zn(s) = -0.764$V

   $E^0_{cell} = E_{cathode} - E_{anode} = -0.125-(-0.764) = 0.639$V

4. **In predicting the feasibility/occurrence of a redox reaction:** For a redox reaction to be spontaneous or occur readily, the EMF of the cell must be positive. If it comes out to be negative, then the direct, spontaneous reaction cannot take place.

   Eg: 1M $CuSO_4$ cannot be stored in Nickel vessel. Because E^0 $Ni^{2+}|Ni$ = -0.25 V and E^0 $Cu^{2+}|Cu$ = +0.35V. E^0_{cell} = 0.35-(-0.25) = 0.6V. Reaction is spontaneous.

5. **To predict whether a metal reacts with acid to liberate Hydrogen gas:** For a metal to react with acid (H+), the following reaction should take place.

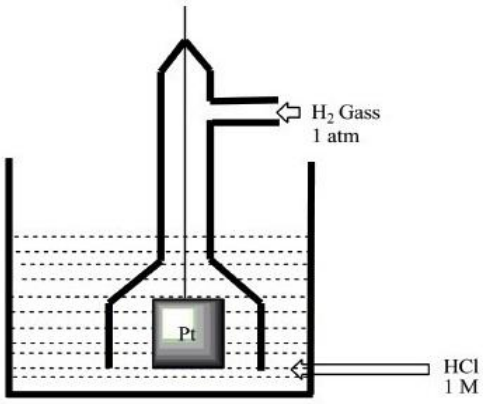
   $M(s) + H^+(aq) \rightarrow M^+(aq) + 1/2H_2(g)$

   The metal undergoes oxidation, and the hydrogen ion by accepting the electron undergoes reduction. If the metal has negative reduction potential and is place above the hydrogen in the electrochemical series, it can react with an acid to liberate hydrogen gas. The metals copper, silver, gold and platinum and placed below the hydrogen are less reactive, and they do not replace hydrogen from water or acid.

2.3 Standard Hydrogen Electrode

Hydrogen Electrode (SHE) or Normal Hydrogen Electrode (NHE):

Standard Hydrogen Electrode (SHE) is the primary reference electrode. Its electrode potential is arbitrarily fixed as zero. SHE is an example of gas electrode. It consists of an inert platinum rod dipped in 1M HCl solution. H2 gas is bubbled at 1 atm pressure at 298 K temperature.
H^+ (aq)(1M)/H_2(g)(1 atm)/Pt.



It can act as anode or cathode depending on the nature of electrode to which it connected. Thus SHE is a reversible electrode.

If the potential of the coupled electrode is less than zero, then SHE acts as cathode. Reduction takes place at SHE.

$$H^+ + e^- \rightarrow H_2$$

If the potential of the coupled electrode is greater than zero, then SHE acts as anode. Oxidation takes place at SHE.

$$H_2 \rightarrow H^+ + e^-$$

Limitations of SHE

1. SHE is difficult to construct. Bubbling H_2 gas at 1 atm around Pt through the solution is difficult. Maintaining activity of H+ at unity is difficult.
2. Pt metal has high adsorption and catalytic activity. Hydrogen on Pt surface is reduction catalyst. Many substances get reduced.

2.4 Calomel Electrode

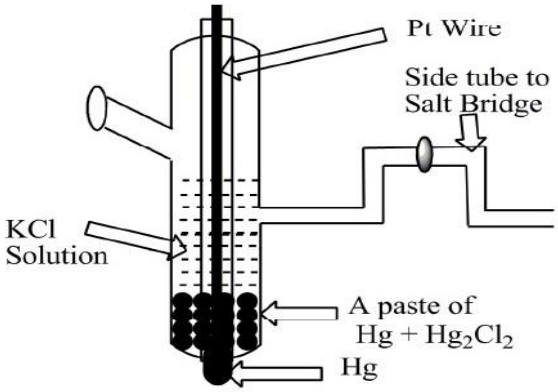
Due to limitations of SHE, secondary reference electrodes are needed. The potential of secondary reference electrodes are calibrated using SHE.

Calomel Electrode is an example of metal-metal insoluble salt electrode.

Composition : Mercury is the metal. Mercurouschloride (Hg_2Cl_2, calomel) is the insoluble salt. Soluble salt with common anion is KCl. Electrical contact is made using Pt wire.

Electrode representation: $Cl^-_{(aq)}$ / Hg_2Cl_2 /Hg

Construction: It is made up of glass tube. Mercury is kept at the bottom. Over which a paste of Hg^+ Hg_2Cl_2 kept. A solution of KCl is placed calomel. A platinum wire is used to make electrical contact. The side tube is used to make contact with other electrodes through a salt bridge.



It is a reversible electrode. It can act as cathode or anode depending on the potential of the electrode to which it is coupled. If the potential of coupled electrode is greater than calomel electrode, calomel electrode undergoes oxidation (Anode). If the potential of coupled electrode is less than calomel electrode, calomel electrode undergoes reduction (Cathode).

The electrode reaction (reduction),

$$Hg_2Cl_2 + 2e^- \rightarrow 2Hg(l) + 2Cl\text{-}(aq).$$

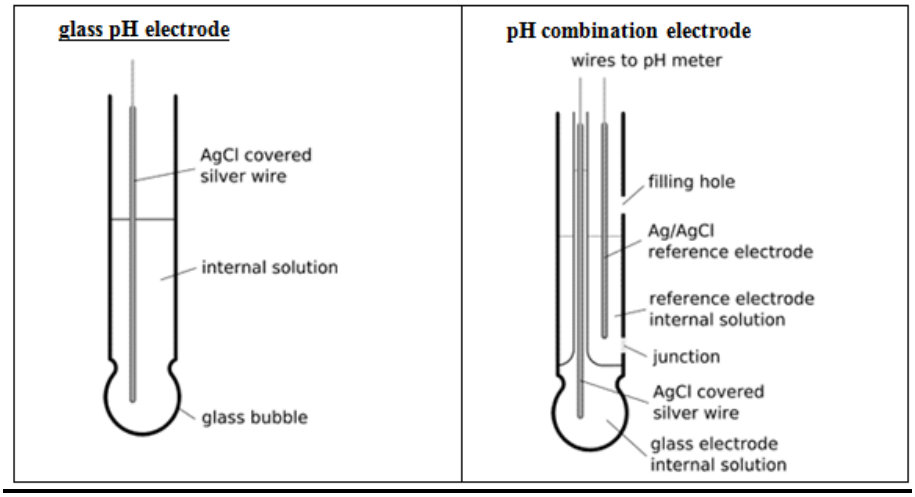
The potential of the electrode is given by,

$$E_{cell} = E^0_{cell} - 2.303\ \frac{RT}{2F}\log([Cl^-]^2) \Longrightarrow E_{cell} = E^0_{cell} - 2.303\ \frac{RT}{F}\log([Cl^-])$$

The potential of Calomel electrode depends on the concentration of Chloride (KCl) solution. If the KCl is saturated then we call it saturated calomel electrode (SCE), If 1N KCl is used then it is Normal calomel electrode (NCE) and if 0.1 N KCl is used then it is deci-normal calomel electrode. Electrode potential decreases in the order 0.1N > 1 N > saturated KCl.

Calomel electrode is convenient to handle and it can be constructed easily. It cannot be used above 50^0C due to instability of Hg_2Cl_2 and mercury is poisonous.

2.5 Glass Electrode



- Most often used pH electrodes are glass electrodes. Typical model is made of glass tube ended with small glass bubble. Inside of the electrode is usually filled with buffered solution of chlorides in which silver wire covered with silver chloride is immersed. pH of internal solution varies - for example it can be 1.0 (0.1M HCl) or 7.0 (different buffers used by different producers).

- Active part of the electrode is the glass bubble. While tube has strong and thick walls, bubble is made to be as thin as possible. Surface of the glass is protonated by both internal and external solution till equilibrium is achieved. Both sides of the glass are charged by the adsorbed protons, this charge is responsible for potential difference. This potential in turn is described by the Nernst equation and is directly proportional to the pH difference between solutions on both sides of the glass.

- The majority of pH electrodes available commercially are combination electrodes that have both glass H^+ ion sensitive electrode and additional reference electrode conveniently placed in one housing.
  This device is essentially a galvanic cell that can be schematically represented as:
  Glass electrode || Reference Solution || *Test Solution* || Glass electrode
  Ag(*s*) | AgCl(*s*) | KCl(*aq*) || 1×10^{-7}M H^+ solution || glass membrane || *Test Solution* || junction || KCl(*aq*) | AgCl(*s*) | Ag(*s*)

Limitations

- The glass electrode has some inherent limitations due to the nature of its construction.
- Acid and alkaline errors are obtained.
- An important limitation results from the existence of *asymmetry potentials* that are present at glass/liquid interfaces.
- The existence of these phenomena means that glass electrodes must always be calibrated before use; a common method of calibration involves the use of standard buffer solutions.

Applications

- Glass electrodes are commonly used for pH measurements.
- There are also specialized ion sensitive glass electrodes used for determination of the concentration of lithium, sodium, ammonium, and other ions.
- Glass electrodes have been utilized in a wide range of applications including pure research, control of industrial processes, analysis of foods and cosmetics, measurement of environmental indicators, and microelectrode measurements such as cell membrane electrical potential and soil acidity.

2.6 Concentration Cell

A concentration cell also produces electrical energy. It consists of two half cells with same type of electrodes but differ in their concentrations. By oxidation or reduction more concentrated half cell is diluted and more diluted half cell is concentrated until both concentrations become identical.

During this process a voltage is created.

Two types

1. **Electrode concentration cell:** Two electrodes are of same material but differ in their concentrations. Both half cells contains same electrolyte with same concentrations.

   Example amalgam cells
   Zn (Hg) C_1 / ZnSO4 // ZnSO4 / Zn (Hg)C_2
   Anode // Cathode

Oxidation occurs at LHS (anode) $Zn\ (C_1) \rightarrow Zn^{2+} + 2e^-$
Reduction occurs at RHS (cathode) $Zn^{2+} + 2e^- \rightarrow Zn\ (C_2)$

$Zn\ (C_1) \rightarrow Zn\ (C_2)$

(no net chemical reaction, only transfer of substance)

$$E_{cell}=E^0_{cell}-\frac{0.059}{n}\log\left(\frac{C_2}{C_1}\right)$$

If $E^0_{cell} = 0$,

$$E_{cell}=\frac{0.059}{n}\log\left(\frac{C_1}{C_2}\right)$$

For E_{cell} to be positive $C_1 > C_2$, that is anode electrode should be of higher concentration and by oxidation the electrode concentration decreases. Cathode should be of lower concentration and by reduction the concentration increases.

2. **Electrolyte concentration cell:** Two electrodes are of same metal having unit activity. Both half cells contains same electrolyte but differ in their concentrations.

Example amalgam cells
$Zn / ZnSO4\ (C_1) // ZnSO4\ (C_2) / Zn$
Anode // Cathode

Oxidation occurs at LHS (anode) $Zn \rightarrow Zn^{2+}(C_1) + 2e^-$
Reduction occurs at RHS (cathode) $Zn^{2+}(C_2) + 2e^- \rightarrow Zn$

$Zn^{2+}(C_2) \rightarrow Zn^{2+}(C_1)$

no net chemical reaction, only transfer of ions)

$$E_{cell}=E^0_{cell}-\frac{0.059}{n}\log\left(\frac{C_1}{C_2}\right)$$

If $E^0_{cell} = 0$,

$$E_{cell}=\frac{0.059}{n}\log\left(\frac{C_2}{C_1}\right)$$

For E_{cell} to be positive $C_2 > C_1$, that is electrolyte in cathode should be of higher concentration and by reduction the concentration decreases. Anode should be of lower concentration and by oxidation the concentration increases.

Applications of concentration cells

1. Determination of valence of ions. For electrolyte concentration cell, $E_{cell} = \frac{0.059}{n} \log\left(\frac{C_2}{C_1}\right)$. If concentrations are known then valence can be determined.

2. Determination of solubility of sparingly soluble salt. AgCl is sparingly soluble salt of Ag

   Construct the following cell
   Ag/AgCl (sat), KCl (0.01M) //$AgNO_3$ (0.01 M)/ Ag
   Anode // Cathode

   In the anodic compartment the Ag electrode is in contact with 0.01 M KCl solution containing a drop of AgNO3 which produces a small amount of saturated AgCl. Cathodic compartment Ag rod dipped in 0.01M $AgNO_3$.

   The emf of the cell is,

   $$E_{cell} = \frac{0.059}{n} \log\left(\frac{0.01}{C}\right)$$

   Where the only unknown quantity is C, which is the concentration of Ag^+ in anode compartment.

   If 0.01 M KCl is completely ionized then $[Cl^-]$ = 0.01 M
   Then K_{SP} of AgCl = $[Ag^+][Cl^-]$ = C x 0.01 $mole^2/litre^2$.
   Hence solubility of AgCl = Square root of K_{SP} =.$\sqrt{C*0.01}$mole/litre.

Batteries

2.7 Explain Dry Cell or Lechlanch Cell

It is called a dry battery as it does not have any liquid electrolyte in it. The zinc vessel serves as the anode. The cathode is a graphite rod in the centre of the cell. It is surrounded by the electrolyte which consists of a paste of NH_4Cl, $ZnCl_2$, and MnO_2. Traces of acetylene black, graphite powder are also added starch is added to make the mixture like a thick paste.

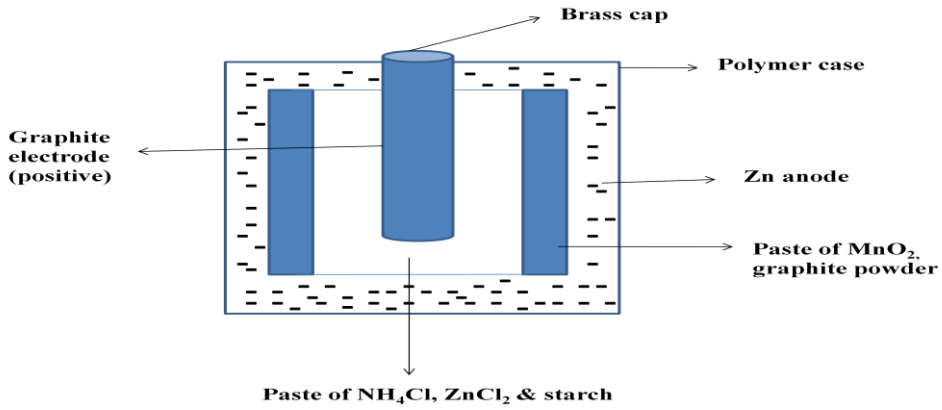
The cell represented as Zn/Zn^{2+}, NH_4/MnO_2, C

The cell reactions are involved are

At anode	$Zn\ (S)$	$\rightarrow$	$Zn^{2+}\ (aq) + 2e^-$
At cathode	$2\ NH_4Cl(aq) + 2MnO_2\ (S) + 2e^-$	$\rightarrow$	$2NH_3(aq) + Mn_2O_3(S) + H_2O + Cl^-(aq)$
Net cell reaction	$Zn\ (S) + 2\ NH_4Cl\ (aq) + 2MnO_2(S)$	$\rightarrow$	$Zn(NH_3)_2Cl_2(s) + Mn_2O_3(S) + H_2O(l)$

The NH_3 liberated reacts with Zn^{2+} to form a complex $[Zn(NH_3)_2]Cl_2$.
The cell is a primary cell and gives a voltage of 1.5 V
Gradually, the products are formed and the cell voltage drops.

Applications: these cells are used in transistors, tape recorders, toys and portable electronic gadgets.



2.8 Explain Lithium-MnO_2 Batteries or Lithium Batteries

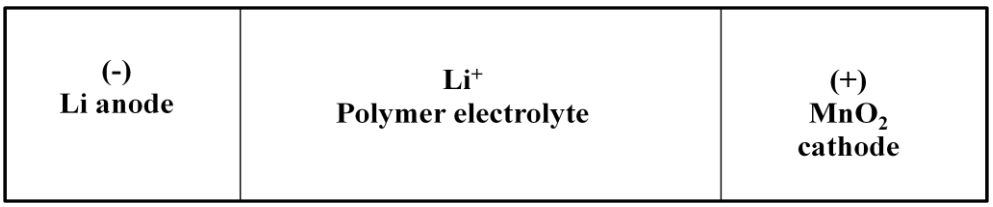
- These are primary batteries that cannot be recharged.
- In these batteries lithium metal or lithium compounds are used as anode.
- The cathode is made of specially treated MnO_2 crystal obtained by special heat treatment.
- Since, lithium is highly reactive with water and non-aqueous solvents, these solvents cannot be used as electrolyte.
- Instead, lithium salt solution in propylene carbonate and dimethoxy salts is used the electrolyte solvents like thionyl chloride containing lithium compounds such as LiCl, LiBr, $LiAlCl_4$, $LiSO_3F_3$, $LiClO_4$ can also be used.
- The cell can be represented as

Li, non-aqueous solvents containing lithium salts/MnO_2

The cell reactions are

At anode	Li	→	$Li^+ + e^-$
At cathode	$MnO_2 + Li^+ + e^-$	→	$MnO_2(Li^+)$
Net cell reaction	$Li + MnO_2$ (S)	→	$MnO_2(Li^+)^{2+}$

Lithium batteries find use in clocks, calculators, toys, digital camera, watches, medical equipments like artificial pacemakers, thermometers and remote car locks etc.



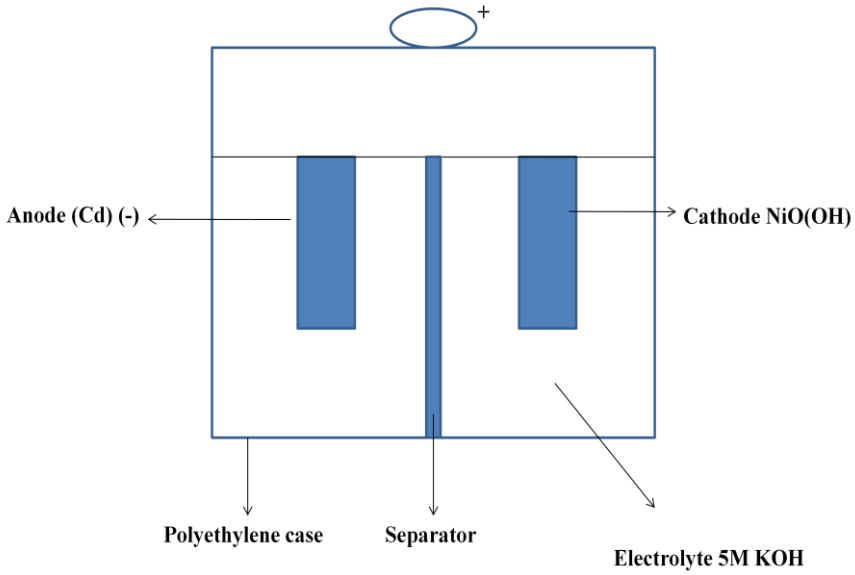
2.9 Explain Ni-Cd Battery

- It consists of a Cd andoe. The cathode is composed of a paste of NiO(OH). The electrolyte used is 5M KOH solution.
- The cell reactions are

At anode	Cd (S)+ $2OH^-$ (aq)	→	$Cd(OH)_2$(S)+ $2e^-$
At cathode	2 NiO(OH)(S)+ $2H_2O$(l) +$2e^-$	→	$2Ni(OH)_2$(S)+$2OH^-$
Net cell reaction	Cd (S)+ 2 NiO(OH) (S)	→	$Cd(OH)_2$(S)+$2Ni(OH)_2$(S)

- Its voltage is ~1.35 V. however, its voltage is fairly constant.
- The reaction products adhere to the electrode surface and the reaction can be easily reversed.
- This type of battery does not undergo deterioration as no gases are produced.
- It is light, portable and rechargeable.

Applications: It is used in calculators, electronic flash units and cordless shavers, transistors etc.



2.10 Write a Note on Nickel Metal Hydride Batteries

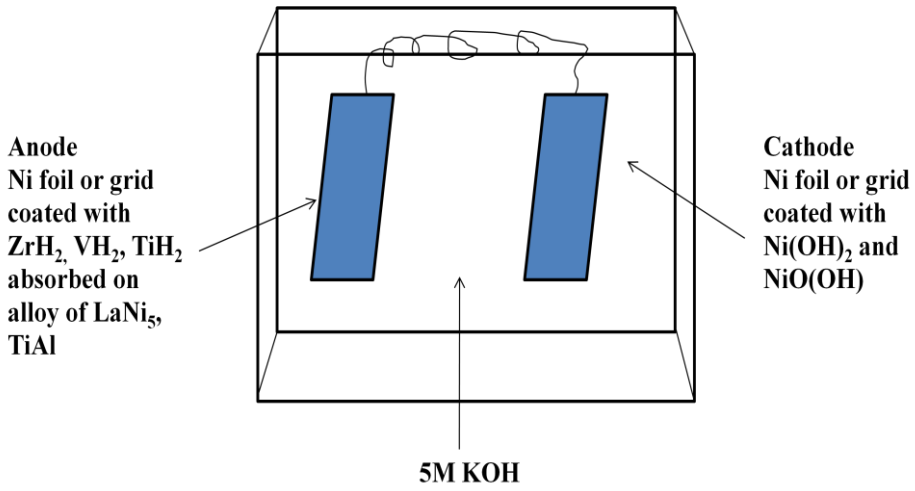
Like nickel cadmium cells, these are rechargeable batteries. However, instead of cadmium anode, zirconium hydride ZrH_2, vanadium hydride VH_2, titanium hydride TiH_2, absorbed on an alloy of $LaNi_5$ & TiAl. These are then coated on the nickel foil or grid. The cathode is also a nickel foil coated with nickel oxy hydroxide. Polypropylene is used as a separator and 5M KOH solution is used as an electrolyte. The electrodes together with the electrolyte are placed in rectangular steel container which is completely sealed.

The cell represented as

MH/KOH (5M)/$Ni(OH)_2$, NiO(OH)

The cell reactions are

At anode	$MH\ (S) + OH^-\ (aq)$	$\underset{\text{charging}}{\overset{\text{discharging}}{\rightleftarrows}}$	$M(S) + H_2O(l) + e^-$
At cathode	$NiO(OH)(S) + 2H_2O(l) + e^-$	$\underset{\text{charging}}{\overset{\text{discharging}}{\rightleftarrows}}$	$Ni(OH)_2(aq) + 2OH^-(aq)$
Net cell reaction	$MH\ (S) + NiO(OH)\ (S)$	$\underset{\text{charging}}{\overset{\text{discharging}}{\rightleftarrows}}$	$Ni(OH)_2(aq) + M(S)$



Applications: these batteries are inexpensive and are used in cameras, medical instruments, electric tooth brushes, razors, mobile phones, pagers etc.

2.11 Write a note on Zinc Air Battery

In zinc air battery, the anode is made of zinc plates. A perforated carbon plate treated with water repellants acts as a cathode. Sodium hydroxide (5M) or potassium hydroxide is used an electrolyte. The anode, cathode and the electrolyte are contained in an ebonite or polymer case. At this anode, zinc reacts with electrolyte (NaOH or KOH) to form zincate ions which decay into zinc oxide and water. The electrons released at the anode travel to the cathode where oxygen of the air accepts the electrons to form hydroxide ions.

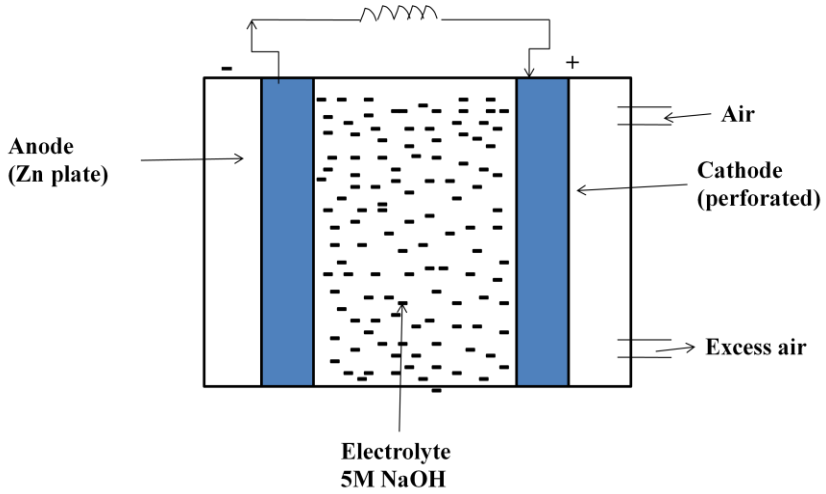
The cell represented as

Zn/NaOH(5M)/Air/C

The cell reactions are

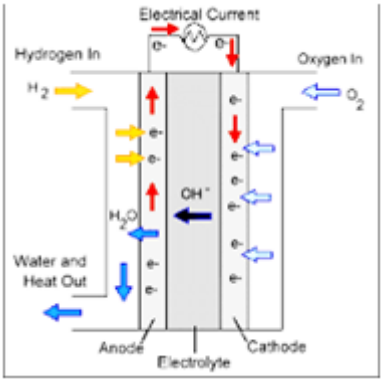
At anode	$2Zn(S)+ 4OH^-(aq)$	→	$2ZnO(S)+ 2H_2O(l)+4e^-$
At cathode	$O_2(g)+ 2H_2O(l) +4e^-$	→	$4OH^-(aq)$
Net cell reaction	$2Zn(S)+ O_2(g)$	→	$2ZnO(S)$

They produce 1.65V. Zinc air batteries have the properties of both fuel cells and batteries. Zinc air batteries are electrically non-rechargeable; however they can be recharged mechanically by converting the zinc oxide produced in the battery into zinc metal. The zinc metal can be are used. They are very cheap and increasingly being used instead of mercury batteries. Their size varies from small button cells that are used in the watches and in hearing aids to large batteries used in cameras and also for the electric propulsion of vehicles.



Fuel cells

2.12 H_2-O_2 fuel cell



Hydrogen - oxygen fuel cells consisting of two porous graphite electrodes, whichis impregnated with an catalyst such as pt/pd/Ag. 25% KOH is used as an electrolyte. Hydrogen and oxygen are continuously supplied to the anode and cathode respectively. The hydrogen is oxidized and the electrons lost in this process migrate to the cathode through the external circuit. Oxygen at cathode receives the electrons and gets reduced to OH- ions. The cell reaction, hydrogen and hydroxyl ions combine to form water which is removed from the cell.

Cell reactions

At anode	:	$H_2 + OH^-$	→	$2H2O + 2e^-$
At catode	:	$½ O_2 + H_2O$	→	$2OH^-$
Overall cell reaction	:	$H_2 + ½ O_2$	→	H_2O

Advantages

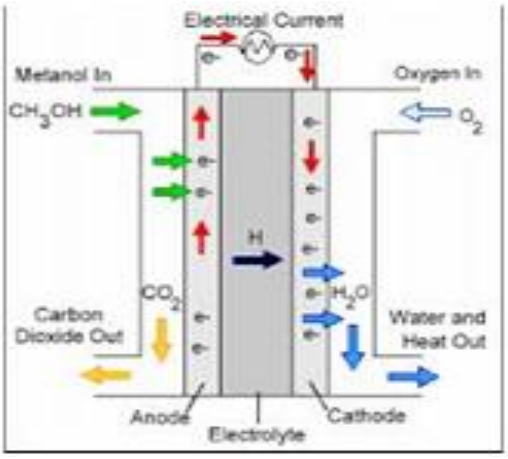
- The energy conversion is very high (75-82%)
- The product H2O is drinking water for astronauts.
- Noise and thermal pollution are low
- Maintenance cost is low for the fuels
- The regenerative H2-O2 fuel cell is an energy storage system for space applications.

Limitations

- The life time of fuel cell is not accurately known
- Initial cost is high
- The distribution of hydrogen is not proper.

Uses: The H_2-O_2 cells are used in Space vehicles, military and mobile powersystems.

2.13 Methanol – O_2 Fuel Cell

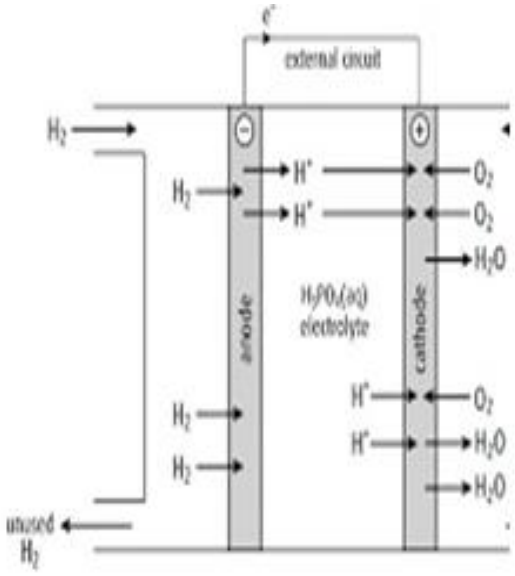


It consists of two electrodes made up of platinum.In between the electrodes KOHis placed as an electrolyte. Methanol and O_2 is supplied at the anode and oxygen is supplied at the cathode. The methanol is oxidized to CO_2& H_2O with the liberation of electrical energy. The cell delivers an emf of 1.20v

Cell reactions

At anode	$CH_3OH + H_2O$	$\rightarrow$	$CO_2 + 6\,H^+$
At cathode	$\frac{3}{2}O_2 + 6\,H^+ + 6\,e^-$	$\rightarrow$	$3H_2O$
NCR	$CH_3OH + \frac{3}{2}\,O_2$	$\rightarrow$	$CO_2 + 2H_2O$

2.14 Phosphoric Acid Fuel Cells

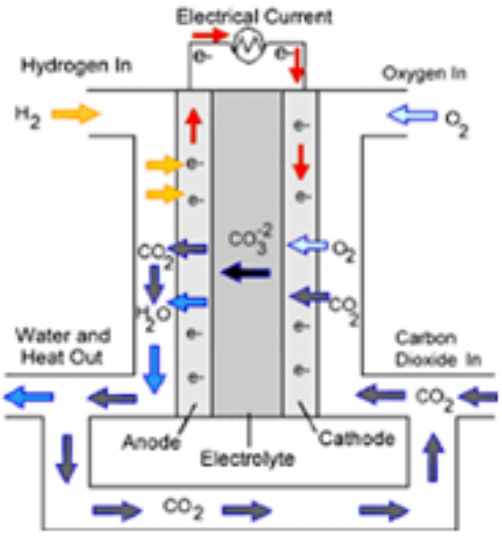


These fuels cells use phosphoric acid as electrolyte.The electrodes are made of porous carbon coated with finely dispersed ptcatalyst. O_2 is used as oxidant. Hydrogen is used as fuels.Hydrogen and oxygen are continuously supplied to the anode and cathode respectively. These operate at a temp 150 to 200°C.

Cell Reactions

At anode	:	$2H_2$	→	$4\ H^+ + 4\ e^-$
At cathode	:	$O_2 + 4\ H^+ + 4\ e^-$	→	$2\ H_2O$
Overall	:	$2\ H_2 + O_2$	→	$2H_2O$

2.15 Molten Carbonate Fuel Cells



These fuel cells consisting of molten carbonates such as lithium carbonate and potassium carbonate (K_2CO_3) and sodium carbonate as electrolyte. The anode is made up of nickel and cathode made up of nickel oxide. Operating temperature is650°C.

Cell Reactions

At anode	:	$CO_3^{2-} + 2H_2$	$\rightarrow$	$H_2O + CO_2 + 4\,e^-$
At cathode	:	$CO_2 + O_2 + 2\,e^-$	$\rightarrow$	CO_3^{2-}
Overall	:	$2\,H_2 + \frac{1}{2}\,O_2 + CO_2$	$\rightarrow$	$2H_2O + CO_2$

Corrosion

Definition

Corrosion: The process of decay of metal by environmental attack is called corrosion.

Effects or disadvantages of the corrosion

1. The valuable metallic properties conductivity, ductility. Strength and malleability etc, are lost.
2. The process of corrosion is very harmful and causing for the wastage of metal. It is estimated that the amount of iron wasted by corrosion each year is about ¼ th of its annual world protection.
3. Life span of the metallic parts of the machines is reduced.

Theories of corrosion (chemical and electrochemical)

2.16 Dry Corrosion or Chemical Corrosion

This type of corrosion occurs mainly through the direct chemical action of environment or atmospheric gases such as oxygen, halogen, hydrogensulphide, sulphur dioxide, etc with metal surfaces in immediate proximity.

There are three main types of dry corrosion.

a. Oxidation corrosion: Direct attack of oxygen on metals in the absence of moisture at high or low temperatures is called oxidation corrosion. Except Ag, Au and Pt all metals are oxidized even though alkali (Li, Na, K, Rb etc) and alkaline earth metals (Be, Ca, Sr etc)are very rapidly even at lower temperatures.

$$2M \rightarrow 2M^{+} + 2e^{-} \text{(Oxidation by loss of electrons)}$$

$$O_2 + 2e^{-} \rightarrow 2O^{2-} \text{(Reduction by gain of electrons)}$$

$$\text{Total reacton } \; 2M + O_2 \rightarrow 2M^{+} + 2O^{2-} \rightarrow 2MO$$

Mechanism: The oxidation of the metal occurs at the surface first, resulting in the formation metal oxide scale, which restricts further oxidation. Further oxidation can continue if the metal diffuse out of the

scale or the oxygen must diffuse in through the scale of the underlying metal.

Nature of the metal oxide formed plays an important role in oxidation corrosion.

1. A stable metal oxide layer tightly adheres to the metal surface is impermeable to the attacking oxygen.
   Eg: the oxide film on Al, Sn, Pb, Cu, Cr, W etc.
2. An unstable oxide formed on the metal surface decomposes back into the metal and oxygen. In such cases oxidation corrosion is not possible.
   Eg: Ag, Au & Pt do not undergo oxidation process.
3. A volatile oxide layer formed during corrosion evaporates as soon as it is formed and leaving the metal for further attack. This leads to rapid and continuous corrosion leading to excessive corrosion to occur.
   Eg: Molybdenum oxide (MoO_3) is volatile.
4. A porous layer of metal oxide contains pores, cracks etc. provide access to oxygen to reach the underlying the surface of the metal.
   Eg: Alkali metals and alkaline earth metals (Li, K, Na, Mg) such type of oxide film.

b. Corrosion by other gases: The gases like SO_2, CO_2, Cl_2, H_2S and F_2 etc, also cause chemical corrosion and the extent of the corrosion mainly depends on the reactivity to the gas on the metal surface. The corrosion product is metal compound. The nature of the metal compound determines the corrosion.

c. Liquid metal corrosion: The chemical action of a flowing liquid metal at high temperature, on a solid metal or alloy produces liquid metal corrosion. The reason for this corrosion is the dissolution of the solid metal by liquid metal or internal penetration of the liquid metal into the solid phase, weakening the solid metal.

2.17 Explain the Mechanism of Wet Corrosion

Wet or electrochemical corrosion: Wet corrosion or electrochemical corrosion takes place under wet or moist conditions through the formation of short circuited electrochemical cells. Wet corrosion is more common than dry corrosion.

a. Evolution of hydrogen: this type of corrosion casually occurs in acidic environments. The rusting of iron takes place in acidic environment in the following way.

At anode the dissolution (oxidation) of iron to ferrous ion with liberation of electrons takes place.

$$Fe \rightarrow Fe^{2+} + 2e^{-} \text{ (oxidation)}$$

The $2e^{-}$ flow from anodic area to cathodic area through the metal where H^{+} ions (of acidic solution) are available and eliminated as hydrogen gas.

$$2H^{+} + 2e^{-} \rightarrow H2 \uparrow \text{ (reduction)}$$

Thus Fe^{2+} ions are liberated at anode and H_2 gas is liberated at the cathode. The overall reaction in acidic medium is given by

$$Fe + 2H^{+} \rightarrow Fe^{2+} + H_2$$

In hydrogen evolution type of corrosion, the anodes are usually large and the cathodes are small areas.

In neutral medium at the cathode formation of OH^{-} takes place along with the liberation of hydrogen.

$$2H_2O + 2e^{-} \rightarrow H_2 \uparrow 2OH^{-}$$

The Fe^{2+} and OH^{-} diffuse towards each other through the conducting medium forming the corrosion product rust in between cathodic and anodic areas.

$$Fe^{2+} + 2OH^{-} \rightarrow Fe(OH)_2 \downarrow \text{ (ferrous hydroxide)}$$

$$4Fe(OH)_2 + O_2 + 2H_2O \rightarrow 4Fe(OH)_3 \rightarrow 2Fe_2O_3.3H_2O$$

(Ferric hydroxide) (rust)

b. Absorption of oxygen: In presence of neutral, aqueous solution of electrolytes like (NaCl solution), in the presence of atmospheric oxygen and some cracks developed in iron oxide film causes this type of electrochemical corrosion. The surface of iron is always coated with a thin film iron osxide. If a crack is developed in this iron oxide film, the anodic areas are created on the surface, while the well coated metal parts act as cathode. It follows that anodic areas are small and the cathodic areas are large.

The following reactions take place taking rusting of iron

At the anodic areas, the metal oxidizes as ferrous ions liberating two electrons which pass through metal to the cathodic area.

$$Fe \rightarrow Fe^{2+} + 2e^{-}$$

At the cathode the electrons were accepted by water and oxygen to generate OH^- ions byreduction.

$$O_2 + H_2O + e^{-} \rightarrow 4OH^{-}$$

The combination of Fe^{2+} and OH^- originated at anode and cathode respectively occurs at cathode, because the mobility of Fe^{2+} ions, is greater than OH- ions. Corrosion occurs at anode and the rust occurs in between cathodic and anodic areas.

The presence of oxygen accelarates the corrosion at anodic area and rust formation at cathodic area.

The Fe2+ ion at anode and which OH^- ions at cathode diffuse towards cathodic and anodic areas respectively and they form rust when they meet.

$$Fe^{2+} + 2OH^{-} \rightarrow Fe(OH)_2 \downarrow \text{(ferrous hydroxide)}$$

If enough oxygen is present ferrous hydroxide is easily oxidised to ferric hydroxide and then to hydrated ferric oxide which is nothing but rust.

$$4Fe(OH)_2 + O_2 + 2H_2O \rightarrow 4Fe(OH)_3 \rightarrow 2Fe_2O_3.3H_2O$$

(Ferric hydroxide) (rust)

2.18 Different types of Wet/ Electrochemical Corrosion

1. **Galvanic corrosion:** When two dissimilar metals are connected and jointly exposed to the corrosive atmosphere, the metal possessing higher oxidation potential or higher in the electrochemical series become anodic and undergoes corrosion.

   Eg: When zinc and copper are connected and exposed to corroding environment or higher position in the electrochemical series, zinc under goes oxidation and corroded where as copper undergoes reduction and protected.

2. **Concentration cells:** The concentration cell corrosion may arise due to (i) difference in the concentration of the metal ions or (ii) difference in the exposure to air/oxygen (differential aeration

corrosion) or (iii) difference in the temperature or (iv) inadequate agitation or (v) different diffusions of metal.

3. **Differential aeration corrosion:** The different ways of creating aeration on the metal surface is listed below

   Partial immersion of metal in a solution
   The Zn rod immersed deep in NaCl --- anodic part
   The Zn rod above NaCl --- cathodic part
   The circuit is completed by the migration of ions through the electrolyte. As a result zinc rod starts corroding in the anodic area immersed in sodium chloride solution. This is also called waterline corrosion. Examples: Ship hulls, storage tanks.

4. **Faulty design of equipment:** An anodic area can be created due ti difference in aeration of the metal parts. When the metal surface is exposed to less oxygen supply due to faulty design of the equipment the metal part less exposed to oxygen becomes anodic and starts corroding. The metal part exposed to oxygen becomes cathodic.

5. **Accumulation of dirt, sand, scale and other contaminations:** When the dirt, sand, scale and other contaminations are deposited on the metal surface, the metal under the deposit is exposed to less oxygen supply and becomes anodic and remaining part of the metal becomes cathodic. Transfer of electrons takes place through the metal and atmospheric moisture acts as a medium for the transfer of ions. Corrosion product is formed between cathodic and anodic areas.

6. **Pitting corrosion:** A cavity, pinholes, pits and cracking of the protective film developed on the metal surface creates the formation of small anodic areas in the less oxygenated parts and large cathodic part takes place in oxygenated parts. Cracking of protective film may be due to (i) surface roughness or non uniform stresses, (ii) scratches or cut edges, (iii) local strain of metal, (iv) alternating stresses, (v) sliding under load and (vi) chemical attack.

7. **Stress corrosion:** Stress corrosion or stress cracking is the combined effect of static tensile stresses and the corrosive environment on a metal. Presence of tensile stresses is a main reason for stress

corrosion. Majorly seen in fabricated articles of alloy like high zinc brasses and nickel brasses.

8. **Intergranular corrosion:** This corrosion occurs along grain boundaries. The main reasons for the corrosion of grain boundaries is
   a. The grain boundaries where the metal is sensitive to corrosive attack,
   b. The corrosive attacking liquid/reagent possess the character of only attacking at the grain boundaries
   c. The precipitation of certain compounds at grain boundary producing a solid solution depleted in one constituent, making the grain boundary anodic.

2.19 The Factors Influencing Corrosion

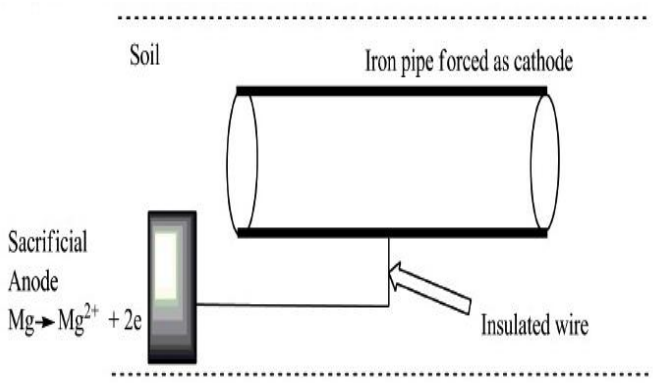
- **Nature of metal:** (1) Position in galvanic series, (2) Over voltage, (3) Relative areas of cathodic and anodic areas, (3) Purity of metal, (4) Physical state of metal, (5) Nature of surface film, (6) Passive character of metal, (8) Volatility of corrosion products, (9) Solubility of corrosion products.

- **Nature of corroding environment:** (1) Temperature, (2) Humidity of air, (3) Presence of impurities in atmosphere, (4) Nature of ions present, (5) Conductance of corroding medium,(6) Amount of oxygen in atmosphere, (7) Velocity of ions which flow in the medium, (8) pH value of the medium.

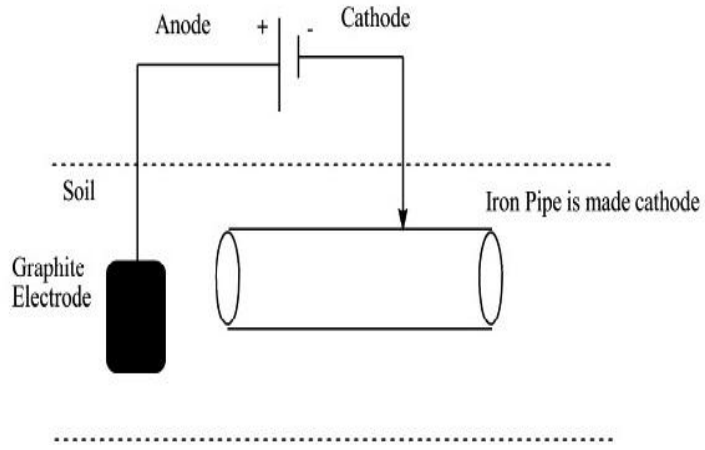
2.20 Cathodic Protection

Force the metal to be protected to behave as a cathode, then corrosion will not takes place.

a. **Sacrificial anodic protection or galvanic protection**: The metal structure to be protected is made cathode by connecting it to a more anodic material. As a result corrosion takes place at anode and the metal structure is saved from corrosion. Metals like Al, Zn, Mg and their alloys are used as sacrificial anodes.



b. Impressed current cathodic protection: The metal structure to be protected is made cathode by connecting it to a negative terminal of a DC source. The positive terminal is connected to an insoluble anode like graphite.



2.21 Cathodic and Anodic Inhibitors

These are substance when added in small quantities to the corrosive environment will decreases the corrosion reaction rate.

a. Anodic inhibitor: These inhibitors reduce the corrosion reaction at the anode by forming a protective layer on the metal surface. The protective layer increases the electrode potential.

Eg: Chromate, Nitrite, molybdate and ortho- phosphates.

b. Cathodic inhibitor: It reduces the cathodic reaction rate. Two major cathodic corrosion reactions are, in acidic solution the evolution of hydrogen ($2H^+ + 2e \rightarrow H_2$) and in neutral and alkaline solution the absorption of Oxygen.

$$O_2 + H_2O + 2e \rightarrow 2OH^-.$$

c. Cathodic inhibition is done by, adding organic inhibitors: Organic inhibitors form a hydrophobic film on the metal surface. It provides a barrier for diffusion of ions. Examples amines, carboxyls, phosphonates etc.

Adding oxygen scavengers: Sodium sulphite, hydrazine etc remove dissolved oxygen from aqueous solution.

$$N_2H_4 + O_2 \rightarrow N_2 + 2H_2O$$

$$2Na_2SO_3 + O_2 \rightarrow 2Na_2SO_4$$

Adding Inorganic Inhibitors: These inhibitors will precipitate on the cathode surface and blocks the cathode reaction sites. Zn^{2+} when added to cathode environment will precipitate as $Zn(OH)_2$ on the cathode metal surface.

d. Vapour Phase Inhibitor: These are used for temporary protection against atmospheric corrosion. These compounds readily vaporize and condense on the metal surface to from a thin layer and prevent corrosion. Examples are dicyclohexylamine-nitrite, cyclo hexyl ammonium carbonate.

2.22 Protective Coating

It is a way to prohibit corrosion. It is continuous barrier between the coated surface and environment. The coating must be chemically inert to the environment and it should prevent diffusion of chemicals in the environment to the base metal. Sometimes coating will also increase mechanical and physical properties.

Two types of protective coating
i) Inorganic (Galvanizing, anodizing, chromate coating and tinning)
ii) iorganic (Paint, varnish etc)

Inorganic Coating: It can be either metallic or non-metallic.
Metallic Coating: A metallic barrier on the surface of metal. It increases corrosion resistance, mechanical strength and change other physical properties. It also used for decoration.

2.23 Galvanization

Hot Dip Galvanizing: Coating Zinc on Iron or Steel (anodic coating) by hot dip process. It is done to prevent iron from rusting.

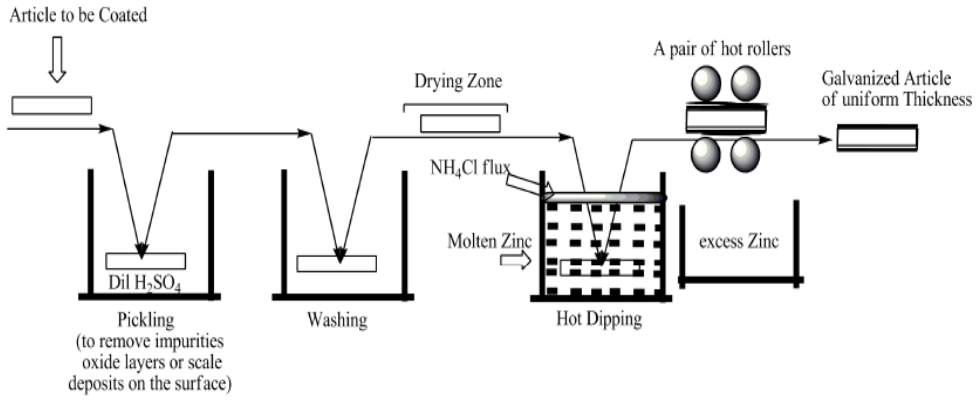
Process

Step 1: The Iron or steel is first cleansed by pickling with dil. H_2SO_4 solution. This treatment removes any impurities, scales or oxide layer formed in the base metal Iron.

Step 2: The article is then washed and dried.

Step 3: It is then dipped in a bath of molten Zinc. The surface of bath is kept covered with a flux of NH_4Cl to prevent oxidation.

Step 4: The dipped article is then passed through a pair of hot rollers to remove excess zinc and to get a coating of uniform thickness.



Roofing sheets, pipes, wires, nails, bolts, nuts etc are galvanized to prevent rusting. Zinc gets dissolved in dil. Acids to form highly poisonous compounds. Hence galvanized utensils cannot be used for storing and preparation of food stuffs.

2.24 Tinning

Hot Dip Tinning

Coating Tin on Iron or Steel (cathodic coating) by hot dip process is known as tinning. It is done to prevent iron from rusting.

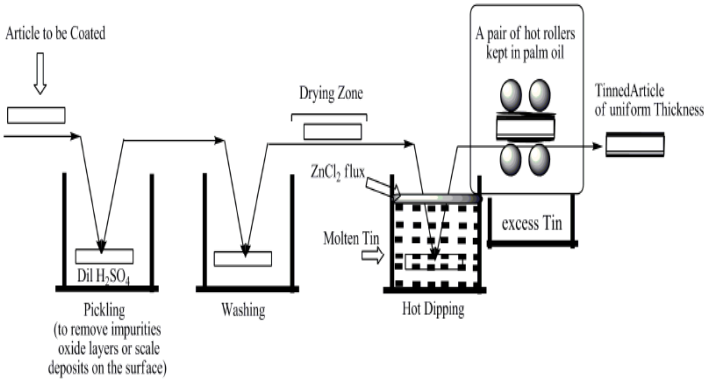
Step 1: The Iron or steel is first cleansed by pickling with dil. H_2SO_4 solution. This treatment removes any impurities, scales or oxide layer formed in the base metal Iron.

Step 2: The article is then washed and dried.

Step 3: It is then dipped in a bath of molten tin. The surface of bath is kept covered with a flux of $ZnCl_2$ to prevent oxidation.

Step 4: The dipped article is then passed through a pair of hot rollers kept in palm oil to remove excess tin and to get a coating of uniform thickness. The palm oil prevents contact with oxygen.

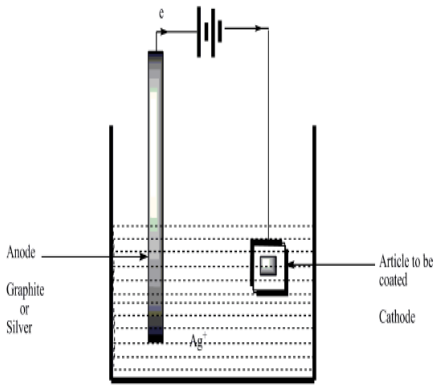
Tinning is widely used for coating steel, copper, brass etc. Tin is non-toxic therefore tinned items can be used to store food items, used to make cooking utensils, refrigeration equipments etc.



2.25 Explain Electroplating or Electro-Deposition Process

Electroplating is a method of application of metallic coating. Coating is done by electrochemical process.

Theory of electroplating: The object to be plated is made cathode of electrolytic cell. The anode is either coating metal or an inert electrode like graphite. The electrolyte is a soluble salt of coating metal.
Example is copper plating.



The article to be electroplated is subjected to acid pickling with dil.HCl or dil.H_2SO_4 to remove any scales, oxides etc. Then it is subjected to remove any scales, oxides etc. The cleaned article is made cathode of the electrolytic cell and the anode can be made of the pure coating metal or graphite. The electrodes were dipped in the salt solution of the coating metal, which acts as an electrolyte. When direct current is passed, the coating metal ions migrate to the cathode and deposit on the base metal article in the form of a thin layer. For brighter and smooth deposits, conditions like low temperatures, medium current density, and low metal ion concentration are used.

For example for electroplating of copper on iron article, the following are maintained

1	Electrolytic bath solution	:	Copper sulphate
2	Temperature maintained	:	40-70^{0}C (low temperature for brighter and smooth surface)
3	Current density	:	20-30 mA/cm^2 (medium current density for high efficiency)
4	Anode	:	Copper pellets (99 % pure) taken in a titanium mesh basket or Graphite
5	athode	:	Base material article

Application of electroplating

- Base metals like Al, Cu, Fe are electroplated with gold or silver to increase their commercial and decorative value.
- In internal combustion engines electroplating with chromium increases the wear resistance.

2.26 Explain Electroless Plating of Ni

The method of deposition of a metal from its salt solution on a catalytically active surface by a suitable reducing agent without using electrical energy is called electroless plating. This process is also called chemical plating or autocatalytic plating.

The metallic ions (M^+) are reduced to the metal with the help of reducing agents(R^+) When the metal (M) is formed, it gets plated over a catalytic surface

$$M^+ + R^- \xrightarrow{\text{catalyst surface}} M + R$$

M^+	+	R^-		M	+ R
electroactive surface		Reducing agent		Plated on the catalyst surface	oxidizing agent

The metal surface to be subjected to electroless plating is prepared by a) Etching (acid treatment) , b) for non conducting surfaces like plastics or printed circuit boards, treatment with stannous chloride and palladium chloride alternatively used. C) To get an active surface electroplating followed by heat treatment is adopted, d) For Ni surface treatment of organic solvents followed by acid treatment used.

Elecetroless plating of Ni: The base object, a plastic material like acrylonitrile – butadiene styrene (ABS) can be coated with nickel.

The base metal surface is activated by stannous chloride followed by palladium chloride to get a layer of Pb and the surface is dried and dipped in a solution of nickel chloride, sodium hyposulphite (reducing agent), sodium acetate buffer, sodium succinate (complexing agent and exalt) pH of the bath maintained is 4.5 and temperature maintained is 93^0C. The following reactions take place and Ni get plated electrolessely on the surface of the base object.

$$Ni^{2+} + H_2PO_2^- + H_2O \xrightarrow{\text{catalyst}} Ni + H_2PO_3 + 2H^+$$

Plated on the surface of object

Properties: Electroless plated Ni objects has better corrosion resistance, deposits are pore free, hard and wear resistant.

Applications

1. Electroless Ni coating are used extensively in electronic application providing a non magnetic underlay in magnetic components.
2. ABS plastics objected are used for decorative applications, like knobs in hi-fi equipments, tops on perfume bottles, and car decorative.

2.27 Explain Electroless Plating of Cu

Electroless Plating of Copper: Before Electroless plating the plastic board is degreased and etched in acid. It is activated by dipping in $SnCl_2$ / HCl at 25^0 c followed by dipping in $PdCl_2$.

Electroless plating is done under the following conditions,

Plating bath solution	:	Copper Sulphate
Reducing agent	:	Formaldehyde
Complexing agent and Exaltant	:	EDTA
Buffer	:	Sodium hydroxide and Rochelle salt [PH=11]
Temperature	:	25^0 c

The plating is carried out at room temperature by dipping the activated plastic board in the bath solution. The metallic copper obtained as a reduced product in the bath deposit on the metal surface.

$$2HCHO + 4OH^- + Cu^{2+} \rightarrow 2HCOO^- + 2H_2O + H_2 + Cu$$

Typical application of Electroless copper plating is printed circuit boards, particularly double sided boards in which plating through holes is required.

2.28 Paints

Organic coatings are inert-barriers, applied as metallic surfaces and other constructional materials for both protection and decoration.

Factors determine the protective value of organic coatings:

1. Protects from corrosion environment.
2. Good surface adhesion.
3. Impermeability to salts, gases and water.

Paint: Paint is a mechanical dispersion mixture one or more pigments in drying oil.

Constituents of paints

1. **Pigment:** Pigment is a solid substituent and an essential constituent of paint. A good pigment must be opaque, chemically inert, non toxic, freely mixable and cheap.

   The functions of pigment are: give strength, provide colour, give pleasant appearance, improves impermeability of paint from moisture and increases weather resistance.
   Eg: White lead, zinc oxide (white), ferric oxide and chrome red (red colour), chromium oxide (green) and carbon black (black)

2. **Drying oil:** Drying oil is film forming constituent of paint. The functions of drying oils are: The main function of drying fiProvides toughness, adhesion, durability and water proof.
   Eg: dehydrated castor oil, soyabean oil and sunflower oil

3. **Thinners:** Function of thinners are reduce viscosity of paint, helps in dissolving additives, increase elasticity of the paint film and helps in drying of the paint.
   Eg: Turpentine, benzene, naptha, xylol and kerosene etc.

4. **Driers:** Driers are oxygen catalaysts and accelerate the drying quality of the paint film.
   Eg: Resinates, tungstates and napthalates of Co, Mn, Pb and Zn.

5. **Fillers:** Fillers help to reduce the cost, increase durability of paint, reduce the cracking of dry paint film and act as carrier for pigment colour.
   Eg: Barytes ($BaSO_4$), tlac, asbestos, ground silica, gypsum and china clay

6. **Plasticizers:** The functions are to provide elasticity to the fil and minimize its cracking.
   Eg: Tricresyl phosphate, triphenyl phosphate anbdn tributy phthalate.

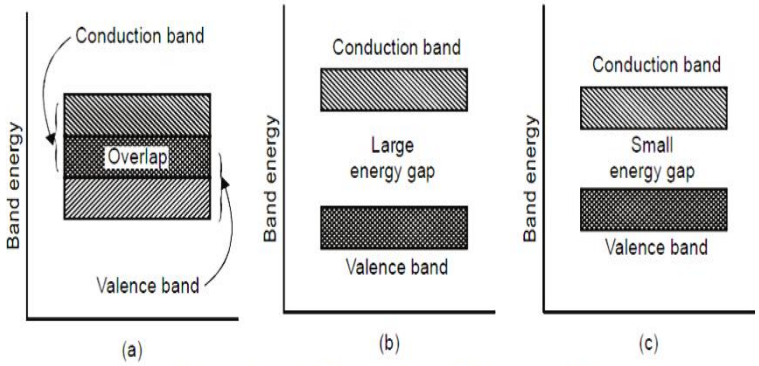
7. **Anti skinning agents:** To prevent gelling and skinning of the paint film.
   Eg: polyhydroxy phenols.

Unit:3

Material Chemistry

3.1 Semiconductors

A semiconductor material is one whose electrical properties lie in between those of insulators and good conductors.
(**Eg:** Ge, Si).



(a) conductor (b) insulator and (c) semiconductor

The semiconductors are classified mainly into two types based on composition of materials.

1. Elemental semiconductors
2. Non elemental semiconductors

1. **Elemental semiconductors:** These semiconductors are made from a single element of fourth group elements of the periodic table.
   They are also known as indirect band gap semiconductors
   Eg: Germanium, silicon

2. **Non elemental semiconductors**

- stoichiometric semiconductors
- non stoichiometric
- controlled vallency semiconductors
- chalcogenphoto **semiconductors**

3.2 Stoichiometric Semiconductors

Stoichiometric semiconducting compounds are intermediate compounds having an average four valence electrons per atom and possess crystal structures and band structures similar to those of Si and Ge. Such intermediate semiconducting compounds may be prepared by the combining

1. Elements of group III (e.gGa, In) and group V(e.gP,As,Sb)
2. Elements of group II(e.gCd,Pb) and group VI(e.g S, Se, Te)

Group III & Group V	**Fermi gap**	**Group II & Group VI**	**Fermi gap**
GaP	2.24	CdS	2.42
GaAs	1.35	CdSe	1.74
GaSb	0.67	PbS	0.37

3.3 Defect Semiconductors/Non Stochiometric Semiconductors

Many non-stoichiometric compounds act as semiconductors. For example, some metal oxides and sulphides containing non stoichiometric defects in their crystal structure. Let us consider iron oxide FeO, which is made up of Fe^{2+} and O^{2-} ions. On heating FeO, some Fe^{2+} ions are oxidized to Fe^{+3}, thereby giving compound with compositions ranging from $Fe_{0.81}O$ to $Fe_{0.94}O$. these oxides are actually a combination of FeO and Fe_2O_3. For example, a crystal compound $Fe_{0.90}$ actually contains $Fe^{2+}_{0.70}\ Fe^{3+}_{0.20}\ O^{2-}$.

Since Fe^{3+} ion is short by one electron with respect to Fe^{2+} ion, so Fe^{3+} in FeO crystal lattice can considered as a hole. Consequently, under the influence of electric field, the positive holes can move from Fe^{3+} to Fe^{2+} ions. Thereby crystal producing electrical conductivity, Therefore, such non-stoichiometric crystals can be represented as $M_{(1-x)}Y$ and these metal ion deficient compounds act as p-type semiconductors. Examples are FeO, Cu_2O, NiO and FeS.

On the other hand, when ZnO is exposed to reducing atmosphere, non-stoichiometric compound such as $Zn_{1.1}O$ ($Zn^{2+}_{0.9}Zn^{+}_{0.2}O$) are produced by partial removal of oxygen. In such compounds, Zn^{+} ion is rich by one electron wrt Zn^{2+}. So each Zn^{+} ion is capable of donating one electron to the conduction band. Hence under the influence of electric field, ZnO act

as n-type semiconductor. These metal excess non stoichiometric semiconductors can be represented as $M_{(1+x)}Y$. examples are ZnO,CdO, ZnS, PbS,CdS,Fe_2O_3,Cr_2O_3.

Limitations

- Difficulty in controlling the conductivity, since it depends both temperature and oxygen pressure applied.

3.4 Controlled-Valence Semiconductors

Defect semiconductors are generally difficult to obtain and the compositions are difficult to reproduce exactly. These difficulties can be overcome by the use of "**controlled-valence"semiconductors**,

For example, consider the material formed by heating a mixture of NiO and a small amount of Li_2O in air at 1200°C. Oxygen is absorbed, and a single phase of composition $Li_xNi_{1-x}O$ is formed

$$\frac{x}{2}Li_2O + (1-x)NiO + \frac{x}{4}O_2 \rightarrow Li_xNi_{1-x}O$$

As Li^+ is smaller in size by virtue can occupy interstitial site and substitute Ni^{2+} ions in the lattice. One Ni^{3+} ion is formed for every Li^+ ion introduced to preserve electrical neutrality. The resulting compound can be represented by the expanded formula **$Li_xNi^{III}_xNi^{II}_{1-2x}O$**. The positive holes in the Ni^{2+} ions (i.e., the Ni^{3+} ions) can move from one nickel ion to another and thus give rise to **p-type semi conduction.** By simply controlling the ratio of lithium to nickel, the electrical conductivity can be varied at will. The conductivity of pure NiO is about 10^{-10} Scm^{-1}, whereas the compound in which 10percent of the nickel atoms have been replaced by lithium atoms has a conductivity of about $10^{-1}Scm^{-1}$

Ni^{2+} O^{2-} $Ni^{(3+)}$ O^{2-} $Ni^{(2+)}$ O^{2-}
O^{2-} Ni^{2+} * Ni^{2+} O^{2-} Ni^{2+}
Ni^{2+} O^{2-} $Ni^{(3+)}$ O^{2-} Ni^{2+} O^{2-}
O^{2-} Li^+ O^{2-} Ni^{2+} O^{2-} Ni^{2+}
[Li^+ ion doped NiO lattice]

The material exhibits **n-type semi conductivity** is **$La_xBa_{1-x}Ti^{III}_xTi^{IV}_{1-x}O_3$**.

- Such semiconductors are employed in thermistors (thermally stable resisters)

3.5 Chalcogen Photo Semi Conductors

Oxygen, sulphur, selenium and tellurium are collectively called chalcogens or ore forming elements because a large number of metal ores are oxides or sulphides.

- They behave as semiconductors or photo conductors either alone or by combining with other elements.
- The conductivities of pure chalcogens increase with increase in atomic mass
- Among these chalcogen semiconductors selenium (or As_2Se_3) is a good photonconductor.
- Selenium is an excellent photo conductor. Its conductivity increases enormously on exposing to light, hence it is used in photo copying process (Xerox)

3.6 Preparation of Semiconductors

For preparing semiconducting devices, the basic materials such as Si and Ge must at least 99.9999% pure. Consequently extreme precautions have to be taken during the preparation of these elemental semiconductor materials.

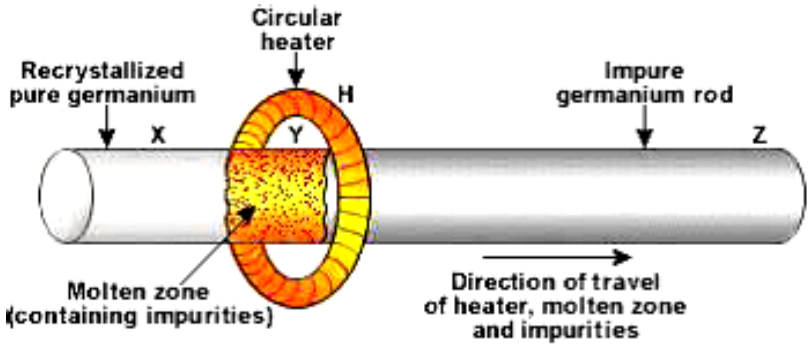
3.7 Distillation

It is based on the separation of materials due to difference in boiling points. The starting raw materials employed are germanium tetrachloride and trichloro silane for the preparation of pure Ge and Si respectively.

For example $GeCl_4$ (boiling point 86.5 ^{0}C) is taken in a series of distillation stills and a layer of HCl placed over the charge, which facilitates in the removal of As (usually present as an impurity). Chlorine gas is then passed through the charge, which is heated electrically. The vapours passing through the fractionating column finally distils over in a receiver, kept in an ice bath. Pure $GeCl_4$ so obtained is then treated with extra pure water to get germanium oxide which is subsequently reduced (in an atmosphere of hydrogen) to get elemental Ge, which is subjected to further purification. Similarly, distilled trichloro silane ($SiHCl_3$) is employed to get silicon of high purity.

3.8 Zone Refining

- This method is particularly used when metals are required in high degree of purity.
- It is based on the principle that when an impure metal in a molten state is allowed to cool, only the metal crystallizes while the impurities remain present in the molten state (mass) or melt.
- The zone refining technique involves melting a narrow zone of long ingot of the material to be processed.
- The impure metal converted into a rod which is heated at one end with a circular heater. A narrow zone of metal is melted. The heater is slowly moved along the rod. The pure metal recrystallizes out of melt while the impurities remain in the melt which moves along with the melted zone of the rod with the movement of heater. The process is repeated several times. The end of rod where impurities have collected is cut off.
- This method is employed for the purification of germanium, silicon, gallium etc, which are used in semi –conductors.

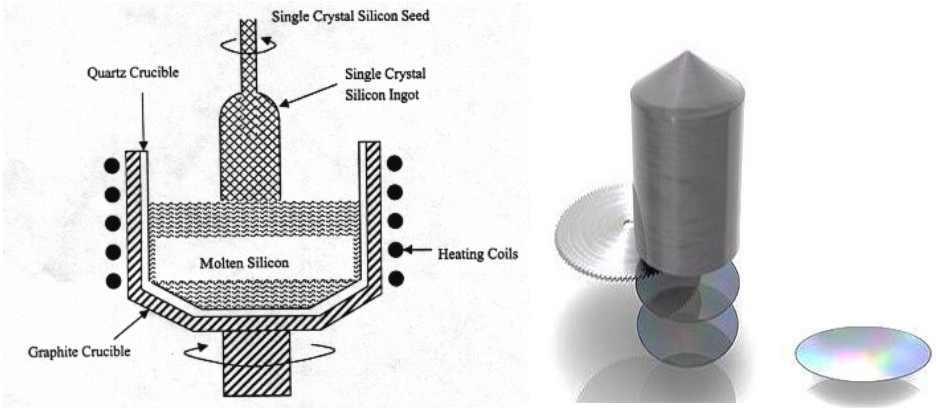


Zone refining of germanium metal.

3.9 Czochralski Crystal Pulling Technique

- Ge or Si obtained by zone refining method is polycrystalline, i.e., there is no regularity in their crystal structure and contain crystals of different sizes. But fabrication of most of the semiconductor devices requires single crystal of semi conducting materials. Consequently, they are grown in the form of single crystals.
- In this process, solid polycrystalline semiconducting material is taken in a quartz chamber and heated by using RF power source, till it melts.
- Then the temperature is lowered to just above the melting point of Ge.

- At this stage, properly oriented single crystal of Ge attached to a puller rod is carefully lowered into the melt, till it is in contact with the surface of the charge.
- Then the crystal is rotated slowly and then gradually pulled out so that the melt in contact with the seed crystal starts solidifying producing the orientation identical as that of the seed crystal.
- Finally, an intrinsic crystal of Ge is obtained.
- This crystal is tested for its conductivity and cut into wafers with the help of a diamond-tipped saw.



{**NOTE:** Doping : For getting a doped material, calculated quantity of dopant (Boron or Phosphorous) is added to the melt before crystal pulling stage. Usually dopant atoms added are of the order of one atom of B or P per 10^8 atoms of Ge or Si. Other methods of doping are: epitaxy, diffusion technique and ion implantation}

3.10 Epitaxy

Epitaxy means the natural or artificial growth of crystals on a crystalline substrate that determines their orientation. It involves in unified crystal growth or deposition of a thin crystal in another substrate, often with added dopants. The material in the epitaxial layer must have a little spacing and structure closed to that of the substrate.

Eg: Si on Ge or Ge on Si.

The Si or Ge wafer (kept in graphite boat) is placed in a long cylindrical quartz tube reactor, which is then heated (by RF induction coil). Then gases containing compounds of Ge or Si mixed with calculated/ appropriate quantities of dopant over the wafer results. The doped wafer so obtained is ready for the fabrication of semiconductor

devices. It may be pointed out that for getting Si epitaxial film. $SiCl_4$ H_2 and N_2 mixture is used. For carrying out n-type doping, this mixture is mixed with a calculated amount of phosphine (PH_3) and for p-type doping Diborane (B_2H_6) is employed.

3.11 Diffusion Technique

It involves in conversion of a region of semiconductor material by solid or gaseous diffusion of impurity atom into the crystal lattice of semiconductor material without melting any melting and it consists of heating a P-type dopant. This causes some of the impurity atoms to condense on the surface of water and diffuse thereby resulting in a P-type semiconductor material.

By this technique, the extent of impurity atoms penetration can be controlled even less than a few millions of a centimeter by controlling the concentration of dopant atoms and temperature. This technique is especially employed for preparing n or p type semiconductors, used in very high frequency devices. This doping technique can also be used to prepare n-p-n type transistor with a base width of even 5 microns.

3.12 Ion Implantation Technique

This technique involves in bombarding the semiconductor material with an electrically controlled beam of high energy ions of B or P of 10000 ev. This causes the implantation of some dopant atoms into the crystal lattice of semiconductor.

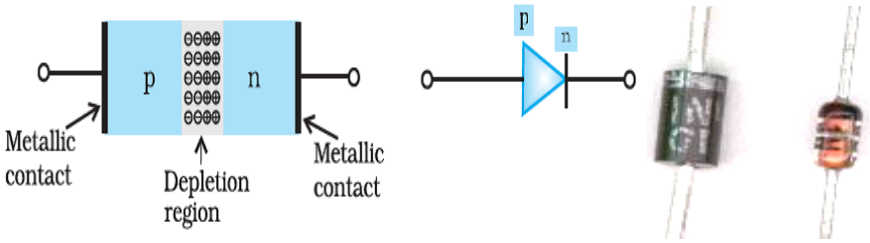
Main advantages of this technique are

1. Implantation at low temperature.
2. Extremely precise control of the dopant amount on desired specific semiconducting material.
3. Obtaining deep or shallow regions of very precise and required penetration depths having high or low concentration of dopant. This technique is especially employed for the fabrication of high frequency devices.

3.13 Semiconductor Devices

PN junction diode

- A p–n junction is formed by joining P-type and N-type semiconductorstogether in very close contact.
- Diode is a two-terminal electronic component that **conducts electric current in only one direction.**



- When an external voltage V is applied across a semiconductor diode suchthat p-side is connected to the positive terminal of the battery and n-sideto the negative terminal, it is said to be forward biased.

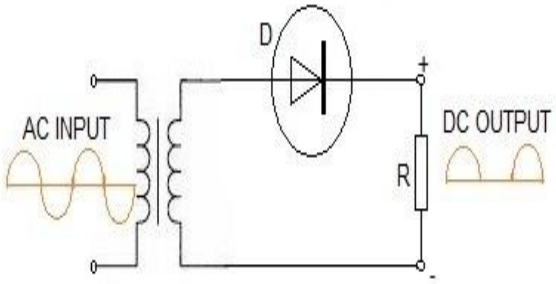


- When an external voltage (V) is applied across the diode suchthat n-side is positive and p-side is negative, it is said to bereverse biased



PN junction diode as Rectifier: A device which converts alternating current or voltage into direct current or voltage is known as rectifier. The process of converting AC into DC is called rectification.

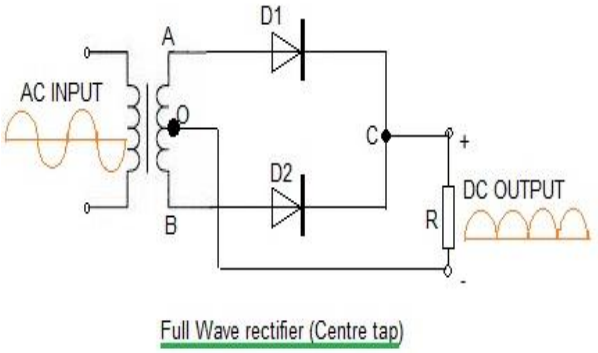
Half-Wave Rectifier: A half-wave rectifier converts the half cycle of applied AC signal into DC signal. Ordinary transformer may be used here.



Half Wave Rectifier

Full-Wave Rectifier: A full-wave rectifier converts the whole cycle of applied AC signal into DC signal. Centre tap transformer is used here.

[Half-wave rectifier converts only one-half of AC into DC while full wave rectifier rectifies both halves of AC input.]

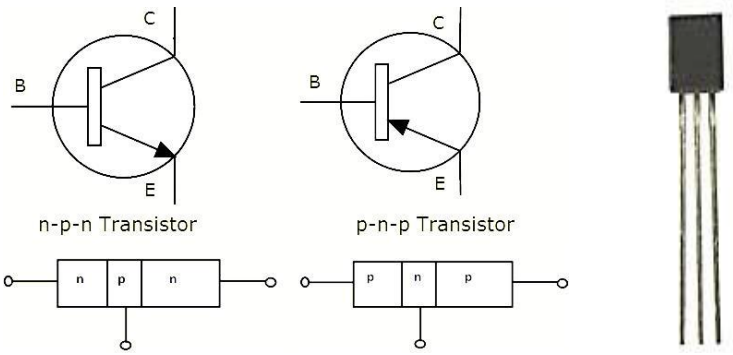


Full Wave rectifier (Centre tap)

Transistor: A transistor is an arrangement obtained by growing a thin layer of one type of semiconductor between two thick layers of other similar type semiconductor.

- A bipolar junction transistor (BJT) is a three-terminal electronic device constructed of doped semiconductor material.
- It may be used in amplifying or switching applications.
- Bipolar transistors are so named because their operation involves both electrons and holes.
- Charge flow in a BJT is due to bidirectional diffusion of charge carriers across a junction between two regions of different charge concentrations.
- 3 adjacent regions of doped Si (each connected to a lead)
  1. Base. (thin layer,less doped).
  2. Collector.
  3. Emitter

Types of transistors



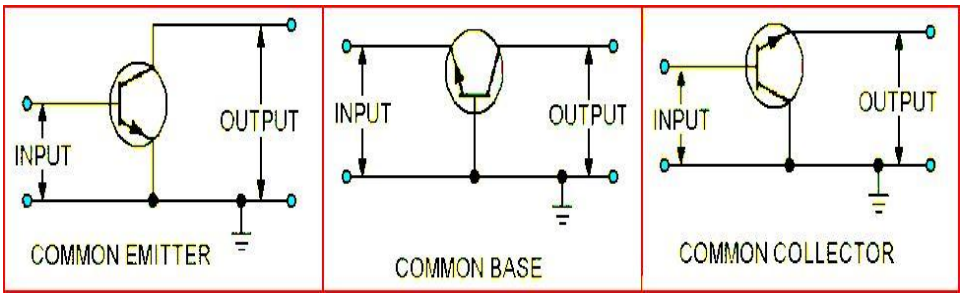
- The left side semiconductor is called emitter, the right side semiconductor is called collector and the thin middle layer is called base.
- Emitter is highly doped and base is feebly doped.
- A transistor can be utilized as an amplifier and oscillator but not a rectifier
- Maximum amplification is obtained in common-emitter configuration.

Depending upon the terminals which are used as a common terminal to the input and output terminals, the transistors can be connected in the following three different configurations.

Common emitter configuration: In this configuration emitter terminal is connected as a common terminal. The input is applied between the base and emitter terminals. The output is taken between the collector and base terminals.

Common base configuration: In this configuration base terminal is connected as a common terminal. The input is applied between the emitter and base terminals. The output is taken between the collector and base terminals.

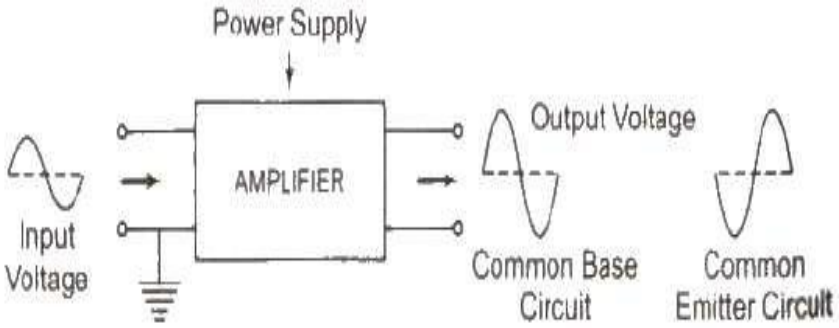
Common collector configuration: In this configuration collector terminal is connected as a common terminal. The input is applied between the base and collector terminals. The output is taken between the emitter and collector terminals



Transistor as an Amplifier

An amplifier is a device which is used for increasing the amplitude of variation of alternating voltage or current or power.

The amplifier thus produces an enlarged version of the input signal. The general concept of amplification is represented in figure. There are two input terminals forthe signal to be amplified and two output terminals for connecting the load; and' a means of supplying power to the amplifier.



3.14 Magnetic Materials

Any materials that can be magnetized by an applied by an applied external magnetic field is called a magnetic material.

The magnetic materials are classified into two categories

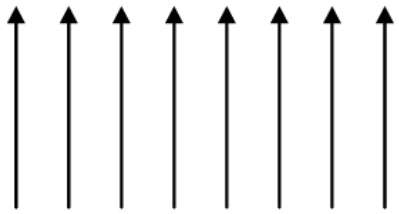
1. The materials **without** permanent magnetic moment
Eg: 1. Diamagnetic materials.

2. The materials **with** permanent magnetic moment.
Eg:

- **Paramagnetic materials:**In certain materials, net magnetic moment is zero though each atom or molecule possesses a permanent magnetic moment in the absence of an external magnetic field. But when an external magnetic field is applied the magnetic dipoles tend to align themselves in the direction of the magnetic field and the material becomes magnetized. This effect is known as paramagnetism.

- **Anti-Ferromagnetic materials**: In this material, the spins are aligned in anti-parallel manner due to unfavorable exchange interaction among them, resulting in zero magnetic moment. Even when the magnetic field is increased, it has almost zero induced magnetic moment.
  - Ferromagnetic materials
  - Ferri-magnetic materials.

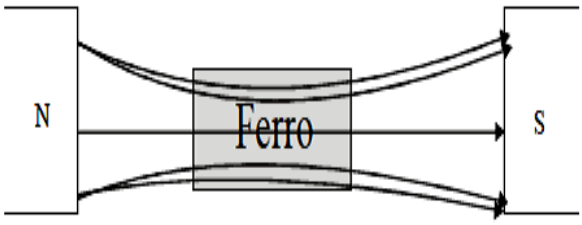
Ferromagnetic materials: Ferromagnetism is due to the presence of more unpaired electrons. Even in the absence of external field, the magnetic moments align parallel to each other. So that it has large magnetism. This is called spontaneous magnetization.



Examples: Nickel, iron, Cobalt, Steel, etc.

Effect of magnetic field

If a small external magnetic field is applied the magnetic moments align in the field direction and become very strong magnets.

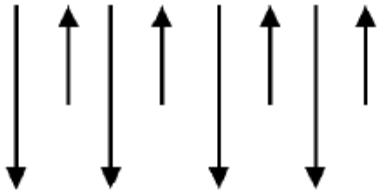


Properties of ferromagnetic materials

- All the dipoles are aligned parallel to each other due to the magnetic interaction between the two dipoles.
- They have permanent dipole moment. They are strongly attracted by the magnetic field.
- They exhibit magnetization even in the absence of magnetic field. This property of ferromagnetic material is called as spontaneous magnetization.
- They exhibit hysteresis curve.
- On heating, they lose their magnetization slowly.

Ferri-magnetic materials: Ferri-magnetic materials or Ferrites are much similar to Ferromagnetic materials. The magnetic dipoles are aligned anti-parallel with unequal magnitudes. If small value of magnetic field is applied, it will produce the large value of magnetization. Ferri-magnetic materials are widely used in high frequency applications and computer memories.

Eg: Ferrous Ferrites and Nickel Ferrites



Properties

- Ferrites have net magnetic moment.
- Above Curie temperature, it becomes paramagnetic, while it behaves ferromagnetic material below Curie temperature.
- The susceptibility of ferrite is very large and positive.
- Spin alignment is antiparallel of different magnitudes
- Mechanically it has pure iron character.
- They have high permeability and resistivity.
- They have low eddy current losses and low hysteresis losses.

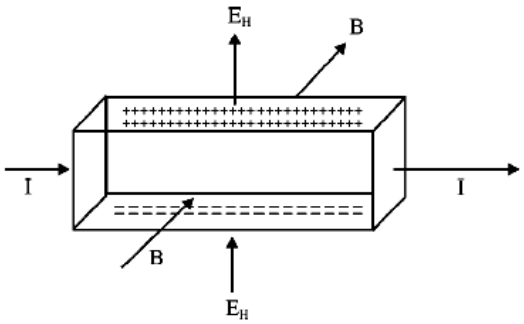
Applications

- They are used to produce ultrasonic waves by magnetostriction principle.
- Ferrites are used in audio and video transforms.
- Ferrite rods are used in radio receivers to increase the sensitivity.
- They are also used in power limiting and harmonic generation.
- They are used in computers and data processing circuits.
- Ferrites are used in paramagnetic amplifiers so that the input can be amplified with low noise figures.

3.15 Hall Effect

Measurement of conductivity will not determine whether the conduction is due to electron or holes and therefore will not distinguish between p-type and n-type semiconductor.Therefore Hall Effect is used to distinguish between the two types of charge carriers and their carrier densities and is used to determine the mobility of charge carriers.

When conductor (metal or semiconductor) carrying a current is placed in a transverse magnetic field, an electric field is produced inside the conductor in a direction normal to both the current and the magnetic field. This phenomenon is known as "**Hall effect"** and the generated voltage is called "**Hall voltage".**



Applications of Hall Effect

- It is used to determine whether the material is p-type or n-type semiconductor. (ie) if RH is negative then the material n-type. If the RH is positive then the material p-type.
- It is used to find the carrier concentration
- It is used to find the mobility of charge carriersμe, μh.
- It is used to find the sign of the current carrying charges.
- It is used to design magnetic flux meters and multipliers on the basis of Hall voltage.
- It is used to find the power flow in an electromagnetic wave.

3.16 Insulators

Insulating materials are those substances which resist the flow of heat, sound and electricity through them.

Electrical insulators: Also called dielectrics, these materials resist the flow of electric current. Perfect insulators do not exist but some materials such as glass, paper and Teflon which have high resistivity are regarded as good electrical insulators.

Eg: Rubber, glass, pure water, oil, air, diamond, dry wood, dry cotton, plastic

Properties of electrical insulating Material

1. It must be mechanically strong enough to carry tension and weight of conductors.
2. It must have very high dielectric strength to withstand the voltage stresses in High Voltage system.
3. It must possess high Insulation Resistance to prevent leakage current to the earth.
4. The insulating material must be free from unwanted impurities.

5. It should not be porous.
6. There must not be any entrance on the surface of electrical insulator so that the moisture or gases can enter in it.
7. There physical as well as electrical properties must be less affected by changing temperature.

Important electrical insulators and their applications

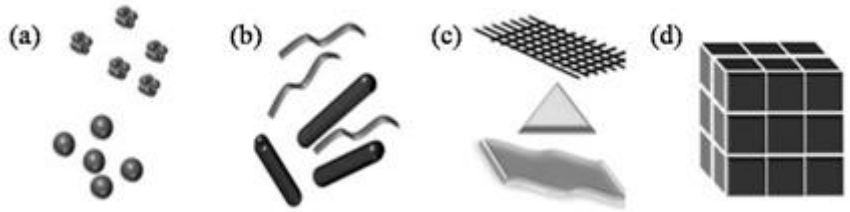
	Insulators	**Engineering applications**
Gases	Air	Air is used as an insulator to insulate high voltage transmission lines
	CO_2	used as an insulator in certain types of fixed capacitors
	N_2	N2 under high pressure is used as dielectric in certain types of electrical capacitors.
	Electronegative gases like SF_6& CCl_4	Used in capacitors and cables etc.,
liquids	Mineral oils	Used in cables, capacitors, transformers, switch gears etc.
	Vegetable oils like castor, palm, olive peanut	Used in formulation of insulating varnishes which are used in transformers and motor coils.
Solids	Paper and press boards made from wood	Used for windings and cable coil insulation.
	Fibrous insulators like silk, wool, jute..	Used in electrical machines and cables
	Polymers-polyethylene,polystyrene,Phenolics, epoxies and silicones	Used as high frequency insulator in radio, TV and communication circuit cables and also in power cables.
	Teflon	Used as capacitor dielectric and insulating material for all most all windings
	glass	Popularly used in transmission and distribution system.
	Ceramics and porcelain	This is widely used insulator. it is free from porosity.
	mica	Sheet mica is used prilimarily in electrical equipments and appliances in the form of washers, spacers, sleeves, tubes. etc.,

3.17 Nano Materials

Introduction

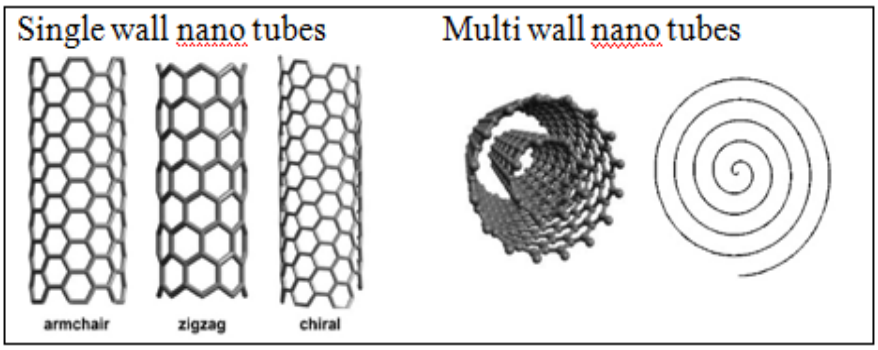
The word nano itself means **dwarf** in Greek. One nanometer is 10^{-9} m. 'The materials having structured components with at least one dimension less than 100 nm are known as nanomaterials'. Nano structured materials are classified as Zero dimensional, one dimensional, two dimensional, three dimensional nanostructures

a. 0-D spheres or clusters. Eg:- nano particles
b. One dimensional nanomaterials. Eg:- nano fibres, wires
c. Two dimensional nanomaterials. Eg:- Graphene,
d. Three dimensional nanomaterials. Eg:-Fullernes, CNT, quantum dots, quantum drivers.



Carbon nanotubes are a graphite sheets consisting of covalently bonded carbon atoms in hexagonal type arrangement. The sheet is rolled up into a cylinder with the ends closed by hemispherical graphite domes. Carbon nanotubes can be devided into two basic classes namely

- **Single wall nanotubes(SWNT) :** Single-walled carbon nanotubes can be formed in three different designs: Armchair, Chiral, and Zigzag. The design depends on the way the graphene is wrapped into a cylinder.
- **Multi wall nanotubes (MWNT):** MWNT's include structures formed in coaxial arrangement of several (2-100) graphite cylinders and their external diameter ranges from 10 to100nm
- There are two structural models of multi-walled nanotubes: – Russian Doll model – Parchment model. In the Russian Doll model, a carbon nanotube contains another nanotube inside it. In the Parchment model, a single graphene sheet is rolled around itself multiple times, resembling a rolled up scroll of paper.



- The way the graphene sheet is wrapped is represented by a pair of indices (*n*,*m*) called the chiral vector, where the integers *n* and *m* represent the number of unit vectors along two directions in the honeycomb crystal lattice of graphene.
- If *m* = 0, the nanotubes are called "zigzag".
- If *n* = *m*, the nanotubes are called "armchair".
- Otherwise, they are called "chiral".
- An ideal multi-wall carbon nanotube can similarly be visualized as concentrically rolled-up sheets of graphene.

3.18 Sol-Gel Method of Preparation of Nano Materials.(6M)

- It is a wet chemical synthesis of nano materials.
- It is a bottom up approach
- Sol-gel method of synthesizing nano-materials is very popular and is widely employed to prepare oxide nano materials.
- Sol-gel process involves hydrolysis followed by condensation
- Metal alkoxides are used as precursors in sol-gel process which disperse in acid or water to form sol. Gel is obtained from this sol by removal of water.

$$MOR + H_2O \rightarrow MOH + ROH \text{ (hydrolysis)}$$
$$MOH + ROM \rightarrow M\text{-}O\text{-}M + ROH \text{ (condensation)}$$

The sol-gel process can be characterized by a series of distinct steps.

1. Formation of stable solutions of the alkoxide or solvated metal precursor.

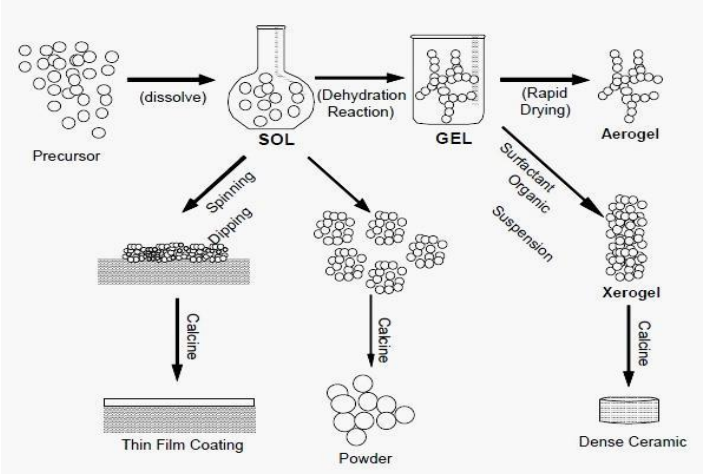
2. **Gelation:** It resulting from the formation of an oxide- or alcohol-bridged network (the gel) by a polycondensation reaction that results in increase in the viscocity of the solution.

3. **Aging of the gel (Syneresis)**: Gradually, the gel solidifies and the smaller particles to form larger particles. The gel transforms into a solid mass, accompanied by contraction of the gel network and expelling of solvent from gel pores.

4. **Drying of the gel**: Where water and other volatile liquids are removed from the gel network. If isolated of solvent is achieved by thermal evaporation, the resulting monolith is termed a xerogel. If the solvent (such as water) is extracted under supercritical or near super critical conditions, the product is anaerogel.

5. **Dehydration:** During which surface- bound M-OH groups are removed, there by stabilizing the gel against rehydration. This is normally achieved by calcining the monolith at temperatures up to 800°C.

6. **Densification and decomposition**: In this process the pores of the gel network are collapsed, at a temperature > 800°C and remaining organic species are volatilized. The typical steps that are involved in sol-gel processing are shown in the schematic diagram below.



Advantages

- The main advantage is that one can get mono sized nanoparticles can be prepared.
- It is cheap and low temperature technique.
- Widely used for synthesizing nonmetallic inorganic materials such as glasses, ceramic at very low temperature

Disadvantages

- ✓ Production rate of nano powders are very low
- ✓ Difficulties in controlling growth of the particles

3.19 Chemical Reduction Method of Preparation of Nano Materials

- The most common approach for synthesis of silver nanoparticles is chemical reduction by organic and inorganic reducing agents.
- In general, different reducing agents such as sodium citrate, DMF, $NaBH_4$, tollen's reagent are used for reduction of Ag^+ in aqueous or non- aqueous solutions.
- These reducing agents reduce Ag^+ and leads to formation of metallic Ag which is followed by formation of clusters.
- These clusters lead to formation of metallic colloidal Ag particles.
- Protective agents such as poly(vinyl alcohol), poly(vinyl pyrrolidine), poly(ethylene glycol) used to stabilize dispersive nanoparticles.

3.20 Brunauer Emmett Teller Method (BET)

Measured Properties: Specific surface area (m^2 /g)

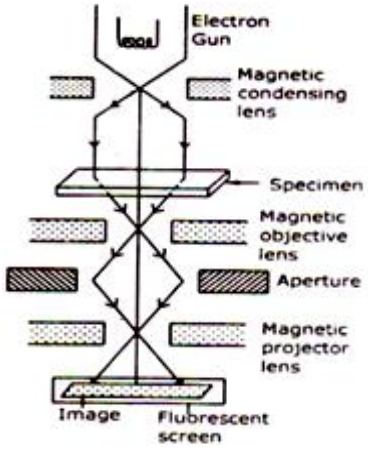
How it Works: The BET method is based on the adsorption of an inert gas on a surface. The sample is weighed and placed into a vacuum chamber, which is then evacuated. The sample is degassed and cooled down to a constant temperature by means of an external bath containing a cryogen like liquid nitrogen. A gas (typically N_2) is admitted and adsorbs on the sample surface. From the amount of gas adsorbed at a given pressure an adsorption isotherm is derived. The BET theory is then used to determine the amount of gas necessary to form a monolayer on the surface, also called the monolayer capacity. From the monolayer capacity, the known surface requirement of a single adsorbate molecule and the mass of the sample, the specific surface area in m^2/g can be calculated.

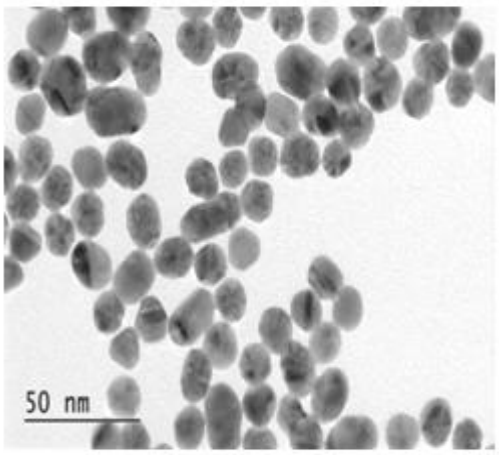
3.21 Transmission Electron Microscopy (TEM)

Measured Properties: Crystal size, Analysis of crystal, aggregate and agglomerate size distribution is possible using TEM in combination with image analysis.

Principle: Electrons are made to pass through the specimen and the image is formed on the fluorescent screen, either by using the transmitted beam or by using the diffracted beam.

How it Works: Transmission electron microscopy (TEM) is a technique whereby a beam of electrons is transmitted through a thin specimen, interacting with the specimen as it passes. An image is formed from the interaction of the electrons transmitted through the specimen; the image is magnified and focused onto an imaging device such as a fluorescent screen, for viewing the sample, or onto photographic film, for a record of the sample. Areas of dense material (such as solid nanoparticles) absorb the electrons and appear as dark areas on the viewing screen. Less dense material (such as coating around the nanoparticles) allows the electrons to pass through more freely and appear as lighter areas.



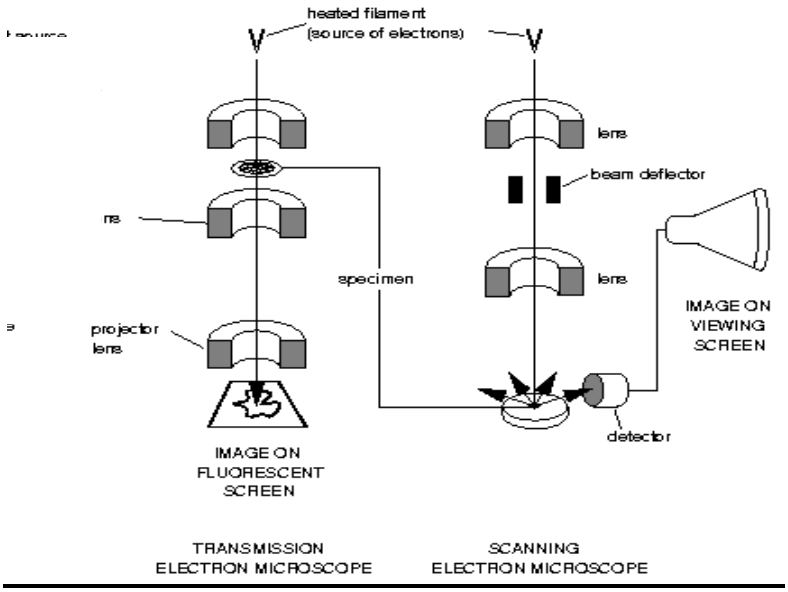


3.22 Scanning Electron Microscope (SEM)

SEM: A focused electron beam (2-10 keV) scans on the surface, several types of signals are produced and detected as a function of position on the surface. The space resolution can be as high as 1 nm.

Different type signal gives different information

a. Secondary electrons: surface structure.
b. Backscattered electrons: surface structure and average elemental information.
c. X-rays and Auger electrons: elemental composition with different thickness-sensitivity.



The SEM has a large depth of field (4mm – 0.4mm), which allows a large amount of the sample to be in focus at one time and produces an image that is good representation of the three-dimensional sample. The SEM also produces images of high resolution (1-10 nm), which means that closely features can be examined at a high magnification (10x – 3000000x).

The combination of large depth of field, high resolution and high magnification and compositional and crystallographic information makes the SEM one of the most heavily used instruments in research areas and industries, especially in semiconductor industry.
Signals were used in SEM are secondary electrons and backscattered electrons.

Secondary electrons (SE) were used to: mainly topography

- Low energy electrons, high resolution
- Surface signal dependent on curvature

Backscattered electrons (BSE) were used to: mainly chemistry

- High energy electrons
- "Bulk" signal dependent on atomic number

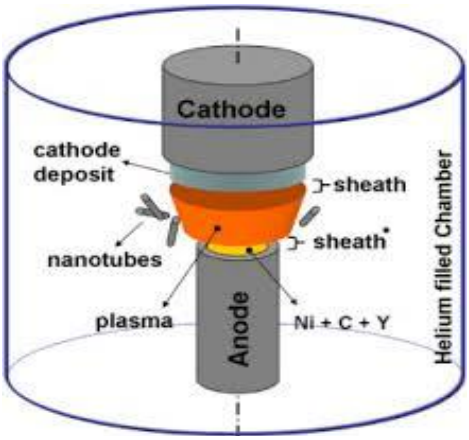
X-rays: to know chemistry

- Longer recording times are needed

3.23 Carbon Nanotubes: Preparation & Applications

1. The synthesis of CNTs by arc discharge method.

- The carbon arc discharge method, initially used for producing C_{60}fullerenes.
- It is the most common and perhaps easiest way to produce carbon nano tubes as it is rather simple to undertake.
- This method creates nanotubes through arc-vaporization of two carbon electrodes.
- In this two carbon rods are placed end to end, separated by approximately 1mm, in an enclosure that is usually filled with inert gas (helium, argon) at low pressure (between 50 and 700mbar).
- A direct current of 50 to 100 A driven by approximately 20 V creates a high temperature discharge between the two electrodes.
- The discharge vaporises one of the carbon rods and forms a small rod shaped deposit on the other rod usually cathode.
- However,it produces a mixture of components and requires separating nanotubes from the soot and the catalytic metals present in the crude product.
- Producing nanotubes in high yield depends on the uniformity of the plasma arc and the temperature of the deposit form on the carbon electrode.

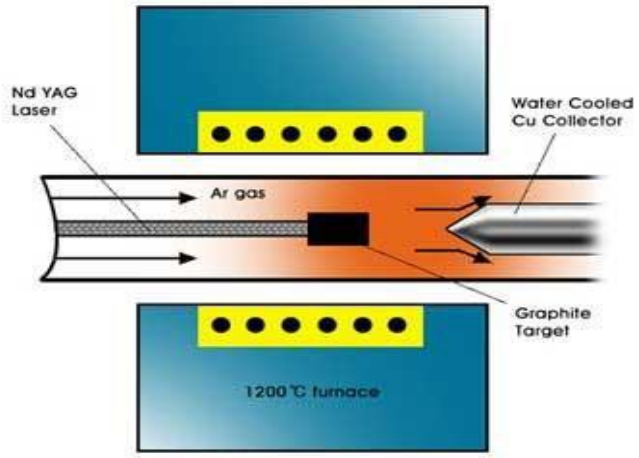


2. Laser ablation method of producing CNT.

- Both MWNTs and SWNTs can be produced with this technique. In order to generate SWNTs, metal particles as catalysts must be added to the graphite targets similar to the arc discharge technique.
- In laser ablation process a pulsed or continuous laser used to vaporise a

target containing 1.2 % of cobalt/nickel with 98.8 % of graphite composite in an oven at 1200°C.

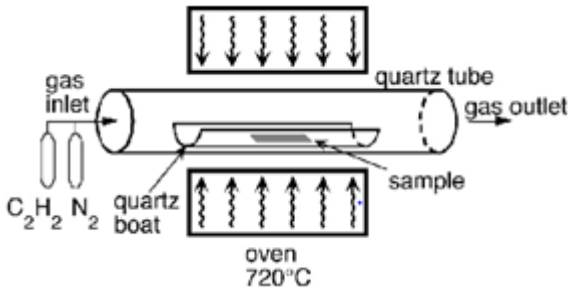
- The oven is filled with helium or argon gas in order to keep the pressure at 500 Torr.
- The vaporized species are then directed to a water-cooled copper collector and self-assemble from carbon vapors and condense on the walls of the cool surface by a flow of inert gas.
- The catalysts also begin to condense and attach to carbon clusters and prevent their closing into cage structures.
- From these initial clusters, tubular molecules grow into single-wall carbon nano tubes.



3. CVD method of preparing carbon Nano tubes.

- Chemical vapour deposition (CVD) synthesis is achieved by putting a carbon source in the gas phase.
- By using an energy source, such as plasma or a resistively heated coil, to transfer energy to a gaseous carbon molecule.
- Commonly used gaseous carbon sources include methane, carbon monoxide and acetylene.
- The energy source is used to "crack" the molecule into reactive atomic carbon.
- Then, the carbon diffuses towards the substrate, which is heated and coated with a catalyst (usually a first row transition metal such as Ni,Fe or Co) where it will bind.
- Carbon nanotubes will be formed if the proper parameters are maintained.
- Excellent alignment, as well as positional control on nanometre scale, can be achieved by using CVD.

- Control over the diameter, as well as the growth rate of the nanotubes can also be maintained.
- The appropriate metal catalyst can preferentially grow single rather than multi- walled nanotubes.



4. Applications of CNTs

- CNTs find use in water purification
- They are used in stain- resistant textiles
- Electrically conducting CNT films are used in LCD's, touch screens, computers, cell phones and photovoltaic devices.
- Used to prepared high tensile strength fibers
- CNTs are used for hydrogen storage owing to their capillary effect.
- CNT fibers are used to prepare bullet proof jackets
- CNTs are used for treatment of cancer
- CNTs are used to build ultra-capacitors in which energy storage capacity is much higher than the traditional capacitors.
- CNT may be able to replace steel in suspension and other bridges.
- CNTs can be used for drug delivery within the body by placing the drugs within the tubes or by attaching the drug to the sides of the tubes.

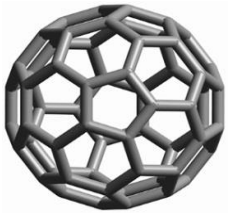
5. Properties of CNTs

- These are allotropes of carbon with a cylindrical nanostructure
- The chemical bonding of nanotubes is composed entirely of sp^2 bonds
- **Mechanical Properties:** Young's modulus of carbon nano tubes is 10 times greater than that of steel. CNTs have very structural defects in their walls, and hence, do not fracture on bending. The tensile strength of CNTs is about 20 times that of steel.
- **Electrical Properties:** The electrical properties of CNTs vary between metallic to semiconducting materials. The very high electrical conductivity of CNT is due to the minimum defects in the structure.

- **Thermal Conductivity:** The thermal conductivity of CNT is very high due to the vibration of covalent bonds due to minimum defects in the structure.

3.24 Fullerenes: Types, Preparations, Properties &Applications

- A fullerene is any molecule composed entirely of carbon, in the form of a hollow sphere, ellipsoid, tube, and many other shapes.
- Spherical fullerenes are also called buckyballs, and they resemble the balls used in football (soccer).
- Cylindrical ones are called carbon nanotubes or bucky tubes.
- Fullerenes are similar in structure to graphite, which is composed of stacked graphene sheets of linked hexagonal rings; but they may also contain pentagonal (or sometimes heptagonal) rings.



- The first fullerene molecule to be discovered is Buckminster fullerene (C_{60}).
- Fullerenes are cage like structure having formula C60, C70, C78 etc.,

Types of fullerene

1. **Spherical fullerenes:**
   Buckyball clusters: smallest member is C_{20} (unsaturated version of dodecahedrane) and the most common is C_{60};
   nano"onions": spherical particles based on multiple carbon layers surrounding a buckyball core, proposed for lubricants;

2. **cylindrical fullerenes:**
   Nano tubes: hollow tubes of very small dimensions, having single or multiple walls; potential applications in electronics industry;
   Mega tubes: larger in diameter than nanotubes and prepared with walls of different thickness; potentially used for the transport of a variety of molecules of different sizes

4. **linked "ball-and-chain" dimers:** two buckyballs linked by a carbon chain;

Preparation methods of fullerens

- Laser ablation
- Hydro carbon combustion
- Low pressure He sputtering

Applications of fullerenes

1. As a catalyst for hydrogenation, since these have ability to transfer H_2
2. Act as potential inhibitor to HIV.
3. Takes part in catalytic processes.
4. Used in rechargeable batteries, in microelectronic devices, solar cells.
5. Used in cosmetics for skin protection.
6. Used as MRI agents in medical world.
7. Used as proton exchange membrane for fuel cells.
8. Used as lubricants since act as miniature ball bearings.
9. Used to prepare soft ferro magnets with zero remanence (TDAE C_{60})(tetrakis dimethyl amino ethylene)

3.25 Graphene

Graphene is an allotrope of carbon that exists as a two-dimensional planar sheet. In other words grapheme is a single atomic graphite layer. Gaphene is technically a non-metal but is often referred to as a quasi-metal due to its properties being like that of a semi-conducting (partial conductance) metal. Each carbon atom is covalently bonded (SP^2 hybridized) to three other carbon atoms in a hexagonal ray, leaving one free electron per each carbon atom.

Types of graphene: Graphene oxide, graphene nanoplatelets, graphene nanoribbons and graphene quantum dots, graphene ink or graphene masterbatches.

Preparations of graphene: There are 3 main ways to synthesize

a. Chemical vapor deposition
b. Chemical or plasma exfoliation from natural graphite
c. Mechanical cleavage of natural graphite.

Properties

1. **Electronic properties**: Gaphene has a delocalized pi-electron system across the entirety of its surface, the movement of electrons is fluid. Due to overlapped of pi-electrons no band gap in graphene. Electron mobility is very high (15,000 $cm^2V^{-1}s^{-1}$).

2. **Thermal properties**: The repeating structure of graphene makes it an ideal material to conduct heat in plane. Graphene can exhibit two types of thermal conductivity – in-plane and inter-plane.

3. **Mechanical strength**: Graphene is one of the strongest materials ever discovered with a tensile strength of 1.3 x 10^{11} pa and also lightweight (0.77 mg/m^2). The mechanical strength of graphene is unmatched and as such can significantly enhance strength in many composite materials.

4. **Flexibility/Elasticity:** The repeating sp^2 hybridized backbone of graphene molecules allow for flexibility, as there is rotation around some of the bonds, whilst still providing enough rigidity and stability that the molecule can withstand changes in conformation and support other ions. Elasticity of graphene is 1-5 Nm^{-2} with a Young's modulus of 0.5TPa.

Applications

1. **Sensors:** Due to the planar and consistent arrangement of atoms in a graphene sheet, every atom within the sheet is exposed to the surrounding environment it can be applicable in sensor applications like bio-sensors, diagnostics, field effect transistors, DNA sensors and gas sensors with high degree of sensitivity.

2. **Batteries:** Graphene can be incorporated into either the anode or the cathode in various battery systems to increase the efficiency of the battery and improve the charge/discharge cycle rate. Due to its versatile nature, graphene has been incorporated into lithium-ion batteries, lithium-sulphur batteries, supercapacitors and fuel cells.

3. **Electron emission displays:** Graphene uses as ideal material for use in electronic emission displays as it exhibits a high aspect ratio.

4. **Strictural composites:** Graphene is being incorporated into materials to make the existing material stronger and more lightweight. It can be used in aviation industry.

5. **Catalyst supports:** Catalysts in the form of metal ions can sit in these cavities and be supported. In addition to providing mechanical support, the excellent charge carrier ability of graphene assists the charge transfer reactions involving the catalyst.

6. **Polymer Masterbatches:** Graphene can be incorporated into polymeric materials to form graphene-polymer composite materials. Incorporating graphene into polymers can also give polymers electrical conductivity properties.

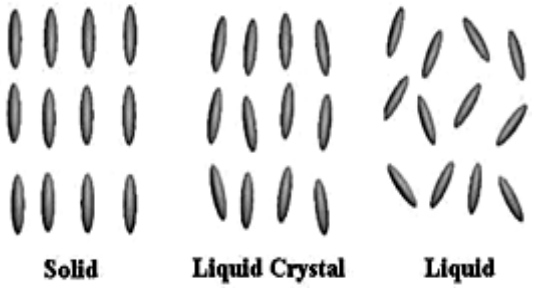
7. **Functional Inks:** Graphene can be used in functional inks for electronic, heat resistant and anti-corrosion purposes. By incorporating graphene into ink formulations, the conductivity properties associated with graphene influence the ink, causing it to become conductive.

3.26 Liquid Crystals: Introduction, Types, Applications

Liquid crystals (LCs) are matter in a state that has properties between those of conventional liquid and those of solid crystal.

Properties of LCs

- For instance, a liquid crystal may flow like a liquid, but its molecules may be oriented in a crystal-like way
- A state that occurs between a solid & a liquid
- Possess properties, characteristics of both liquids & crystalline solids
- Also possess properties not found in either liquids or solids
- May response to external perturbations & some changes color with temperature.
- Molecules in LCs generally elongated rod-like and disc like structure.
- Liquid crystals are anisotropic materials- their properties depend on the direction in which the measurement made.
- LCs are temperature sensitive.



They can be divided into
1. Thermotropic Phase
2. Lyotropic Phase

Types of liquid crystals

1. Thermotropic liquid crystals

- When a solid is heated to a certain temperature, it changes to liquid crystal then such liquid crystals are called thermotropic LCs
- If the temperature rise is too high, thermal motion will destroy the delicate cooperative ordering of the LC phase, pushing the material into a conventional liquid phase.
- At too low temperature, most LC materials will form a conventional crystal

There are three distinct phases in which thermotropic crystal exist.

A. Nematic phase
B. Smectic Phase
C. Cholesteric Phase

A. Nematic phase

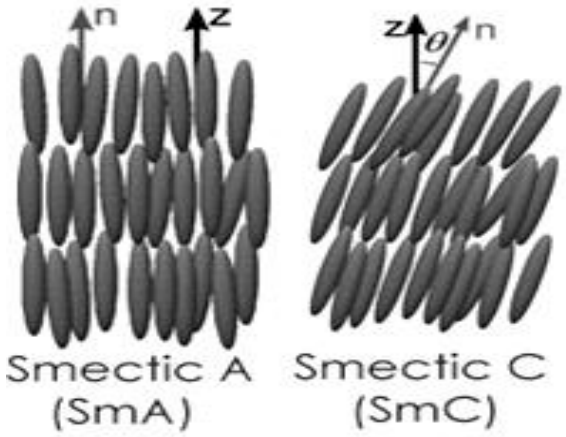


- One of the most common LC phases is the nematic.
- In a nematic phase, the rod-shaped organic molecules have no

positional order, but they have long-range directional order with their preffered axes called director roughly parallel.

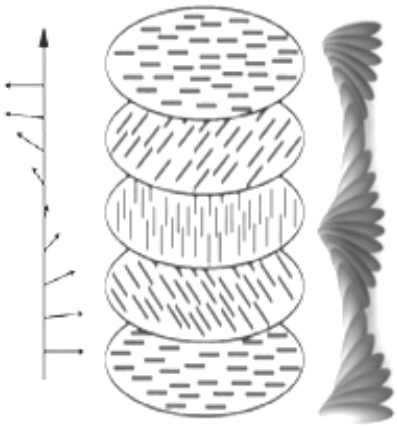
- Nematics have fluidity similar to that of ordinary (isotropic) liquids but they can be easily aligned by an external magnetic or electricfield.
- The alignment of molecules is temperature sensitive-as the temperature increased, the degree of orientation of nematic crystals decreases and they change into isotropic liquids.
- **Eg:** p-methoxycinnamic acid, p- azoxyanisole

B. Smectic phases



- The smectic phases, which are found at lower temperatures than the
- nematic form.
- Well-defined layers that can slide over one another in a manner similar to that of soap.
- The smectics are thus positionally as well as directionally ordered along one direction.
- More order and higher viscosity
- **Smectic-A:**In smectic-A,The molecules are aligned perpendicular to the layer plane
- **Smectic-C:** In smectic-C, the arrangement of molecules is similar to smectic-A except that the molecules are slightly tilted.
- **Eg:** Common soap exhibit smectic state at 200 oC

C. Cholesteric phase



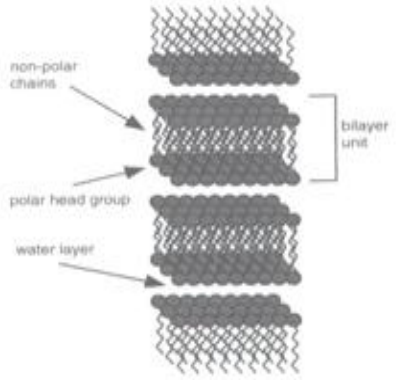
- Also known as chiral nematic, since these can rotate plane polarized light to right or left.
- Many derivatives of cholesterol exhibit this phase. Hence it is called cholesteric phase.
- In cholesteric phase, there is orientational order & no positional order, but, director is in helical order.
- Like the nematic phase, the molecules in this type of crystal are also parallel to each other but arranged in layers.
- The structure of cholesteric phase depends on the PITCH, the distance over which the director makes one complete turn
- One pitch - several hundred nanometers
- Pitch is affected by:-Temperature, Pressure, Electric & magnetic fields.
- **Eg:** chloesteryl benzoate.

2. Lyotropic liquid crystals

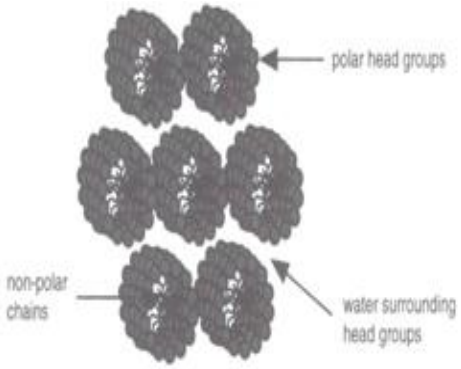
- The orientational behavior of lyotrophic crystal is a function of concentration and solvent.
- These belongs to amphiphilic compounds
- Amphiphilic molecules possess both polar and non-polar regions in the same molecule.
- In concentrated solutions, lyotropic liquid crystals can be formed.
- **Eg:** sodium laurate in water (soap) and various phospholipids present in cell memberane
- Solvent and temperature influence the stability of this phase.
- Lyotropic LCs are two types

Lamellar lyotropic phase: This phase consists of a layered arrangement of amphiphilicmolecules.

Hexagonal lyotropic phase: The hexagonal lyotropic liquid crystal phases have a molecular aggregate ordering which corresponds to a hexagonal arrangement.



Lamellar phase LC's

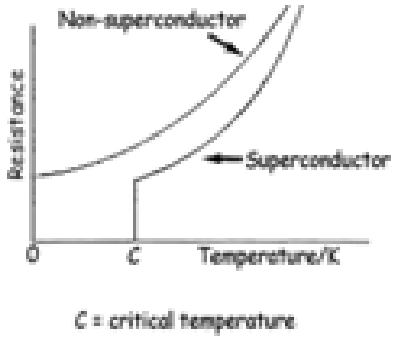


Hexagonal phase LC's

Applications of Liquid Crystal

- Liquid crystals can be used to detect radiations like UV, IR, and Microwave, ultra sonic and other ionizing radiations.
- Liquid crystals can be used to detect pollutants in the atmosphere.
- Liquid crystal displays are used in watches, calculators, laptops computers, sign boards.
- Liquid crystals can be used in thermo strips (to read body temperature) and disposable thermometers.
- Tumor cells are at elevated temperatures than normal cells. LCs shows different color when it is in contact with tumor cell. Hence LCs is used to detect tumor cells and infections.
- In electronic industry, they find use in locating faults in circuits, condition of batteries.
- An application of liquid crystals that is now being explored in optical imaging and holography for recording.
- Some of the liquid crystals are used in hydraulic break/clutch system due to their high viscosity values.

3.27 Super Conductors: Type-I, Type-Ii, Properties, Applications



C = critical temperature

The phenomenon of sudden disappearance of electrical resistance in a material, when it is cooled below a certain temperature is known as super conductivity.

(Or)

The ability of certain metals and alloys exhibit almost zero electrical resistivity when they are cooled to low temperature is known as superconducting materials.

Eg: Tin (3.2 K), Mercury (4.15 K).
Barium - Lanthanum - Copper - Oxide (BLCO) - 35 K
Yttrium - Barium - Copper - Oxide - ($YBa_2Cu_3O_4$) - 92 K

1:2:3 super conductor

Yttrium - Barium - Copper - Oxide - ($YBa_2Cu_3O_4$) - 92 K

Critical temperature (or) Transition Temperature? (2M)

The temperature at which a normal conductor loses its resistivity and becomes a superconductor is known as critical temperature (or) Transition temperature.

Every superconductor has its own critical temperature at which it passes over into superconducting state.

Depending on the transition temperature, superconductors are classified into two groups are

1. **Low temperature superconductors (LTS):** The superconductors which have low transition temperature (below 30K) are known as low temperature superconductors.

   Eg: Tin (3.2 K), Mercury (4.15 K).

2. **High temperature superconductors (HTS):** The superconductors which have high transition temperature (above 30K) is known as high temperature superconductors.

   Example: Barium - Lanthanum - Copper - Oxide (BLCO) - 35 K
   Yttrium - Barium - Copper - Oxide - (Y Ba2 Cu3 O4) - 92 K.

2.28 Properties of Superconductors

1. **Zero Electrical Resistance:** The electrical resistivity drops to zero at the transition temperature.

2. **Effect of magnetic field:** Below T_C, the superconducting property disappears when strong magnetic field is applied. i.e., it returns to normal state.

3. **Diamagnetic property:** The superconductor is a perfect diamagnetic.

4. **Effect of heavy Current:** The superconducting property disappears when a heavy current flows, since current flow will set up a magnetic field.

5. **Isotope effect:** The presence of isotopes in superconductors changes the transition temperature of the superconductors.

   Eg: The atomic mass of mercury varies from 199.5 to 203.4, and hence the transition temperature varies from 4.185 K to 4.146 K.

6. **Thermal properties:** Entropy and specific heat decreases at transition temperature and the thermoelectric effect disappear in the superconducting state.

7. **Effect of Impurities:** The impurities change the magnetic properties of the superconductors

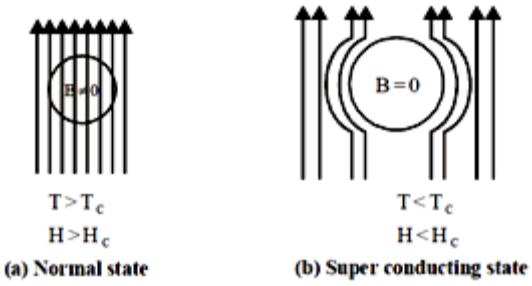
8. **Effect of pressure:** If pressure increases, the critical temperature also increases

9. The transition temperature is unchanged with the frequency variation.

3.29 Meissner Effect

When a superconducting material is placed in a magnetic field, under the condition T≤Tc and H≤Hc the flux lines are excluded from the material. Thus the material exhibits perfect diamagnetism. This phenomenon is called as Meissner effect

We know that diamagnetic material have the tendency to expel the magnetic lines of force. Since the super conductor also expels the magnetic lines of forces it behaves as a perfect diamagnet. This behavior is first observed by Meissner and hence called as Meissner effect.



When a superconducting material is placed in a magnetic field of flux density 'B' the magnetic lines of force penetrates through the material as shown in fig. a

Now, when the material is cooled below its transition temperature then the magnetic lines of force are ejected out from the material as shown in fig. b

Application of Meissner effect

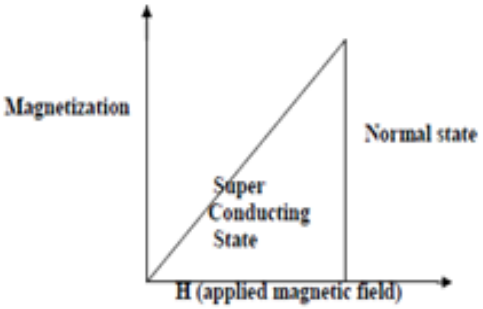
1. It is a standard test to prove whether the material is a perfect superconductor or not.
2. This effect is used for magnetic levitated train.

3.30 Type-I and Type –II Super Conductors

There are two types of super conductors based on their variation in magnetization, due to external magnetic field applied.

- Type I superconductor or soft super conductor
- Type II superconductor or hard superconductor

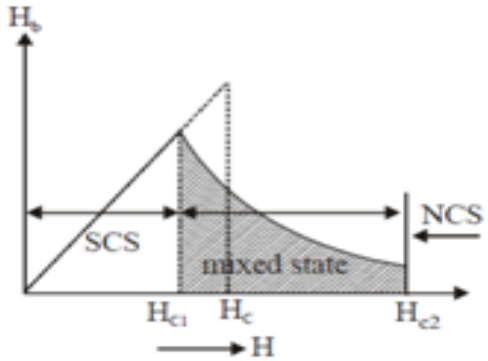
Type I Superconductors



When the super conductor is kept in the magnetic field and if the field is increased the superconductor becomes normal conductor abruptly at critical magnetic field as shown in fig. These types of materials are termed as Type I superconductors**.**

Below critical field, the specimen excludes all the magnetic lines of force and exhibit perfect Meissner effect. Hence, Type I superconductors are perfect diamagnet, represented by negative sign in magnetization.

Type II Superconductors



When the super conductor kept in the magnetic field and if the field is increased, below the lower critical field Hc1, the material exhibit perfect diamagnetism (i.e) it behaves as a super conductor and above Hc1, the magnetization decreases and hence the magnetic flux starts penetrating through the material. The specimen is said to be in a mixed state between Hc1 and Hc2. above Hc2 (upper critical field) it becomes normal conductor as shown in fig.

3.31 Distinguish Between Type–I And Type – II Superconductors

Sn No	Type I (or) Soft superconductors	Type II (or) Hard superconductors
1.	It exhibits complete Meissner Effect.	It does not exhibit a complete Meissner Effect
2.	They are completely diamagnetic.	They are not completely diamagnetic
3.	Ex : Tin, Lead, Mercury, etc.,	Ex : Nb-Zr, Nb- Ti, Nb-Sn, Va- Ga, etc.,
4.	The value of H_C is always too low. It is about 0.1 tesla.	The values of H_C is high and it is about 30 tesla.
5.	They have only one critical magnetic field value.	They have two critical magnetic field values. Lower critical magnetic field [H_{C1}] and Higher critical magnetic field [H_{C2}].
6.	No mixed state exists.	Mixed state is present.
7.	These are called Soft superconductors.	These are called Hard superconductors
8.	It requires low magnetic field to destroy the superconductivity.	It requires large magnetic field to destroy the superconductivity.
9.	The material loses magnetization suddenly	The material loses magnetization gradually

3.32 Applications of Superconductors

1. Electric generators can be made by using superconductors with smaller size, less weight and low energy consumption.
2. Superconductors can be used for the transmission of power over very long distances without any power loss.
3. Superconductors can be used in switching Devices.
4. The superconductors can be used in sensitive electrical instruments.
5. It can be used as a memory or storage element in computers.
6. These are used to design **Cryotron**, **Maglev**, Josephson Devices and **SQUID.**
7. Superconductors are used to design rectifiers, logic gates, modulators etc.
8. Ore separation can be done by using machines made of superconducting magnets.
9. Superconductors are used for the detection of brain tumor, defective cells, etc.,
10. Superconducting magnetic field may be used to launch satellite into orbit directly from the earth without use of rockets.
11. These are used in NMR (Nuclear Magnetic Resonance) imaging equipment which is used for scanning purposes.
12. DC superconducting motors are used in ship propulsion and in large mills.

13. Superconducting solenoids are used in magneto hydrodynamic power generation to maintain the plasma in the body.

Mag Lev: Magnetic levitated train is the train which cannot move over the rail; rather it floats above the rail, under the condition, when it moves faster**.**

Cryotron: Cryotron is a magnetically operated current switch.

Squid: Superconducting Quantum Interference Device

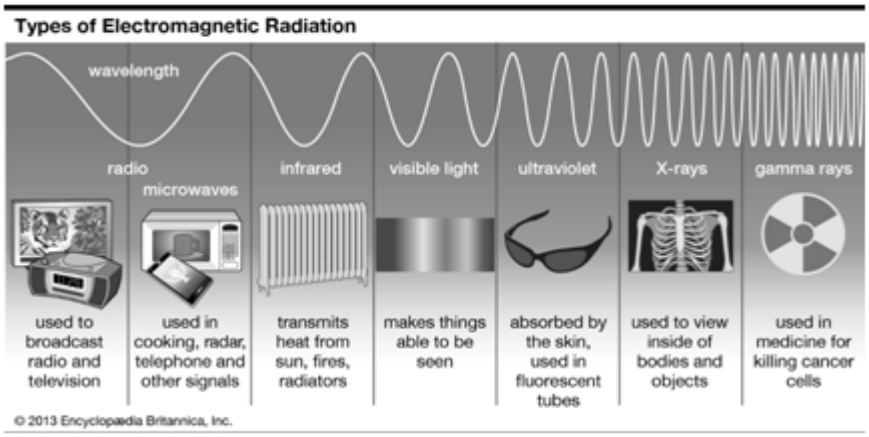
Unit IV

Spectroscopic Techniques & Non Conventional Energy Sources

Spectroscopic Techniques

4.1 Electromagnetic Spectrum

- The electromagnetic spectrum (EM spectrum) is made up of radio waves, microwaves, infrared waves, visible light, ultraviolet light and even X-rays and gamma rays.
- All of these electromagnetic waves are similar in a lot of important ways and can travel through empty space.
- The speed of these waves at empty space is about $3x10^8$ m/s.
- Electromagnetic waves can travel extremely fast in vacuum but differ in their wavelength and frequency.
- The frequency of a wave is the number of times it repeats in one second, and frequency is usually measured in units of Hertz (1Hz = 1cycle/second).
- The wavelength is the distance between successive wave peaks. For electromagnetic waves, wavelength is usually measured in units of nanometers (1 nm = $1x10^{-9}$ m).



UV spectroscopy

UV spectroscopy is an important tool in analytical chemistry. The other name of UV (ultra-violet) spectroscopy is Electron spectroscopy. It

involves the promotion of the electrons form the ground state to excited state (higher energy state).

4.2 Laws of Absorption

1. **Beer's law**: The intensity of a beam of monochromatic light decrease exponentially with the increase in concentration of absorbing substances.

$$-\frac{dI}{dC} \propto I$$

$$I = I_0 \bar{e}\, KC$$

2. **Lambert's law**: When a beam of light is allowed to pass through a transparent medium, the rate of decrease of intensity with the thickness of medium is directly proportional to the intensity of the light**.** **A = ε. C. l**
   Where
   ε. = molar absorptivity or absorption coefficient;
   C = concentration; l = path length

Limitations of laws

1. The real limitation of the beer's law successfully in describing the absorption behavior of dilute solutions only.
2. As degree of interaction depends upon the concentration, the occurrence of this phenomenon causes deviations from linear relationship between absorbance and concentrations.

4.3 Instrumentation of UV Spectroscopy

UV spectroscopy obeys the Beer – Lambert law. Instrumentation and working of the UV spectrometers can be studied simultaneously.

UV spectrometers consists the following parts-

Light source: Tungsten filament lamps and Hydrogen – Deuterium lamps are most widely used and suitable light sources which can cover the whole UV/Vis region.

Monochromator: Monochromators generally composed of prism and slits. The most of the spectrophotometers are double beam

spectrophotometers. The radiation emitted from the primary light source is dispersed with the help of rotating prisms.

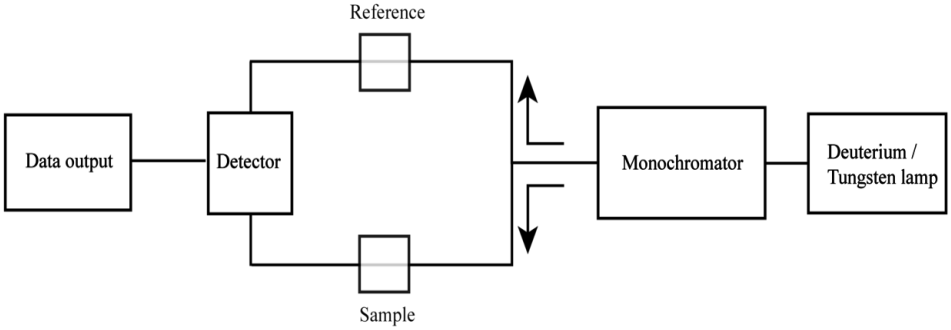
- The various wavelengths of the light source which are separated by the prism are then selected by the slits such the rotation of the prism results in a series of continuously increasing wavelength to pass through the slits for recording purpose.
- The beam selected by the slit is monochromatic and further divided into two beams with the help of another prism.

Sample and reference cells: One of the two divided beams is passed through the sample solution and second beam is passes through the reference solution. Both sample and reference solution are contained in the cells. These cells are made of either silica or quartz. Glass can't be used for the cells as it also absorbs in the UV region.

Detector: Generally two photocells serve the purpose of detector in UV spectroscopy. One of the photocell receives the beam of sample and second detector receives the beam from the reference. The intensity of the radiation from the reference cell is stronger than the beam of sample cell. This results in the generation of pulsating or alternating currents in the photocells.

Amplifier: The alternating current generated in the photocells is transferred to the amplifier. The amplifier is coupled to a small servo meter. Generally current generated in the photocells is of very low intensity, the main purpose of amplifier is to **amplify the signals** many times so we can get clear and recordable signals.

Recording devices: Most of the time amplifier is coupled to a pen recorder which is connected to the computer. Computer stores all the data generated and produces the spectrum of the desired compound.



4.4 Theory of Electronic Spectroscopy

In U. V spectroscopy molecule undergo **electronic transition involving σ, π and n electrons**.

Four types of electronic transition are possible.
1. σ → σ* transition,
2. n → σ* transition,
3. n → π* transition and
4. (iv) π → π* transition**.**

1. σ → σ* transition

- An electron in a bonding σ orbital of a molecule is excited to the corresponding anti-bonding orbital by the absorption of radiation.
- To make a σ → σ*transition, it required **Large energy.**
  Eg: Methane
- Methane contain only single C-H bonds it undergoes only σ → σ*transition. It gives absorption at 125 nm.

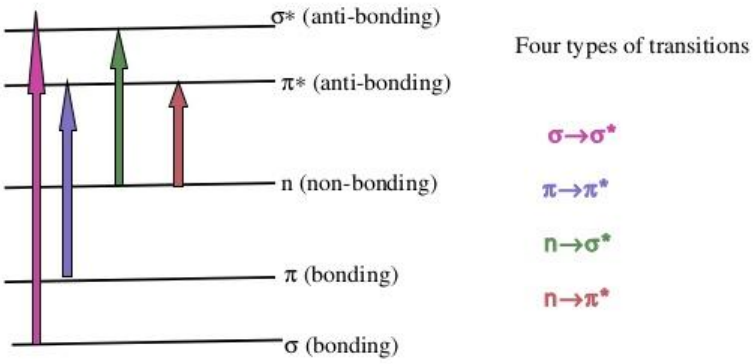
2. n → σ* transition

- In this transition, saturated compounds containing atoms with unshared electron pairs are undergo n → σ*transition.
- It require less energy than the σ → σ*transition.
- Most of the absorption peaks appearing below 200 nm.
- In the presence of polar solvents the absorption maximum tend to shift shorter wavelength
  Eg: Water, ethanol
- In this the peaks in U.V region relatively small, Ex: Methyl chloride, oxygen, Nitrogen.

3. n → π* & π → π* transitions

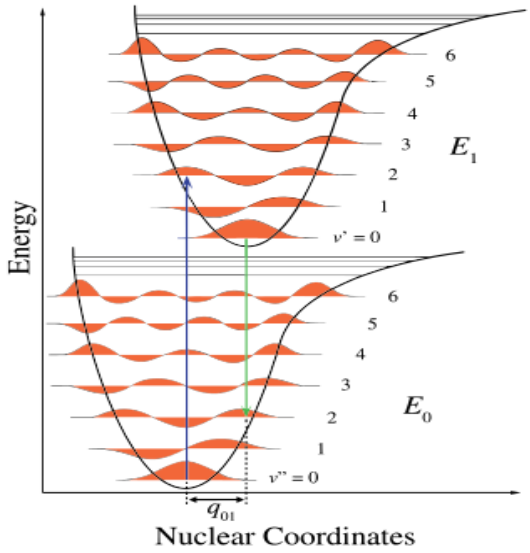
- Most organic compounds are undergo transitions for n → π* & π → π*.
- Because energies required for processes bring the absorption peaks into spectral region.
- Both transitions require the presence of an unsaturated functional group to the π orbitals.

- **Eg:** For π → π* transition: Alkenes, carbonyl compounds and alkynes. For n → π* transition: carbonyl compounds



4.5 Franck – Condon Principle

The Franck – Condon principle is a rule in spectroscopy and quantum chemistry that explains the intensity of vibronic transitions. Vibronic transitions are the simultaneous changes in the electronic and vibrational energy levels of a molecule due to the absorption or emission of a photon of the appropriate energy. The principle states that during an electronic transition, a change from one vibrational energy level to another will be more likely to happen if the two vibrational wave functions overlap more significantly.



Franck – Condon Factors: If electronic excitation is much faster than nuclei move, then wave functionand cannot change. The most likely transition is the one that has most overlap with excited state wave function.

4.6 Chromophores and Auxochromes

Chromophore: Chromophore is defined as any isolated covalently bonded group that shows a characteristic absorption in the ultraviolet or visible region (200 – 800 nm). Chromophores can be divided into two groups.

a. Chromophores which contain 'p' electrons and which undergo $\pi \rightarrow \pi^*$ transitions. Ethylenes and acetylenes are the examples of such chromophores.
b. Chromophores which contain both 'p' and nonbonding (n) electrons. They undergo two types of transitions $\pi \rightarrow \pi^*$ and $n \rightarrow \pi^*$. Carbonyl, nitriles, azo compounds, nitro compounds etc. are the example of such chromophores.

Auxochromes: An auxochrome can be defined as any group which does not itself acts as a chromophore bit whose presence brings about a shift of the absorption band towards the longer wavelength of the spectrum.

-OH, -OR, $-NH_2$, -NHR etc. are the examples of auxochromic groups.

4.7 Intensity Shifts

There are four types of shifts observed in the UV spectroscopoy.

1. **Bathochromic effect:** This type of shift is also known as red shift. Bathochromic shift is an affect by virtue of which the absorption maximum is shifted towards the longer wavelength due to the presence of an auxochrome or change in solvents.
   $n \rightarrow \pi^*$ transitions of carbonyl compounds observes bathochromic or red shift.

2. **Hypsochromic shift:** This effect is also known as blue shift. Hypsochromic shift is an effect by virtue of which absorption maximum is shifted towards the shorter wavelength. Generally it is caused due to the removal of conjugation or by changing the polarity of the solvents.

3. **Hyperchromic effect:** Hyperchromic shift is an effect by virtue of which absorption maximum increases. The introduction of an auxochrome in the compound generally results in the hyperchromic effect.

4. **Hypochromic effect:** Hyperchromic effect is defined as the effect by virtue of intensity of absorption maximum decreases. Hyperchromic effect occurs due to the distortion of the geometry of the molecule with an introduction of new group.

4.8 Applications

1. **Detection of functional groups:** UV spectroscopy is used to detect the presence or absence of chromophore in the compound. This is technique is not useful for the detection of chromophore in complex compounds. The absence of a band at a particular band can be seen as an evidence for the absence of a particular group.

2. **Detection of extent of conjugation:** With the increase in double bonds the absorption shifts towards the longer wavelength.

3. **Identification of an unknown compound:** The spectrum of unkown compound is compared with the spectrum of a reference compound. If both spectrums coincide then it confirms the identification of unknown substance.

4. **Determination of conjugations of geometrical isomers:** Two isomers can distinguished with each other when one of the isomer has non-coplanar structure due to steric hindrances. The cis-isomer suffers distortion and absorbs at lower wavelength as compared to trans-isomer.

5. **Determination of the purity of the substance:** The absorption of the sample solution os compared with the absorption of the reference solution. The intensity of the absorption of can be used for the relative calculation of the purity of the sample solutions.

4.9 FT-IR Instrumentation

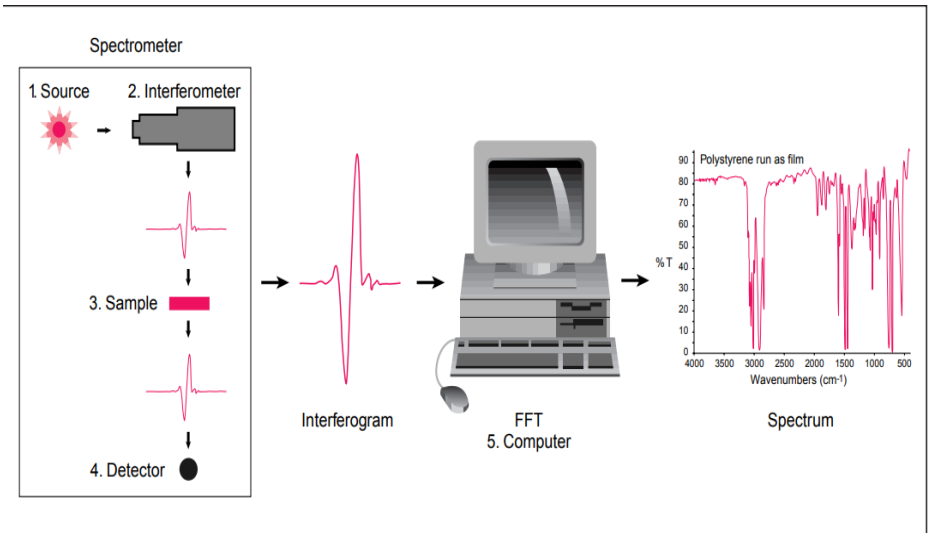
1. FT-IR stands for Fourier Transform Infrared, the preferred method of infrared spsectroscopy. In infrared spectroscopy, IR radiation is passed through a sample.

2. Some of the infrared radiation is absorbed by the sample and some it is passed through (transmitted).

3. The resulting spectrum represents the molecular absorption and transmissions, creating a molecular finger print of the sample.

4. The normal instrumentation process is as follows

   a. **The source:** Infrared energy is emitted from a glowing black-body source. This beam passes through an aperture (slit) which controls the amount of energy presented to the sample.
   b. **The interferometer:** The beam enters the interferometer where the "spectral encoding" takes place. The resulting interferogram signal then exists the interferometer.
   c. **The sample:** The beam enters the sample compartment where it is transmitted through or reflected off of the surface of the sample, depending on the type of analysis being accomplished (able to perform). This is where specific frequencies of energy, which are uniquely characteristic of the sample, are absorbed.
   d. **The detector:** The beam finally passes to the detector for final measurement. The detectors used are specially designed to measure the special interferogram signal.
   e. **The computer:** The measured signal is digitized and sent to the computer where the Fourier transformation takes place. The final infrared spectrum is then presented to the user for interpretation and any further manipulation.



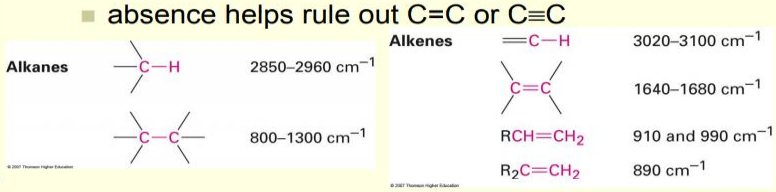
4.10 IR of Some Organic Compounds

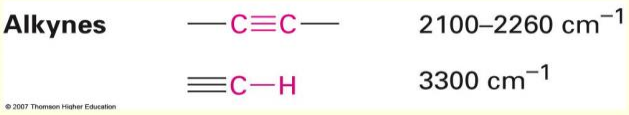
Table of IR absorptions common. Note: strong, medium, weak refers to the length of the peak.

Approximate Frequency (cm^{-1})	Description	Bond Vibration	Notes
3500 - 3200	Broad, round	O – H	Much boarder, lower frequency (3200 – 2500) if next to C=O
3400 – 3300	Weak, triangular	N – H	Stronger if next to C=O
3300	Medium-strong	=C – H (sp C – H)	
3100 – 3300	Weak - medium	=C – H (sp^2 C – H)	Can get bigger if lots of bonds present
3000 – 2900	Weak - medium	=C – H (sp^2 C – H)	Can get bigger if lots of bonds present
2800 and 2700	medium	C – H in O=C – H	Two peaks; " alligator jaws"
2250	Medium	C=N	
2250 – 2100	Weak - medium	C=C	Stronger if near electronegative atoms
1800 – 1600	Strong	C=O	Lower frequency (1650-1550) if attached to O or N Middle frequency if attached to C, H Higher frequency (1800) if attached to Cl
1600 – 1450	Weak – medium	C=C	Lower frequency (1600-1450) if conjugated (i.e., C=C-C=C)
1450	Weak - medium	H –C – H bend	
1300 -1000	Medium - strong	C – O	Higher frequency (1200 -1300) if conjugated (i.e., O=C=O or C=C-O)
1250 – 1000	Medium	C – N	
1000 – 650	Strong	C=C – H bend	Often several if benzene present

IR of Hydrocarbons

- **Alkanes, Alkenes, Alkynes**
- C-H, C-C, C=C, C≡C have characteristic peaks based on bond strengths
  - absence helps rule out C=C or C≡C

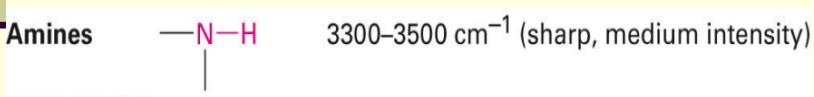




IR of Alcohols and Amines

- O–H 3400 to 3650 cm^{-1}
  - Usually broad and intense
- N–H 3300 to 3500 cm^{-1}
  - Sharper and less intense than an O–H

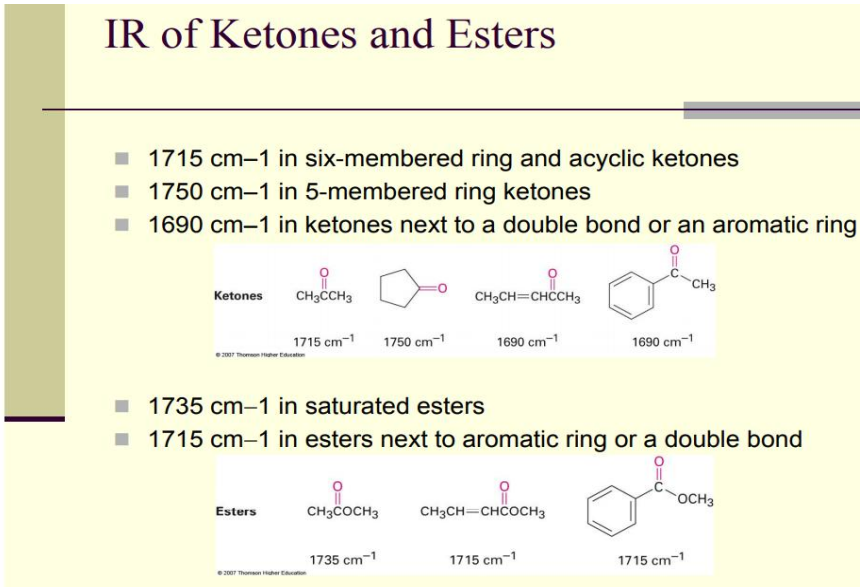


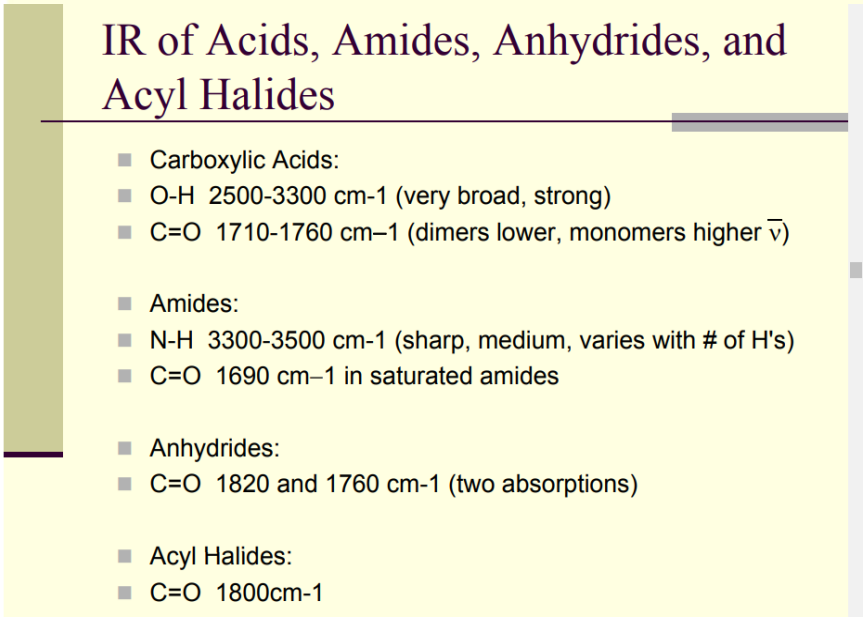


IR of Carbonyl Compounds: Aldehydes

- Strong, sharp C=O peak 1670 to 1780 cm^{-1}
- Exact absorption characteristic of type of carbonyl compound (ald, ket, ester, acid, amide, etc)
  - 1730 cm^{-1} in saturated aldehydes
  - 1705 cm^{-1} in aldehydes next to double bond or aromatic ring







4.11 Applications of FT-IR

1. It is a valuable technique for monitoring air quality, testing water quality and analyzing soil to address environmental and health concerns caused by increasing pollution levels.
2. This technique can provide fast, easy and consistent analysis for drugs, chemical evaluation, fibers and residues.
2. It helps in basic drug research and structural elucidation, formulation development and validation, quality control processes and packaging testing.
3. It is used to quickly and definitively identify compounds such as compounded plastics, blends, fillers, paints, rubbers, coatings, resins and adhesives.

4. It provides high – quality spectral data to accelerate your research, routine QA/QC testing or investigative needs.
5. It can be used in remote sensing.

4.12 Magnetic Resonance Imaging Procedure

1. Magnetic resonance imaging (MRI) is a spectroscopic imaging technique used in medical settings to produce image of the inside of the human body.

2. MRI based on the principles of nuclear magnetic resonance (NMR), which is a spectroscopic technique used to obtain microscopic chemical and physical data about molecules.

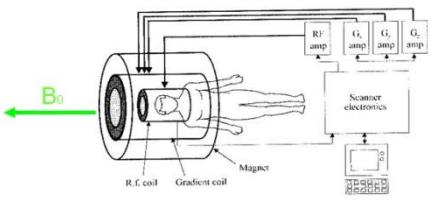
3. The magnetic resonance imaging is accomplished through the absorption and emission of energy of the radio frequency (RF) range of the electromagnetic spectrum.

4. The main components in MRI scanner are

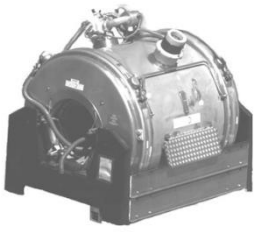
   b. A magnet which produces a very powerful uniform magnetic field.
   c. Gradient magnets which are much lower in strength.
   d. Equipment to transmit radio frequency (RF).
   e. A very powerful computer system, which translates the signals transmitted by the coils.

Equipment





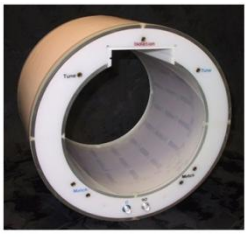
Magnet



Gradient Coil



RF Coil



5. The most important component of MRI scanner is the magnet. The magnets currently used in scanners today are in the 0.5 tesla to 2.0 tesla range (5,000 to 20,000 gauss). Higher values are used for research. (Earth magnetic field = 0.5 gauss)

6. There are three types of magnets used in MRI systems: Resistive magnets, Permanent magnets, super conducting magnets (the most commonly used type in MRI scanners).

7. Principle of MRI

    a. Spinning atoms (hydrogen) face outside magnetic fields,
    b. Energy absorption by RF coils.
    c. Resonance of hydrogen atoms measured by RF antenna which is received the signal is sinusoidal in shape.
    d. Imaging by the computer which receives mathematical data, which is converted through the use of a Fourier transform into an image.

8. Working

    a. Put subject in big magnetic field.
    b. Transmit radio waves into subjects.
    c. Receive radio waves that have been re-transmitted by the subject.
    d. Store the data & repeat step ‘b’.
    e. Process raw data to reconstruct image.
    f. Getting high resolution image.

9. When placed in large magnetic field, hydrogen atoms have a strong tendency to align the in the direction of the magnetic field.

10. Inside the bore of the scanner, the magnetic field runs down the centre of the tube in where the patient is placed, so the hydrogen protons will line up in either the direction of the feer of the head.

11. The majority will cancel each other, but the net number of protons is sufficient to produce an image.

12. The MRI machine applies radio frequency (RF) pulse that is specific to hydrogen. The RF pulses are applied through a coil that is specific to the part of the body being scanned.

13. The pulse directed to a specific area of the body causes the protons to absorb energy and spin in different direction, which is known as resonance.

14. Frequency (Hz) of energy absorption depends on strength of external magnetic field. The resonance frequency, ω_0, is referred to as Larmor frequency. This frequency is needed to excite transverse magnetization (precession).

    Larmor equation: $\boldsymbol{\omega_0 = \frac{\gamma \beta_0}{2\pi}}$; γ is gyromagnetic ratio, β_0 is the magnetic field, and ω_0 is the Larmor frequency.

    The gyromagnetic ratio is given by the equation, $\boldsymbol{\gamma = \frac{-eg}{2m}}$**.** Here, 'm' is the mass of system and 'g' represents the systems g- factor.

    The g-factor is a constant which represents relationship between angular momentum of system to intrinsic magnetic moment of system. The g-factor does not have any unit.

15. Imaging: When the RF pulse is turned off the hydrogen protons slowly return to their natural alignment within the magnetic field and release their excess stored energy. This is known as relaxation.

16. Measuring the MR signal
    a. The moving proton vector induces a signal in the RF antenna.
    b. The signal is picked up by a coil and sent to the computer system. The received signal is sinusoidal in nature.
    c. The computer receives mathematical data, which is converted through the use of a Fourier transform into an image.

4.13 Magnetic Resonance Image Applications

1. MRI (magnetic resonance image) scans are used as an extremely accurate method of detecting disease throughout the body.
2. Head trauma, Brain Aneurysms, Stroke, Tumors of the brain and Spine tumors are all abnormalities that are detected by a MRI scan.

4.14 CT Scan Procedure and Applications

1. CT scan means Computed Tomography scan and the tomography word taken from Greek, the word tomo means "slice" and graphy means "write".

2. Computed Tomography is the process of generating a two-dimensional image of slice or section through a 3-dimensional object (a tomogram).

3. Computed tomography (or computerized axial tomography) is an examination that uses X-ray and computer to obtain a cross – sectional image of the human body.

4. **Parts of a CT scan machine**
   a. Gantry (scanner system)
   b. Patient's table
   c. Computer (operation control)

5. **Scanner system**
   a. **X ray tube:** Generates X-ray beams
   b. **Detector system:** Detects the X-rays passing through the patient's body.
   c. **Collimators:** It narrows the beams of X-rays.
   d. **Filters:** These are used to filter some rays from entering the patient's body that may be harmful.

6. There are 2 types of scanners.
   a. **Conventional:** Scan is taken slice by slice and after slice the scan stops. Eg. From top of abdomen till the pelvis. This needs the patient to hold their breath.
   b. **Spiral scan:** continuous scan taken in spiral fashion. It is quick.



7. Principle

a. When X-rays pass through the human body, some of the rays are absorbed and some pass through the body to produce an image.
b. In plain X-ray imaging, the film directly absorbs penetrated X-rays.
c. In CAT (computer aided tomography) scanning, an electronic device called a 'detector array' absorbs the penetrated X-rays, measures the X-ray amount and transmits the data to a computer system.
d. A sophisticated computer system, in turn, calculates and analyzes data from each detector in each level, and finally reconstructs multiple, two-dimensional, cross-sectional images. Images displayed on screen.
e. CT number measured in Hounsfield unit (HU). CT number or Hounsfield scale is a measure of radio density. Hounsfield scale (linear transformation reference being distilled water).

Tissue	**CT number (HU)**
Bone	+1000
Liver	40-60
White mater	-20 to -30
Grey mater	-37 to -45
Blood	40
Muscle	10-40
Kidney	30
Water	0
Fat	-50 to -100
Air	-1000

8. Applications

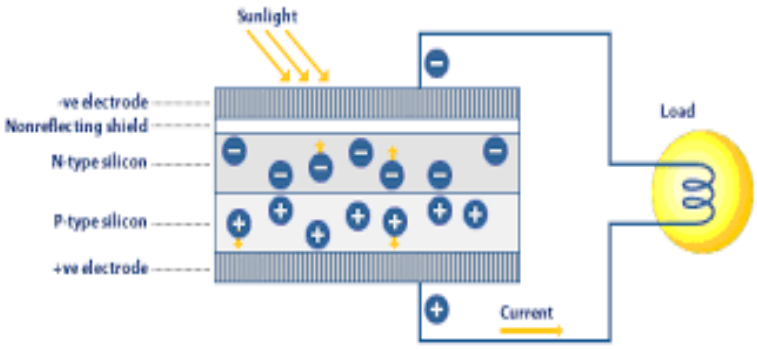
1. CT scans can detect bone and joint problems, like complex bone fractures and tumors.
2. If patients have a condition like cancer, heart disease, emphysema or liver masses, CT scans cam spot it or help doctors see any changes.
3. They show internal injuries and bleeding, such as those caused by a car accident.
4. They can help locate a tumor, blood clot, excess fluid or infection.
5. Doctors use them to guide treatment plans and procedures, such as biopsies, surgeries and radiation therapy.
6. Doctors can compare CT scans to find out if certain treatments are working. For example, scans of a tumor over time can show whether it's responding to chemotherapy or radiation.

4.15 Photovoltaic Cells

Photo conversion: Photo conversion of the solar energy can be done by photo voltaic cells.

PV-cell/ photo voltaic cell

- Solar cell consists of a p-type semiconductor (such as Si doped with B) and n-type semiconductor (such as Si doped with P). They are in close contact with each other to form p-n junction.
- When the solar rays fall on the outer layer of n-type semiconductor, the electrons in the valence band get promoted to the conduction band by absorbing light energy.
- Since the conduction electrons can easily cross the p-n junction into the p-type semiconductor, a potential difference between two layers is created. This potential difference cause flow of electrons (i.e.: an electric current). The potential difference and hence current increases as more solar energy falls on the surface of the outer layer.
- When this p and n layers are connected to an external circuit, electrons flow from n-type to p-type, thereby the current is generated.

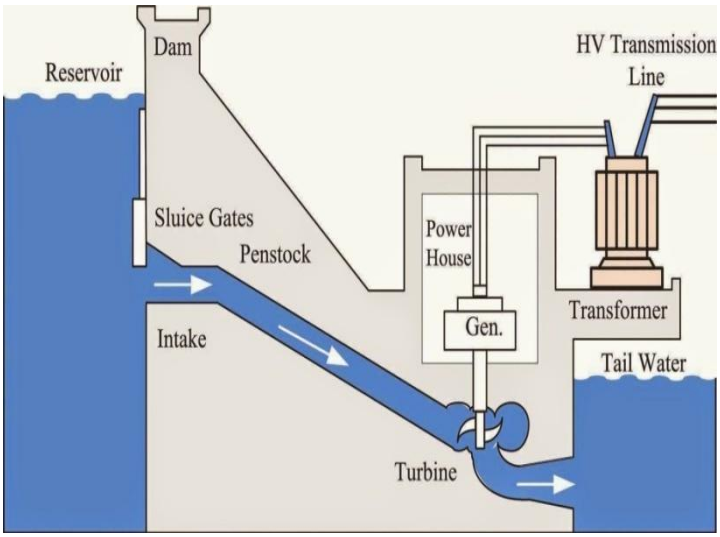


Uses of solar cell

- For producing electricity in artificial satellites, space probes, etc.
- Used to run street lights, water pumps, radio and T.V sets in remote areas.
- For operating electronic / quartz watches, calculators and small instruments.

4.16 Hydropower

- It is the energy generated from flowing and falling water.
- Water is stored in dams built across flowing river. This raises the water level thereby increasing its potential energy.
- The water falls on hydraulic turbines which convert pressure and kinetic energy of water into rotational kinetic energy.
- The rotating shaft of the turbine drives the generator which transforms mechanical energy into electrical energy.
- Electrical energy is then transmitted using transmission line



Advantages

- At present hydroelectricity is the only renewable energy source used to generate electricity on large scale
- It does not cause pollution and is more reliable than wind or solar or wave energy.

Disadvantages

- It involves construction of dams which is very expensive.
- Construction of dams leads to submerging of low laying areas, loss of forest and agricultural land and need to rehabilitate the people.
- The impounded water also increases the seismicity of the region.

4.17 Geothermal power

- The word geothermal comes from the Greek words geo (earth) and thermal (heat). So geothermal energy is the heat producing from inside the earth.
- The energy harnessed from the hot rocks present inside the earth is called geothermal energy.

Geothermal Power Plants

- A geothermal power plant uses steam obtained from these geothermal reservoirs to generate electricity.
- Wells are drilled at the appropriate locations to bring this geothermal energy up to the surface
- A mixture of steam and water is collected from the production well.
- Steam separators are employed to separate the steam and use it to operate turbines.
- These turbines run the generators and hence, electricity is generated.
- The condensed steam and the water collected from the production well are injected back into the reservoir through the injection well.

Types of Geothermal Power Plants

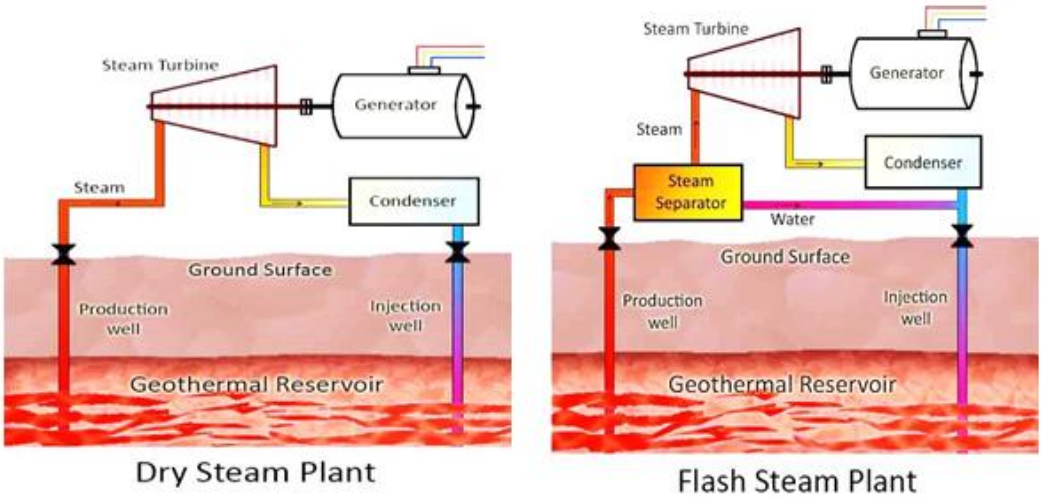
1. Dry Steam Plant

- This is the simplest and oldest type of geothermal plant.
- It directly uses steam from the reservoir to operate the turbine.

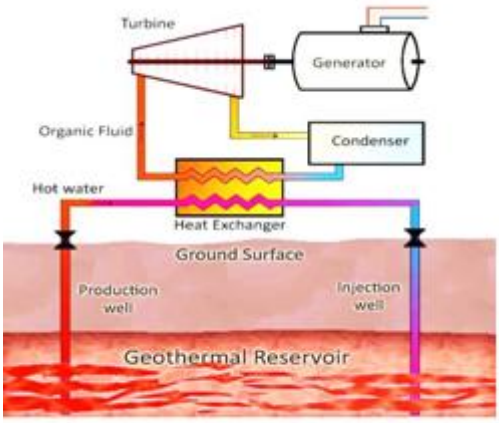
- The steam is collected from the production well and used to operate low-pressure turbines. Hence, the working fluid is steam.
- The used steam is then condensed and injected back through the injection well.

2. Flash Steam plant

- Flash steam power plants are the most commonly employed geothermal plants today.
- They operate on the geothermal reservoirs having water temperature greater than $180^{O}C$
- The high pressure hot water from the reservoir flows up through the production well due to its own pressure.
- The pressure decreases as the water flows upwards and, hence, some of it gets converted into steam.
- The steam is separated from the water by steam separator and sent to drive the steam turbine.
- The unused water as well as the condensed steam is injected back through the injection well.



3. Binary cycle plant



Binary Cycle Plant

- In these plants, hot water from geothermal reservoir is used to heat up an another organic fluid having a lower boiling point.
- Thus, here, the working fluid is the secondary organic fluid and not the water from reservoir.
- The heat energy from the water is transferred to the working fluid in the heat exchanger. As a result, the working fluid vaporizes, and then drives the turbines.
- The spent fluid passes through the condenser and the cycle repeats.
- The water is injected back into the reservoir through the injection well.

Advantages

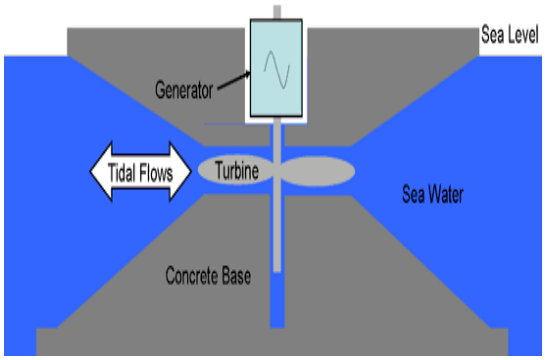
- It is large enough to meet the energy requirement for nearly 3,50,000 years
- Unlike solar and wind energy, geothermal energy is available all the time and does not depend on the weather conditions

Disadvantages

- The technology use to harness is economical
- Continuous extraction of heated water from the ground can lead to subsidence of land in that area.
- Geothermal water is accompanied by dissolved gases that lead to air pollution.

4.18 Tidal and Wave Power

- Ocean tides, produced by gravitational forces of sun and moon, contain enormous amount of energy.
- The "high tide" and "low tide" refer to the rise and fall of water in the oceans.
- The tidal energy can be harnessed by constructing a tidal barrage/dam to hold water during high tides.
- The dam provided with a series of sluice gates to permit the entry of water into the basin during high tides.
- During high tide, water will flow from sea to tidal basin through turbine, thus producing electricity. During low tide, water will flow from tidal basin to sea through turbine producing electricity.
- As the water raises again during high tide, the basin is filled again thus repeating the cycle



Advantages

- It does not cause pollution and it is inexhaustible
- Sustainable energy source
- Reduces the fossil fuel dependence

Disadvantages

- It is generated in regions far away from the load centers there by increasing the cost of transmission

4.19 Ocean Thermal Energy Conversion

- The temperature difference exists between the surface and deep sea water.
- This temperature difference between the surface and deep sea water is utilized to generate electricity.

- The energy harnessed by the use of temperature difference between deep ocean cold water and warm surface water is called OTEC.

Condition

- The temperature difference should be of 20˚C or more is required between surface water and deeper water.

Types of OTEC plants

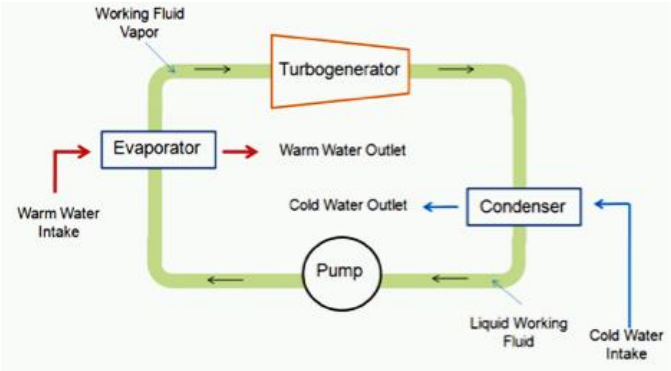
Based on steam generation

- Closed cycle / anderson cycle OTEC plant
- Open cycle / claude cycle OTEC plant
- Hybrid OTEC plant

Closed OTEC plant

- In this type of plant, the working fluid having low boiling point such as ammonia, Freon or propane is evaporated using the worm surface water of the sea.
- The vapors thus formed rotate the turbine thereby generating electricity.
- These vapors are then condensed in a condenser using cold water from deep sea
- It has a large potential for electricity generation

Schematic diagram of closed OTEC



Open OTEC plant

- In this warm seawater from the surface is the working fluid
- In this type of plant, the warm surface water of ocean is evaporated at very low pressure in a flash evaporator which is maintained at partial vacuum enables water to boil at a low temperature

- The vapors thus formed rotate the turbine thereby generating electricity.
- These vapors are then condensed in a condenser using cold water from deep sea.

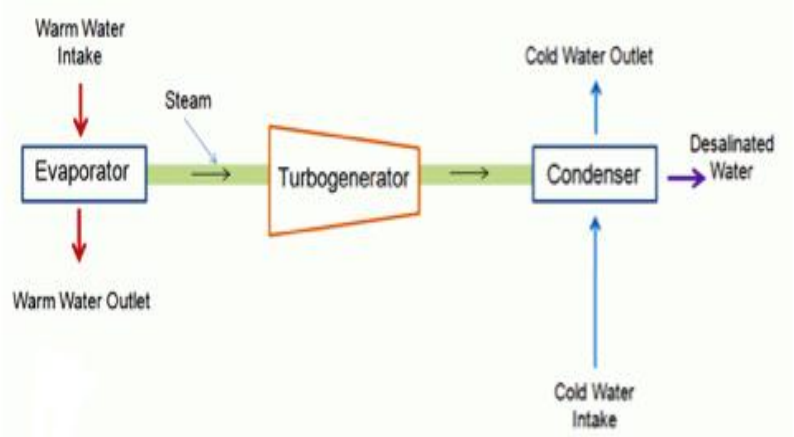
Advantages

- The condensate is desalinated fresh water that can be used as potable water and for irrigation.

Disadvantages

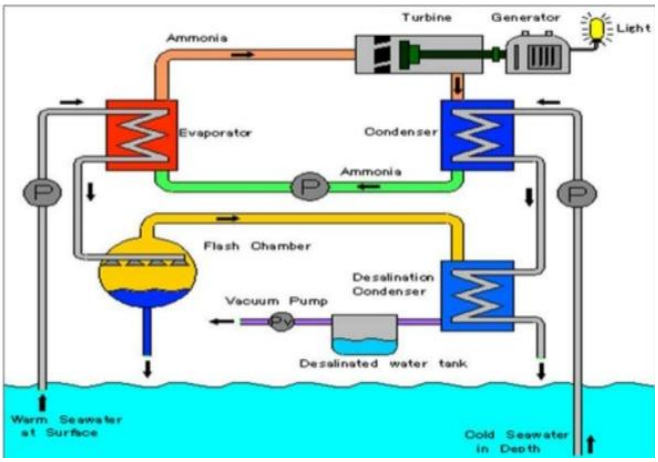
- It is more expensive as it requires expensive degassing and vacuum pumps

Schematic diagram of open OTEC



Hybrid OTEC plant

- The hybrid OTEC cycle combines the both open cycle and closed cycle.
- Like the open cycle, the warms surface ocean water is evaporated at very low pressure in a flash evaporator and this low pressure steam is used to vaporize the low boiling point fluid such as ammonia.
- The vapors thus formed rotate the turbine thereby generating electricity and the steam is condensed to produce potable water.



Advantages

- The OTEC plant supplies energy steadily without being affected by weather.
- It is continuous, renewable and pollution free.

Disadvantages

- The installation is expensive and may cause adverse environmental impact on the aquatic ecosystem
- Capital investment is high
- Due to small temperature difference in between the surface water and deep water, conversion efficiency is low.

Unit V
Advanced Topics/Cocepts in Chemistry

Part-A
Introduction to Computational Chemistry

Introduction

Chemists have been doing computations for centuries, but the field we know today is "Computational Chemistry" is a product of the digital age.

Computational chemistry is used to study the fundamental properties of atoms, molecules and chemical reactions, using quantum mechanics and thermodynamics.

Computational chemists also use simulations to explore the basic physical process underlying phenomena such as superconductivity, energy storage and corrosion or phase changes.

Nobel Prize in Chemistry has been awarded to

- Martin Karplus, Micahel Levitt, Arich Warshel in 2013 for "The development of multi-scale models for complex chemical systems"
- Walter Kohr and John. A. Pope in 1998 for "The development of the density – functional theory and computational methods in quantum chemistry".
- Paul Crutzen, Mario Molina and F. Sherwood Rowland in 1995 for "Constructing mathematical models that used thermodynamic and chemical laws to explain how ozone forms and decomposition in the atmosphere"

5.1 Advantages of Computational Chemistry

- Calculations are easy to perform, where as experiments are often difficult.
- Calculations are becoming low costly, where as experiments are becoming more expensive.
- Calculations can be performed on any system, even those that does not exist, where as many experiments are limited to relatively stable molecules.

- Calculations are safe, where as many experiments have an intrinsic danger associate with them.

Disadvantages of Computational chemistry

1. Calculations are too easy to perform, many black – box programs are available to the uninitiated.
2. Calculations are very expensive in terms of the amount of time required.
3. Calculations can be performed on any system, even those that is does not exist.

5.2 Computed Molecular Properties

- Bond energies, Reaction energies, structure of ground – excited – transition states.
- Atomic energies, electrostatic potentials, vibrational frequencies, Transition energies, intensities of UV and IR spectra.
- Dipole moments, polarizabilities and hyper polarizabilities, reaction pathways and mechanisms.

5.3 Computational Methods

Computational models are mathematical models used to numerically study the behavior of complex systems by means of complex simulation. A computational model can be used to make predictions of the systems behavior under different conditions, often for cases in which intuitive (sensitive) analytical solutions are not available.

There are two broad areas within computational chemistry denoted to the structure of molecules and their reactivity.

1. Molecular mechanics
2. Electronic structure theory

Molecular mechanics: Molecular mechanics simulations use the laws of classical physics to predict the structure and properties of molecules. Molecular mechanics calculations do not explicitly treat the electrons in molecular systems. Instead, they perform computations based upon the interactions among nuclei.

There are many different molecular mechanics methods. Each one is characterized by a force field. The electronic effects are implicitly included in force field via its parameterization.

5.4 Electronic Structure Methods (Theoretical Model)

Electronic structure methods use the law of quantum mechanics rather than classical physics as the basis for their computations. Quantum mechanics states that the energy and related properties of a molecule may be obtained by solving the Schrodinger equation.

$$\hat{H}\psi=E\psi$$

Ψ is wave function; it is a function of the positions of all the fundamental particles in the system.

$\hat{H}$ is Hamiltonian operator. It is operator associated with the observable energy.

E is the total energy of the system

$$\hat{H}=\hat{T}+\hat{V}$$

$\hat{T}$ is kinetic energy operator.

$$\hat{T}=\hat{T}_e+\hat{T}_n$$

$$\hat{T}_e = -\frac{1}{2}\sum_i \Delta_i^2$$

$$\hat{T}_n = -\frac{1}{2M_A}\sum_A \Delta_A^2$$

Δ^2 is the laplacian given by

$$\Delta^2 = \frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}$$

$\hat{V}$ is the potential energy operator.

$$\hat{V}= \hat{V}_{nn}+\hat{V}_{ne}+\hat{V}_{ee}$$

$\hat{V}_{nn}$ - nuclear – nuclear repulsion term

$$\hat{V}_{nn}= \sum_{A>B}\frac{Z_A Z_B}{R_{AB}}$$

$\hat{V}_{ne}$ – nuclear – electron repulsion term

$$\hat{V}_{ne} = -\sum_{iA} \frac{Z_A}{R_{iA}}$$

$\hat{V}_{ee}$ - electron – electron repulsion term

$$\hat{V}_{ee} = \sum_{i<j} \frac{1}{r_{ij}}$$

For any but the smallest, totally symmetric systems, however, exact solutions to the Schrodinger equation are not practical, electronic structure methods are characterized by their various mathematical approximations to the solution.

There are two major classes of electronic structure methods

a. **Semi – empirical methods:** These methods use parameters derived from experimental data to simplify the computation. They solve an approximate from of the Schrodinger equation that depends on having appropriate parameters available for the type of chemical system.

b. **Ab-Initio methods**: Unlike either molecular mechanics or semi-empirical methods these methods not use experimental parameters in their computations. Instead, their computations are based solely on the laws of quantum mechanics – the first principles referred to in the name ab-initio – and on the values of a small number of physical constants.

5.5 Hartree – Fock Theory

Hartree equation is $\psi(r_1,r_2)=\psi_1(r_1).\psi_2(r_2)$. This function is not anti symmetric.

Anti symmetries wave function is called Hartree – Fock wave function.

It can be written as slater determinant.

$$\psi = \frac{1}{\sqrt{N!}} \begin{vmatrix} \psi_1(r_1) & \psi_2(r_1) & \psi_3(r_1) & . & . & . & \psi_n(r_1) \\ \psi_1(r_2) & \psi_2(r_2) & \psi_3(r_2) & . & . & . & \psi_n(r_2) \\ . & . & . & . & . & . & . \\ . & . & . & . & . & . & . \\ . & . & . & . & . & . & . \\ \psi_n(r_n) & \psi_n(r_n) & \psi_n(r_n) & . & . & . & \psi_n(r_n) \end{vmatrix}$$

Linear combination of atomic orbitals $\psi_i(r_i) = \sum_\mu C_{\mu i}\chi_\mu(r_i)$

χ_μ is atomic orbitals or **basis function**.

$C_{\mu i}$ Molecular orbital coefficients

$$E_{HF} = \sum_i H_i + \frac{1}{2}\sum_{ij}\left(J_{ij} - K_{ij}\right)$$

H_i = kinetic energy of the electrons and the nuclear attraction energy.

$$H_i = \langle \psi_i | \hat{h} | \psi_i \rangle$$

$$= \sum_{\mu v} C_{\mu i} C_{vi} \left\langle \chi_\mu \middle| \hat{h} \middle| \chi_v \right\rangle$$

J_{ij} – coloumbic repulsion between the electrons
K_{ij} – exchange of electronic coordinates

Restricted HF theory: The spatial part of the spin orbital are the same (RHF) $\phi_i^\alpha = \phi_j^\beta$

Unrestricted HF –theory: The spatial part of the spin orbital are different (UHF) $\phi_i^\alpha \neq \phi_j^\beta$

5.6 Electron Correlation Energy

The correlation energy is sensitive to changes in the number of electron pairs. The correlation energy is always negative. There are two components to the correlation energy.

a. Dynamic correlation is the energy associated with the pair of the electrons as they try to avoid one another. This is important is bond breaking process.
b. Static correlation arises from deficiencies in the single determinant wave function and is important in systems with stretched bonds and low – lying excited states.

Electron correlation gives rise to inter – electron cusp. (cusp = a point of transition between two different systems)

Computing the correlation energy is the single most important problem in quantum chemistry.

There exist branches of methods to compute the correlation energy, each with their own strengths and weakness.

Eg: Configuration interaction (CISD, CISDCT); Mϕller – plesset perturbation theory (MP2, MP3) & Density functional theory (DFT)

5.7 Basis Function

The atom centered functions used to desire the atomic orbitals are known as basis function and collectively form basis set. Larger basis set give a better approximation to the atomic orbitals as the place fewer restriction on the wave function. Larger basis set attract a higher computational cost.

Eg: STO – 3G, 3 – 21 G, 6 – 31 G, 6 – 311 G, 6 – 311 G (d,p)

5.8 Density Functional Theory (DFT)

The electron density is a fundamental quantity in quantum chemistry.

$$\rho(r_1)= \int \psi^*(r_1,r_2,\ldots\ldots r_n)dr \,.\, \psi(r_1,r_2,\ldots\ldots r_n)dr_1\ldots\ldots dr_n$$

$\rho(r)dr$ = probability of finding an electron in the volume element dr.

It is a function of three variables (x, y, z) and is therefore easy to visualize.

A density functional takes the electron density and returns a number.
$N[\rho] = \int \rho(r)dr$

Density functional theory (DFT) focuses on functions that return the energy of the system.

$$E = E_T + E_V + E_J + E_K + E_L$$

E_T = kinetic energy, E_V = electron – nucleus attraction, E_J = coloumb term, E_K = Hartree –Fock exchange using the DFT density, E_L = Energy correlation

B3LYP (Becke, 3-parameter, Lee–Yang–Parr) method is the most popular density functional that is used and yields very good structural and thermo chemical properties.

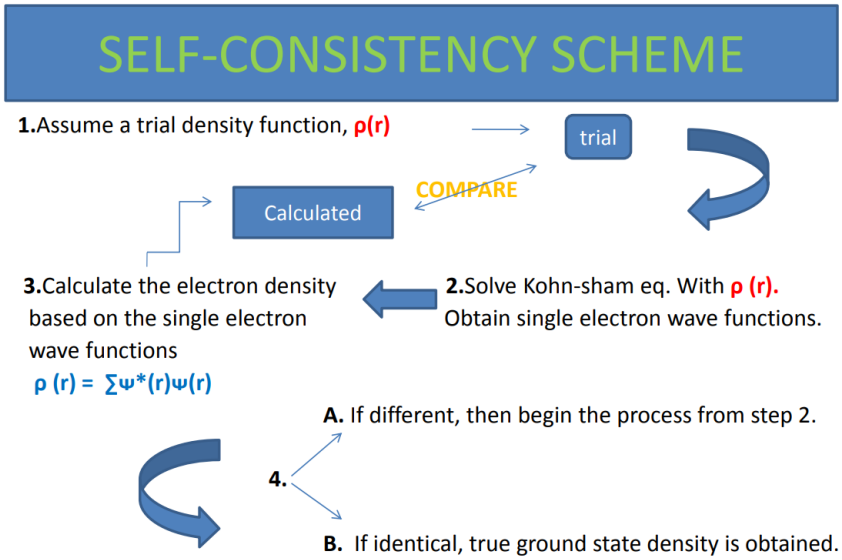
Kohn – Sham Equation

$$\left[\frac{-h^2}{2m}\nabla^2+V_{ion}(r)+V_H(r)+V_{XC}(r)\right]\psi_i(r)=\varepsilon_i\psi_i(r)$$

$V_H(r)=e^2\int\frac{n(r')}{|r-r'|}d^3r'$: Hartree potential of the electron charge density

$V_{XC}(r)=\frac{\delta E_{XC}[n(r)]}{\delta n(r)}$: exchange –correlation potential

$E_{XC}[n(r)]$: exchange – correlation functional



5.9 Applications of DFT calculations

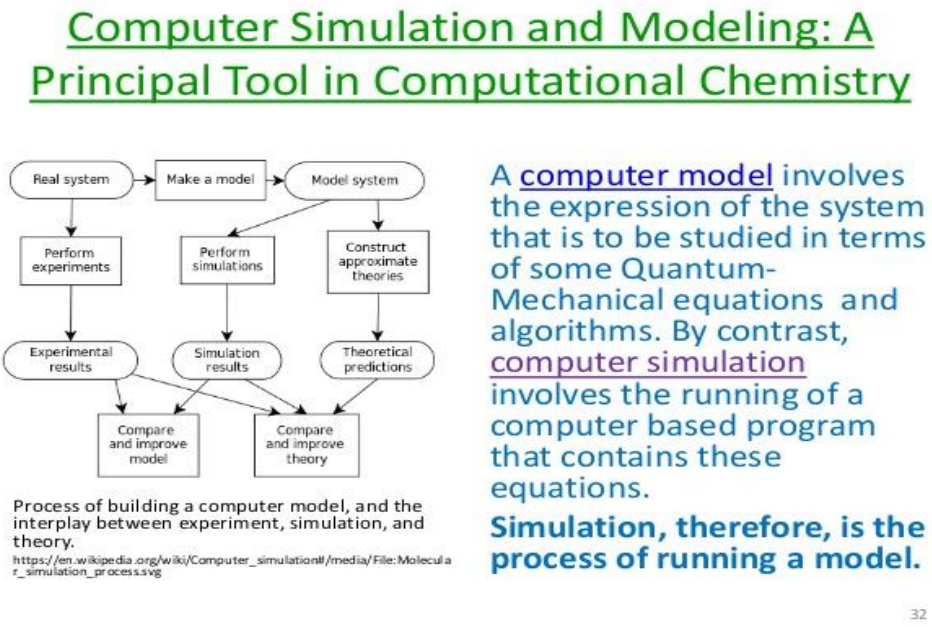
Molecular geometries: Bond angles, Bond lengths, Dihedral Angles, Transition – State Structures (KIE) etc.

Energy changes: Atomization energies, Heats of formation and heat of reaction, energy changes related with isothermal reactions and bond – separation energies, isomerization energies, energy difference between conformers, rotational barriers, activation energies etc.

Other Properties: Dipole moments, vibtational frequencies, entropies, Gibbs free energy of salvation, NMR shielding constants; etc. & Hydrogen bonding.

5.10 Advantages and Disadvantages of DFT

Advantages of DFT	Disadvantages of DFT
Low cost computational cost. Good accuracy for structures and thermo chemistry. The density is conceptually simpler than ψ.	Can fail spectacularly and unexpectedly No systematic way of improving the results. Integral require numerical quadrature grids.



5.11 Molecular Modeling

Importance of Molecular Modelling

The 1998 Nobel Chemistry Prize was awarded to Pople and Kohn for their work in Computational Chemistry and Molecular Modelling.

The 1999 Nobel Chemistry Prize was awarded to Ahmed Zewail for his work in developing spectroscopic methods for studying reactions and in particular transition states, an essential aspect of molecular modelling.

Modelling is an essential component of the **Grand challenge** of **Dial-a-Molecule**.

What is molecular modeling?

Molecular modelling is a collection of (computer based) techniques for deriving, representing and manipulating the structures and reactions of molecules, and those properties that are dependent on these three dimensional structures.

Computational modelling methods used nowadays as standard tools by organic chemists for searching for, rationalising and predicting structure and reactivity of organic, bio-organic and organometallic molecules.

The emphasis will be on helping to develop a feel for the correct "tool" to use in the context of a typical problem in structure, activity or reactivity, by describing the limitations and strengths of each method.

The techniques described include

1. Molecular visualisation
2. Methods for Equilibrium and transition state geometry location
3. Introduction to Molecular mechanics methods
4. Applications of Semi-empirical, ab initio and density functional molecular orbital methods
5. Methods for topological analysis of wavefunctions

Applications

1. Molecular modelling methods are now used routinely to investigate the structure, dynamics, surface properties, and thermodynamics of inorganic, biological, and polymeric systems.
2. The types of biological activity that have been investigated using molecular modelling include protein folding, enzyme catalysis, protein stability, conformational changes associated with biomolecular function, and molecular recognition of proteins, DNA, and membrane complexes.

5.12 Docking Studies

In the field of molecular modeling, **docking** is a method which predicts the preferred orientation of one molecule to a second when bound to each other to form a stable complex. Knowledge of the preferred orientation in turn may be used to predict the strength of

association or binding affinity between two molecules using, for example, scoring functions.

The associations between biologically relevant molecules such as proteins, peptides, nucleic acids, carbohydrates, and lipids play a central role in signal transduction. Furthermore, the relative orientation of the two interacting partners may affect the type of signal produced (e.g., agonism vs antagonism). Therefore, docking is useful for predicting both the strength and type of signal produced.

Molecular docking is one of the most frequently used methods in structure-based drug design, due to its ability to predict the binding-conformation of small molecule ligands to the appropriate target binding site.

Characterisation of the binding behaviour plays an important role in rational design of drugs as well as to elucidate fundamental biochemical processes.

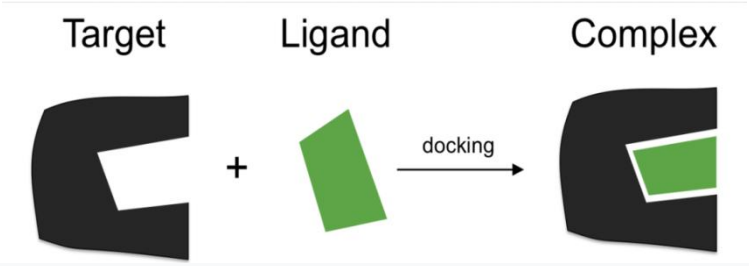


Fig: Docking of a small molecule (green) into the crystal structure of the beta-2 adrenergic G-protein coupled receptor (PDB: 3SN6)

Docking approaches: Two approaches are particularly popular within the molecular docking community. One approach uses a matching technique that describes the protein and the ligand as complementary surfaces. The second approach simulates the actual docking process in which the ligand-protein pairwise interaction energies are calculated.[9] Both approaches have significant advantages as well as some limitations. These are outlined below.

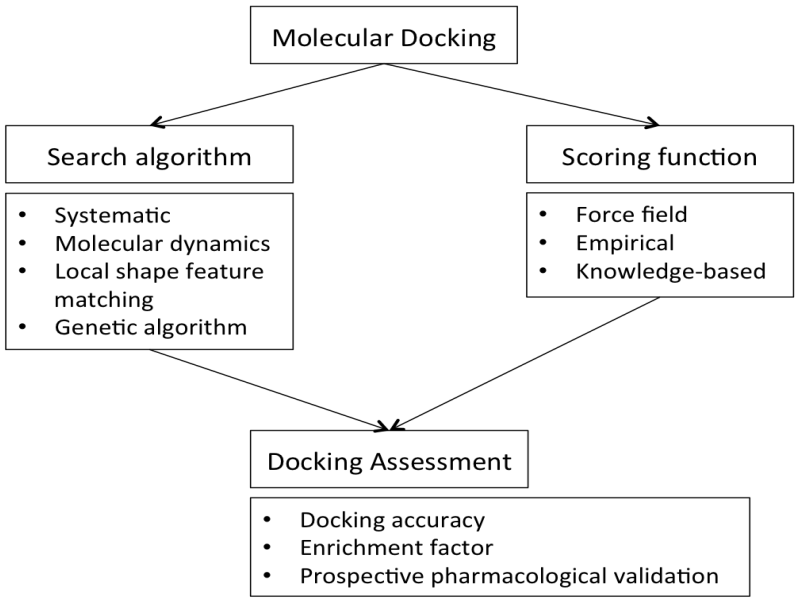
1. Shape complementarity

Geometric matching/ shape complementarity methods describe the protein and ligand as a set of features that make them dockable. These features may include molecular surface / complementary surface descriptors. In this case, the receptor's molecular surface is

described in terms of its solvent-accessible surface area and the ligand's molecular surface is described in terms of its matching surface description. The complementarity between the two surfaces amounts to the shape matching description that may help finding the complementary pose of docking the target and the ligand molecules.

2. Simulation

Simulating the docking process is much more complicated. In this approach, the protein and the ligand are separated by some physical distance, and the ligand finds its position into the protein's active site after a certain number of "moves" in its conformational space. The moves incorporate rigid body transformations such as translations and rotations, as well as internal changes to the ligand's structure including torsion angle rotations. Each of these moves in the conformation space of the ligand induces a total energetic cost of the system. Hence, the system's total energy is calculated after every move.



Part-B
Molecule Motors

Molecule motors are molecules that are capable of rotary motion around a single or double bond.

Single bond rotary motors are generally fueled by chemical reactions whereas double bond rotary motors are generally fueled by light.

The rotation speed of the motor can also be tuned by careful molecular design.

Molecular machines

Molecular – level machines

1. Motors operate as a result of a chemical reaction.
2. The nuclear displacement must be of large amplitude.
3. The motor consists of a mobile and a stationary part (attached on a surface or relative to the bulkiest part of the molecule)
4. Machines contain auxiliary components, eg: Gear: Star shaped molecules and Brake: Steric crowding
5. Interesting chemical reactions for nuclear motions

    a. Reversible reactions or reactions which are part of cycle.
    b. Isomerization involving – N =N - , - C = C- bonds
    c. Acid –base and redox reactions for the making/ breaking of bonds.
    d. Metal – ligand reactions causing the formation or disruption of coordination bonds.

5.13 Characterization of Molecular Level Machines Through

1. The kind of energy supply light (hν), electrochemical energy (e^-) → waste free, easy to control – chemical energy (ATP, H^+, EDTA….) → waste, influence of the reaction product on the equilibrium.
2. The kind of movement performed linear movements, rotary motions, assembly/ disassembly.
3. The kind of operation control (spectroscopy).
4. The possibility of repeating the operation at will
5. The time scale needed for cycle (for picoseconds to days)
6. The function performed.

5.14 Applications of Molecular Machines

1. These are used as catalysts in novel chemical reactions.
2. These are chemical sensors to recognize oxoanions over halides. This novel approach to molecular recognition opens up the way to develop new sensory devices.
3. Polymer based rotaxenes used in corneal repair surgery of rabbits without any inflammation and rejection. This shows the future possibilities in clinical applications of humans.
4. Researchers reported that even a single catenane moiety to a polymer chain already influenced the properties of the polymer. Future design of mechanically interlocked polymers could therefore rely on less catenane moieties for getting the desired physical properties.
5. It suggests that a use for this rotaxene in drug delivery applications mainly in anti-cancer drugs. This anti-cancer drug specifically targets tumor cells and penetrate these cells to inhibit cell growth and to induce cell death.
6. Rotaxene probe could find application in human medicine in for example fluorescence-guided surgery. The probe is used to mark all the cells infected with bacterial and a surgeon can use the correct imaging techniques to find and remove all the cells.
7. These systems are finding future application in switches based on an amino acid backbone, in which the bio-properties of the peptide can be regulated by tuning the pH value of the solution.

5.15 Prototypes

1. Linear motions in rotaxanes

Due to their peculiar structure, at least two interesting molecular motions can be predict in rotaxanes, those are (i) Translation , (ii) Rotation Translation means shuttling of the ring along the axle.



Rotation means rotation of the macrocyclic ring around the axle.



Hence, rotaxanes are good prototypes for the construction of the both rotary and linear molecular motors.

If two identical recognition sites for the ring i.e., 'stations' are located within the dumbbell component, the result is a degenerate, conformational equilibrium state in which the ring spontaneously shuttles back and forth along the axle.

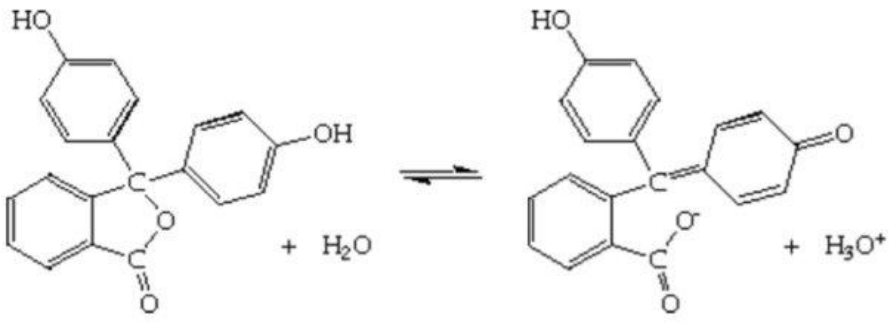
When two recognition sites on the dumbbell component are different, a rotaxane can exist as two different equilibrating conformations, the molecular ring moves by Brownian motion.

In appropriately designed rotaxanes, the switching process can be controlled by reversible chemical reactions (protonation – deprotonation, reduction – oxidation, isomerization) caused by chemical, electrochemical or photochemical simulation.

2. Acid – base controlled molecules shuttle (switch)

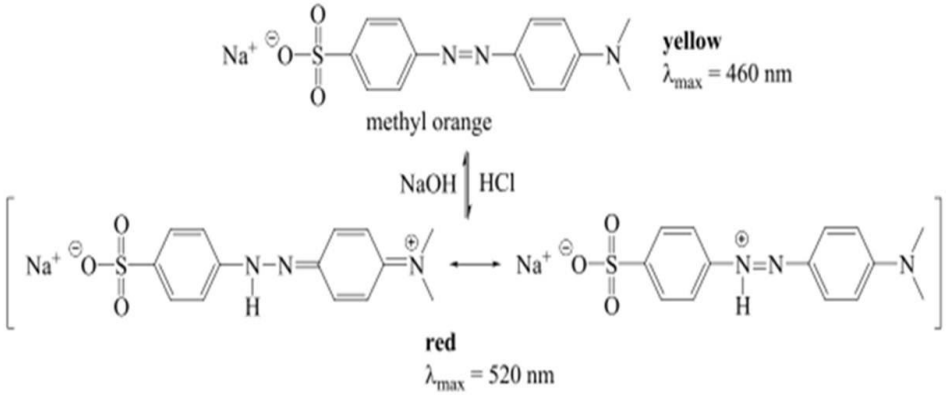
A molecular shuttle or switch is a molecule that can be reversibly shifted between two or more stable states. The molecules may be shifted between the states in response to environment encouragement, such as change in pH, light, temperature, an electric current, microenvironment, or in the presence of ions and other ligands. The oldest forms of synthetic molecular switches are pH indicators, which display distinct colours as a function of pH. When an acid or base is added, the equilibrium between the two forms will be displaced.

Eg 1: Phenolphthalein



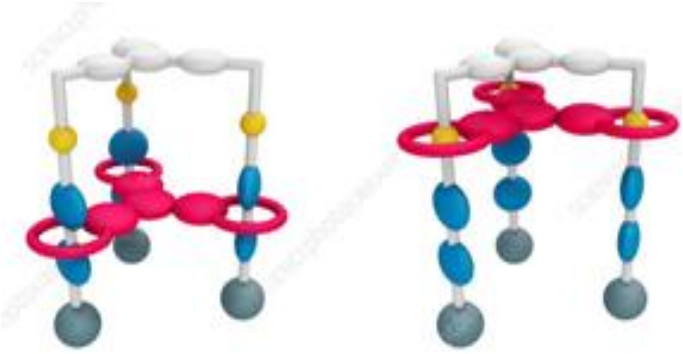
Colourless (acid) Pink (base)

Eg 2: Methyl orange



3. Molecular elevator

Molecular model of a molecular elevator, a type of molecular machine. The elevator is made from small organic molecular subunits. The platform like component is interlocked with a tripodal riglike component. Switching the pH of the medium around the molecular elevator from acidic (right) to alkaline (left) makes the platform move between two recognition units on the tripod.



Base Acid

4. An autonomous light – powered molecular motor

A widely studied class are light powered molecular motors are photochromic compounds which are able to switch between electronic configurations when irradiated by light of a specific wavelength. Each state has a specific absorption and maximum. Members of this class include azobenzenes, diarylethenes, spiropyrans and phenoxy napthacene quinones.

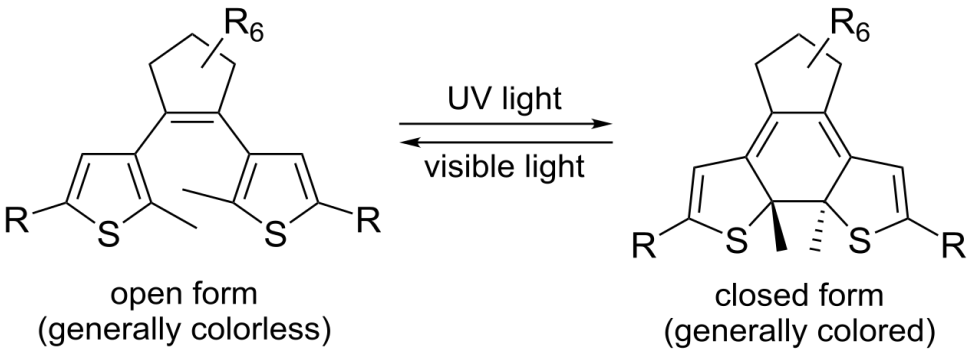


Fig. Switching in dithienyl ethane

Chiroptical molecular motors are a specific subgroup with photochemical switching taking place an enantiomeric pairs (optical isomers). Hindered alkenes such as the one represent below change their helicity(planar chirality) as response to irradiation with right or left-handed circularly polarized light. (r-cpl and l-cpl)

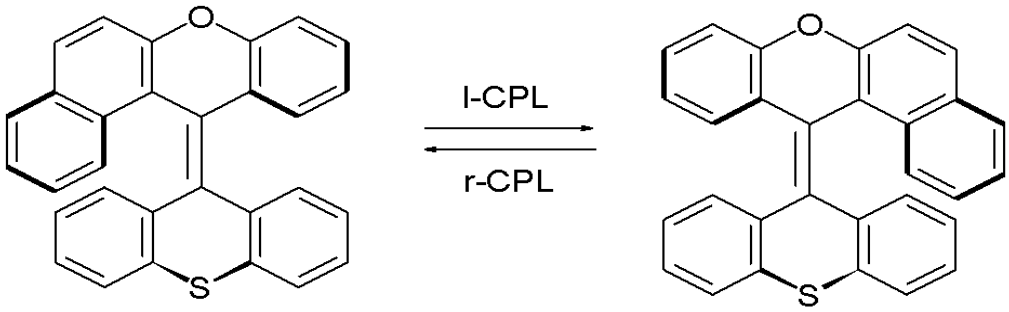
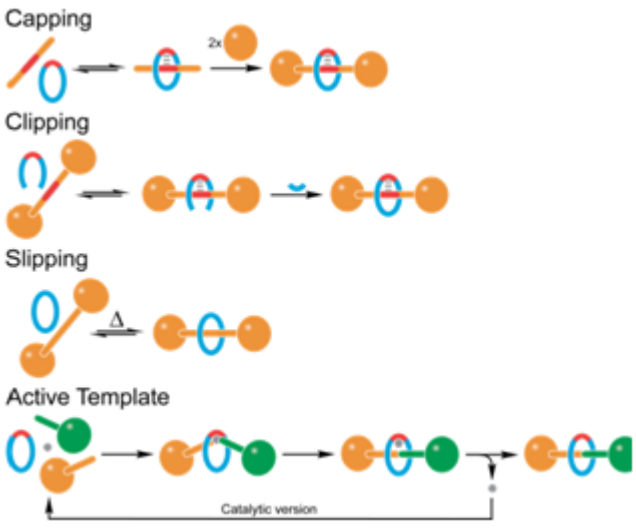
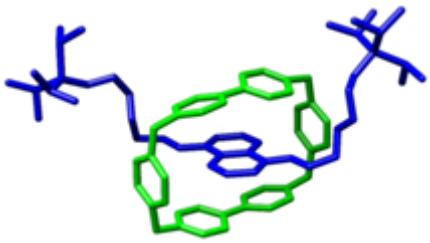


Fig. Switching in hindered alkene

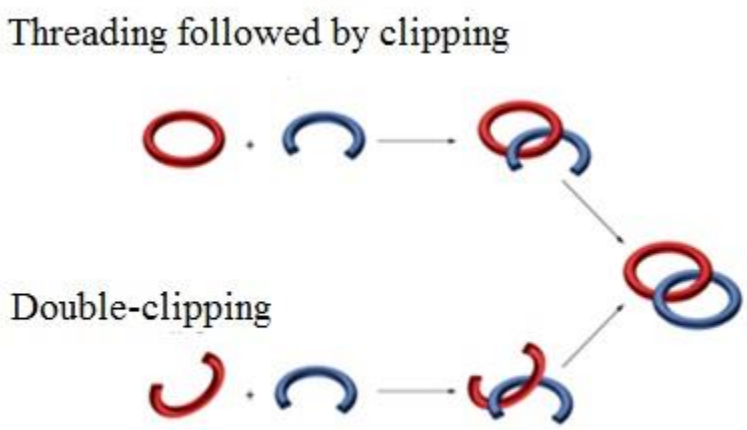
5.16 Preparation of Rotaxane



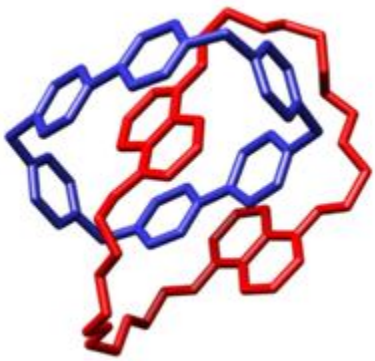
Structure of Rotaxane



5.17 Preparation of Catenane



Structure of Catenane



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